Armoede Commuity Development Trust
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Armoede Disaster Risk Reduction

and

Resilience Implementation Plan

2025–2030

Building resilience and safeguarding community development through local leadership and coordinated action.

ARMOEDE COMMUNITY DEVELOPMENT TRUST (ACDT)
Ga-Sekhaolelo Village
Ward 6
Mogalakwena Local Municipality
Waterberg District
Limpopo Province
Republic of South Africa

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Declaration of Adoption

This Disaster Risk Reduction and Resilience Implementation Plan for Armoede (2025–2030) has been developed in accordance with the Disaster Management Act 57 of 2002, the National Disaster Management Framework (2005) and the principles of cooperative governance. The plan reflects the real risk conditions, capacities and development priorities of the Armoede community and is based on participatory risk assessment, community consultation and technical disaster risk logic.

This plan is authentic and original to the community of Armoede (Ga-Sekhaolelo) and expresses local disaster risk priorities informed by lived experience, vulnerability analysis and community resilience strategies. All information contained herein has been verified through community engagement sessions and field-based hazard and risk analysis.

The plan is a working implementation framework and will be reviewed annually to ensure alignment with evolving risk conditions, development changes and community priorities. The Armoede Community Development Trust serves as the official custodian and coordination authority of this plan on behalf of the community.

We, the undersigned, hereby adopt the Armoede Disaster Risk Reduction and Resilience Implementation Plan (2025–2030) as the official disaster risk management framework for Armoede. We endorse this plan as a community-led, development-focused approach that reduces risk, safeguards livelihoods and strengthens resilience. We commit to its implementation, review and continuous improvement.

Signed on behalf of the Armoede Community

Name (in full):
Position: Chairperson: Armoede Community Development Trust (ACDT)
Signature: **Date:**

Name (in full):
Position: Representative: Ga-Sekhaolelo Traditional Council
Signature: **Date:**

Name (in full):
Position: Ward Committee Representative – Ward 6
Signature: **Date:**

Acknowledgements

The Armoede Community Disaster Risk Reduction and Resilience Implementation Plan (2025–2030) was developed through the active participation and commitment of the people of Armoede (Ga-Sekhaolelo). This plan is a result of community knowledge, lived experience and collective determination to reduce disaster risk and safeguard our future.
Special appreciation is extended to:

- The Ga-Sekhaolelo Traditional Council, for governance guidance and support
- The Armoede Community Development Trust (ACDT), for leadership and coordination
- The Disaster Management Volunteer Force (DMVF), for operational support and field assessments
- Community men, women, elders and youth, whose voices shaped this plan
- Schools, clinic staff and water committee members, for their contributions to risk identification
- Partner institutions and technical advisors, who provided valuable input during development

This plan belongs to the community of Armoede. It reflects our resilience, our priorities and our shared commitment to protect life, livelihoods and development.

Dedication

When the storms come, a rooted community does not fall. It bends, learns, and rises stronger.

This Disaster Risk Management Plan is dedicated to the people of Armoede, whose resilience, courage, and unity continue to inspire collective action for a safer and more sustainable future.

It honours the elders who have preserved wisdom through generations, the youth who carry the torch of renewal, and the families and volunteers who stand together when adversity strikes.

Special recognition is extended to the members of the Armoede Community Development Trust (ACDT) and the Disaster Management Volunteer Force (DMVF) for their unwavering commitment to safeguarding lives, livelihoods, and local heritage.

May this Plan serve not only as a guide for managing disaster risks but as a living testament to the strength of community, the power of preparedness, and the shared vision of development that protects and uplifts all.

Armoede Community Development Trust (ACDT)

October 2025

Acronyms and Abbreviations

Acronym	Meaning
ACDCF	Armoede Community Disaster Coordination Forum
ACDT	Armoede Community Development Trust
DMVF	Disaster Management Volunteer Force
DRM	Disaster Risk Management
DRMP	Disaster Risk Management Plan
DRR	Disaster Risk Reduction
EMS	Emergency Medical Services
FBO	Faith-Based Organisation
IFRC	International Federation of Red Cross and Red Crescent Societies
KPA	Key Performance Area
LM	Local Municipality
MEL	Monitoring, Evaluation and Learning
MoU	Memorandum of Understanding
NDMF	National Disaster Management Framework
NGO	Non-Governmental Organisation
SAPS	South African Police Service
UNDRR	United Nations Office for Disaster Risk Reduction
WDM DMC	Waterberg District Disaster Management Centre

Foreword

Disaster risk in Armoede is not a theoretical concept. It is a daily lived reality that affects the safety, dignity and development of our people. For years, our community has faced recurring challenges—seasonal flooding, soil erosion, household fires, water service interruptions, dust exposure from mining traffic and the quiet suffering caused by poverty and vulnerability. Climate change has intensified these pressures, stretching already fragile coping systems. Despite this, Armoede has never been passive. We endure. We rebuild. We adapt. But resilience without structure is not enough. That is why we have developed this Disaster Risk Reduction and Resilience Implementation Plan (2025–2030)—a practical, community-driven framework to protect lives, livelihoods and our shared future.

This plan is born from the voices and experiences of the people of Armoede. It reflects our challenges as well as our strengths. It acknowledges difficult truths. That development has been slow to reach rural communities like ours, that disaster risk is often ignored until people are displaced or suffer loss, and that our village has long struggled without adequate investment in safe roads, stormwater control, water security and emergency response services. Yet this plan rejects a future defined by vulnerability. Instead, it positions disaster risk reduction as a development priority and a community duty. It affirms that while government support is necessary, the responsibility for reducing risk begins here; at home, with us.

This plan is not a policy document for shelves. It is a working implementation tool. It establishes a clear leadership and accountability model with the Armoede Community Development Trust (ACDT) as the official custodian and coordination authority. It empowers the Disaster Management Volunteer Force (DMVF) as the operational arm of disaster risk management in the community. It recognises the role of traditional leadership, ward structures, schools, the clinic, the water committee and local partners in strengthening resilience. Most importantly, it introduces predictable early warning systems, coordinated response structures, hazard-specific preparedness and realistic recovery strategies.

Partnerships are welcome, but uncoordinated interventions are not. This plan introduces order. It makes it clear that support must align with local leadership and follow approved risk reduction priorities. It turns disaster management from an event-based response into an organised system built on prevention, preparedness and resilience. With this plan, Armoede moves from being reactive and vulnerable to proactive and resilient.

On behalf of the Armoede Community Development Trust, I affirm our commitment to implement this plan with discipline, transparency and unity. We extend our appreciation to every household, youth leader, volunteer, elder and partner who contributed to its development. This is not just a disaster plan. It is a statement of self-determination. It is our roadmap to a safer future.

Together, we will build a resilient Armoede.

_____

Charles Matlou

Chairperson: Board of Trustees
Armoede Community Development Trust (ACDT)

Date: _____ 2025

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1. Introduction

1.1 Background

Armoede (Ga-Sekhaolelo) is a rural resettled village located in Ward 6 of Mogalakwena Local Municipality under the jurisdiction of the Mapela Traditional Authority in the Waterberg District of Limpopo Province. Like many rural communities in South Africa, Armoede is exposed to a range of disaster risks driven by environmental, climatic, social and infrastructural factors. These include seasonal flooding in low-lying homesteads, veld fires, lightning strikes, water scarcity, heatwaves, strong winds, road accidents, outbreaks of communicable diseases, and socio-economic vulnerabilities such as unemployment and poverty.

These risks threaten lives, livelihoods, community assets, livestock, water infrastructure, and local economic activity. In response, the community, under the leadership of the Ga-Sekhaolelo Traditional Council and the Armoede Community Development Trust (ACDT), has formally established a local Disaster Risk Management (DRM) system anchored by the Disaster Management Volunteer Force (DMVF). This Community-Based Disaster Risk Management (CBDRM) Plan provides the framework to manage, reduce and respond to disaster risks in a structured, coordinated and proactive manner.

1.2 Purpose of the Plan

1.2.1 Purpose

The purpose of the Armoede CBDRM Plan is to provide a structured and coordinated approach for the prevention, mitigation, preparedness, response and recovery from disaster risks at community level. It establishes the DRM system for Armoede, formalises DRM arrangements and procedures, and ensures alignment with the Disaster Management Act, 57 of 2002 (as amended) and the National Disaster Management Framework (2005).

1.2.2 Objectives

Although this section is titled "Purpose of the Plan", it also sets out the objectives to give clarity on outcomes the Plan seeks to achieve. The objectives are to:

- Protect lives, property, livelihoods and community assets from disaster risks.
- Identify and prioritise hazards and vulnerabilities affecting Armoede.
- Strengthen prevention and mitigation measures to reduce risk exposure.
- Establish early warning and preparedness systems at community level.
- Formalise roles, responsibilities and communication arrangements for coordinated disaster response.
- Strengthen recovery and rehabilitation processes after emergency events.
- Enable collaboration between the community, government and development partners.
- Support mobilisation of technical resources and funding for risk reduction and resilience.

These objectives ensure that DRM becomes a community-owned, structured and sustainable system rather than a reactive activity.

1.3 Legal and Policy Mandate

1.3.1 Legislative Framework

Disaster Risk Management in South Africa is guided by a clear legal and policy framework. The key instruments that provide the mandate for this Plan are summarised in Table 1.1 below. Before the table is presented, it is important to emphasise that Section 52 of the Disaster Management Act obligates municipalities to develop and implement disaster management

plans that include community participation. Similarly, Enabler 1 of the National Disaster Management Framework (2005) specifically promotes community-based disaster risk management as a cornerstone of local risk reduction.

Table 1.1: DRM Legal and Policy Framework Applicable to this Plan

Legal/Policy Instrument	Relevance to the Armoede DRM Plan
Disaster Management Act, 57 of 2002 (as amended)	Provides the legal mandate for disaster risk reduction, preparedness, response and recovery; Section 52 mandates local planning.
National Disaster Management Framework (2005)	KPA 1–4 and Enabler 1 guide DRM planning, institutional arrangements and community participation.
Constitution of the Republic of South Africa, 1996	Section 41 promotes cooperative governance; Section 152 promotes community involvement in local governance.
Municipal Systems Act, 32 of 2000	Integrates DRM with local development planning and community participation.
Municipal Structures Act, 117 of 1998	Defines ward and local governance roles in disaster response.
Traditional Leadership and Governance Framework Act, 41 of 2003	Protects the role of traditional leadership in communal governance, including DRM.
Sendai Framework for Disaster Risk Reduction (2015–2030)	Aligns local DRM planning to global best practice; emphasises reducing risk and building resilience.

The instruments above establish DRM as a legal requirement and not a voluntary activity. They formalise planning obligations and enable Armoede to cooperate with local and district authorities during disaster events.

1.3.2 DRM Planning Levels in South Africa

In South Africa, DRM planning is structured into four maturity levels to ensure progressive development of local DRM systems. These levels are outlined in Table 1.2 below.

Table 1.2: DRM Planning Levels in South Africa

DRM Plan Level	Description	Scope
Level 1 DRM Plan	Establishes basic risk profile, institutional arrangements and preparedness procedures	Community, Ward, Local Municipality
Level 2 DRM Plan	Introduces risk reduction projects, contingency planning and early warning integration	Local and District Government
Level 3 DRM Plan	Aligns DRM with development planning, includes budgeting and SOPs	District and Provincial structures
Level 4 DRM Plan	Full multi-hazard DRM system with data, GIS, scenarios and monitoring	Advanced municipalities and national systems

These levels demonstrate that DRM capacity is built progressively. Plans evolve from basic emergency readiness (Level 1) to full-scale DRM systems (Level 4).

1.3.3 Positioning of this Plan

The Armoede (Ga-Sekhaolelo) Plan is a Level 1 Community-Based DRM Plan. It establishes:

- Community DRM structures and responsibilities.

- Basic risk identification and priority hazards.
- Community preparedness and activation procedures.
- Linkages with Ward 6, Municipality and District DRM structures.

This Plan provides the foundation for future upgrades to Level 2 and Level 3 plans, which will include formalised risk reduction projects, standard operating procedures, and disaster financing mechanisms.

1.4 Scope and Application

This Plan applies to all residents, institutions, schools, farms, community facilities, organisations, and local economic actors within Armoede (Ga-Sekhaolelo). It covers all phases of DRM, that is prevention, mitigation, preparedness, response, recovery and rehabilitation, and applies to both rapid-onset and slow-onset hazards. The Plan is binding on all DRM role players in the community and forms part of Ward 6 DRM arrangements.

1.5 Institutional Endorsement and Authority

The development and implementation of this Plan are endorsed by the recognised leadership and development governance structures of Armoede, as shown in Table 1.3.

Table 1.3: Institutional Endorsement of the Armoede DRM Plan

Authority	Role
Ga-Sekhaolelo Traditional Council	Provides traditional governance authority and emergency mobilisation power
Armoede Community Development Trust (ACDT)	Development coordination and DRM accountability
Disaster Management Volunteer Force (DMVF)	Operational lead for DRM implementation

This endorsement establishes formal legitimacy, enabling collaboration with government, private sector, and funding partners to implement the DRM system.

1.6 Alignment with Government DRM Structures

The Plan is aligned with South Africa's DRM governance architecture and follows cooperative governance principles. It links with:

- Ward 6 Councillor and Ward Committee
- Mogalakwena Local Municipality DRM Unit
- Waterberg District Disaster Management Centre (DDMC)
- Limpopo Provincial Disaster Management Centre
- National Disaster Management Centre (as needed)

This alignment ensures access to support, escalation pathways and resources during major incidents.

1.7 Methodology Used

This Plan was developed through a participatory and evidence-based approach, combining local knowledge and technical guidance, as summarised in Table 1.4.

Table 1.4: Methodological Approach

Method	Application
Community workshops	Validation of hazards and risks
Household risk	Identification of vulnerable

profiling	households
Field assessments	Mapping of risk zones
Stakeholder consultations	Structured engagement with local leadership
Document review	Compliance with legal and policy frameworks

This approach ensures both community ownership and technical credibility of the Plan.

2. Community Profile

The purpose of this chapter is to provide a clear, evidence-led picture of Armoede (Ga-Sekhaolelo)'s location, history, environment and settlement structure so that subsequent risk analysis is grounded in the real context of the village. The section draws on community inputs, the Armoede Integrated Community Development Plan (ICDP 2020–2035), and applicable municipal planning references.

2.1 Geographic Location and Administrative Context

Armoede (Ga-Sekhaolelo) is a rural resettled village in Ward 6 of Mogalakwena Local Municipality (MLM), within the Waterberg District Municipality (WDM), Limpopo Province. The settlement lies approximately 35–40 km north-west of Mokopane, accessed primarily via district roads that are vulnerable to storm erosion and seasonal wash-aways. The village sits within the Mogalakwena platinum belt and is functionally linked to surrounding mining, agricultural and small-trade economies.

Armoede operates under dual governance—customary leadership through the Mapela Traditional Authority (Ga-Sekhaolelo Traditional Council) and statutory local government through the Mogalakwena Ward 6 system. This duality shapes how land is allocated, how development decisions are taken, and how escalation to municipal and district services occurs.

The governance and oversight layers applicable to Armoede are summarised in Table 2.1, which is referenced here to clarify decision lines and coordination points used elsewhere in this Plan.

Table 2.1: Governance and Oversight Structure Relevant to Armoede

Layer	Primary institution(s)	Core roles in the village context
Customary	Mapela Traditional Authority; Ga-Sekhaolelo Traditional Council	Community leadership, customary land allocation, local order, convening community meetings (dikopano)
Local (Municipal)	Mogalakwena Local Municipality – Ward 6 Councillor & Ward Committee	Service planning and delivery, IDP representation, basic infrastructure, liaison with technical services
District	Waterberg District Municipality	Disaster coordination support, certain shared services, technical support
Provincial/ National	Limpopo Provincial Departments; National sector departments	Policy, programmes and conditional grants; statutory compliance and oversight

The table above shows why coordination protocols are important: responsibilities are distributed across several authorities, each with different mandates. To aid orientation, Figure 2.1 is referenced next to locate the village within the municipal and district geography.

Figure 2.1: Locality Map of Armoede (Ga-Sekhaolelo)



Source: Google Streetmap

Armoede's position, outside immediate service nodes and distant from fire, ambulance and police bases, translates into longer emergency response times during severe weather or incidents. Access constraints in the rainy season exacerbate delays, reinforcing the need for community-level readiness and clear escalation pathways.

2.2 Historical Background of Armoede (Ga-Sekhaolelo)

Armoede (Ga-Sekhaolelo) is a relocation village established as part of the Motlhotlo Community Relocation Programme, following the expansion of mining operations by Anglo American Platinum in the late 1990s and early 2000s. According to the Armoede Integrated Community Development Plan (ICDP 2020–2035), the relocation affected households from the original Ga-Puka and Ga-Sekhaolelo settlements, who were moved from their ancestral land to Farm Armoede in a phased process. The relocation was not a single event but a prolonged and complex social transition that permanently changed the spatial, economic and social structure of the community.

The history of Armoede, summarised in Table 2.2, is not only a story of relocation; it is a development context. It explains why the village faces structural economic vulnerability, why land pressure remains unresolved, and why community institutions like the ACDT, Traditional Council and village committees now play significant roles in local governance. This history is

central to understanding present-day risk drivers and resilience challenges, which are explored in later sections of this Plan.

Relocation discussions began during a period when mining operations were expanding rapidly in Limpopo Province, and the Motlhotlo area was zoned for extraction. The company initiated negotiations with households regarding compensation, land-for-land arrangements, and village establishment. However, negotiations were contested, as community members raised concerns about loss of land rights, burial sites, grazing fields, and independence. The ICDP notes that relocation meetings were marked by strong differences between the mine, traditional authority representatives, community committees and affected households. While agreements were eventually signed, many families felt their concerns were not fully addressed, and this sentiment still influences social cohesion and perceptions of justice in Armoede today.

Table 2.2: Relocation and Armoede Settlement Formation Timeline

Year/Period	Key Event	Impact on Community
Late 1990s	Mine expansion plans announced in Motlhotlo (Ga-Puka and Ga-Sekhaolelo)	Beginning of relocation planning and early tensions
2000–2003	Negotiations and dispute processes intensify	Disagreements over compensation and land rights
2004–2010	Phased relocation to Farm Armoede	Loss of ancestral land and rural livelihood systems
2010–2015	Settlement consolidation and household expansion begins	Informal backyard rooms added due to family growth
2016	Establishment of Armoede Community Development Trust (ACDT)	Formalisation of development representation
2016–present	Ongoing mine-community disputes	Structural cracks, compensation issues, social tension

The physical relocation to Armoede began around 2004 and continued for several years. Households were allocated formal residential stands and standardised houses, and a new engineered village layout was introduced. However, the relocation process also disrupted rural production systems—households lost access to ploughing land, grazing land, water sources and natural resource areas. Traditional spatial organisation based on family clusters and grazing commons was replaced with numbered plots and a dense settlement form. The relocation also fragmented family groups and altered social organisation, as former neighbourhoods from Motlhotlo were not always relocated together spatially in Armoede.

The post-relocation phase exposed deeper development challenges. While the relocation provided formal housing, it did not bring sustainable livelihoods. The local economy became dependent on mine-related employment, welfare grants and informal trade, increasing household dependency and reducing development resilience. According to the ICDP, the loss of land-based livelihoods affected food security, livestock keeping and household independence. Furthermore, some families reported that relocation compensation and benefit agreements were not implemented equally, creating long-term grievance dynamics within the village.

Ongoing mine-community relations continue to define the development context of Armoede. Community records and ICDP inputs indicate recurring grievances related to blasting vibrations causing cracks in houses, dust pollution affecting health, and road damage due to haul trucks. These issues have been raised in community meetings and negotiations over time, resulting in legal, administrative and community engagement processes that are still ongoing. The Armoede Community Development Trust (ACDT) was formally established in

2016 to represent the community in development negotiations and manage community obligations and benefits linked to the relocation agreements.

The historical journey of Armoede has shaped its current vulnerability, spatial structure and social dynamics. Relocation provided physical settlement but removed economic foundations. It increased reliance on external systems and weakened traditional resilience mechanisms such as communal farming, grazing cooperation and family-based social safety systems. Relocation also changed risk exposure—moving the community into a layout prone to drainage disruption, settlement expansion pressure and infrastructure dependency.

The table indicates how relocation altered settlement form, livelihoods and social organisation. Families lost access to agricultural land and grazing, shifting from mixed rural livelihoods to cash-based survival strategies. The transition to a compact layout increased exposure to flood ponding, wind damage and fire spread. Some relocation promises remain contested, and mine-community relations continue to shape local perceptions of fairness and development opportunity. These are not merely historical notes; they explain current settlement pressures and service demands that appear in later sections and influence disaster risk conditions. This historical context is the foundation of community development and risk conditions in Armoede today. It must remain central when planning disaster risk reduction interventions, resilience-building activities and governance structures that reflect the lived experience of this community.

2.3 Physical and Environmental Context

Armoede (Ga-Sekhaolelo) is located within the Savannah Bushveld biome of Limpopo Province, a semi-arid ecological zone characterised by thornveld vegetation, bush shrubs and patches of low grassland. The settlement lies in the transition belt between the Waterberg Mountain range to the west and the Mogalakwena River system to the east. This physical environment shapes the way residents live, build, access water, and use land. It also influences how hazards develop in the area. The environmental characteristics that most influence everyday life and infrastructure performance in the village are set out in Table 2.3, before discussing their practical implications.

Table 2.3: Key Environmental Conditions Affecting Armoede (Ga-Sekhaolelo)

Environmental Factor	Local Expression	Resulting Effect in the Village
Climate and rainfall	Summer thunderstorms; erratic rainfall	Flood ponding, sudden runoff, crop failure risk
Soil type	Sandy-clay shallow soils	Erosion and unstable foundations
Vegetation	Degraded grazing land; bush clearance	Dust exposure, surface erosion
Heat and drought cycles	Increasing seasonal water shortages	Livestock and household water stress
Lightning	Located in high lightning frequency belt (Waterberg)	Injury and house fire risk
Mining environmental effects	Dust fallout, ground vibration, altered drainage	Crack damage to structures, land shifts, respiratory stress

The village experiences a summer rainfall pattern, with most rainfall occurring between November and March, often through short-duration but high-intensity thunderstorms.

According to the South African Weather Service (SAWS), annual rainfall in the Mogalakwena region averages 350–450 mm per year, although rainfall variability has increased in the last decade, with years of extreme dryness followed by sudden storm events. Summers are increasingly hot, and seasonal heatwaves frequently affect the area between October and March, while winters are dry and dusty due to sparse vegetation.

Armoede's soils are mainly sandy-clay with shallow depth in places, which limits vegetation recovery and contributes to high erosion rates around households and access roads. Areas along informal livestock routes and household footpaths show visible gully erosion, worsened by lack of engineered stormwater drainage. During field assessments and community consultations (ICDP 2020–2035), residents reported that rainwater runoff forms unplanned channels through the village, especially where surface flow has been redirected by housing expansions. This has reshaped land levels in several sections of the village, making some homesteads lower than road surface levels, causing recurring water accumulation after heavy rain.

The physical environment in Armoede is also influenced by mining activity in the surrounding landscape. The expansion of Anglo-American Platinum's Mogalakwena operations resulted not only in relocation but also in modified landforms due to mining earthworks. Heavy vehicle movement from haul roads has compacted soils, increasing surface runoff into surrounding settlements. Dust emissions, particularly during dry seasons, contribute to air quality concerns, increasing respiratory strain, especially for children and elderly residents. Community reports indicate that blasting activity has caused soil vibrations and ground shifting, leading to cracking of foundations and walls in some households closest to the mining belt.

Armoede lacks natural buffers such as grassland cover, stormwater channels or wetland absorption zones that usually reduce the impact of storms. Like many resettled villages, the environmental layout was not based on natural watershed planning, meaning development was not aligned with terrain and natural water flow patterns. As a result, even moderate rainfall can trigger yard flooding, road erosion and contamination of pit latrines. Livestock farming, once a key resilience activity for Motlhotlo families, has reduced in Armoede due to grazing land shrinkage, competition for land and lack of watering points.

The interaction between environmental exposure and settlement growth has amplified vulnerability in Armoede. The village faces slow-onset hazards like drought, seasonal hazards like intense rainfall and lightning, and industrial-environmental hazards linked to nearby mining activity. These environmental realities must be understood not just as background conditions but as primary risk drivers, shaping household vulnerability and service disruption throughout the year.

2.4 Settlement Pattern and Land Use

Armoede originated as a planned relocation settlement with a basic formal layout, but over time it has evolved into a complex and mixed settlement pattern shaped by land pressure, household expansion, limited serviced land, and informal spatial adjustments made by residents to meet real family and livelihood needs. Although the original relocation layout was organised in a grid form with surveyed stands and access roads, the spatial structure has changed significantly due to organic growth, social land arrangements, return migration, and gaps in ongoing land-use management. The transformation of the village layout has been gradual but continuous, resulting in patterns of expansion that now extend beyond the boundaries of the original plan. These settlement dynamics are summarised in Table 2.4 below to clarify how land use trends influence both development conditions and exposure to everyday and seasonal hazards.

Household formation has been a key driver of settlement growth. As families grew, households subdivided informally within stands, leading to additional backyard structures made from corrugated iron, bricks, and reclaimed materials. Some of these structures are used to accommodate extended family members, while others are rented out as a survival strategy, creating a form of informal tenure and micro-rental activity. This densification takes place without formal planning guidance or service upgrading, which means stand saturation is increasing while access to space, drainage and infrastructure has remained unchanged.

A second driver of change in land use is return migration. Some family members who left during the relocation period have since returned from urban centres or mining towns due to economic hardship or family responsibilities. With no expansion of planned erven, they often occupy available yard space or settle on undeveloped land at the settlement edge. Because alternative residential sites are limited, this has led to informal peripheral expansion, especially in areas closest to grazing land, along footpath routes and in low-lying open spaces. These emerging residential extensions are not connected to formal road networks or services, resulting in spatial disconnection from the main village grid.

Table 2.4: Land Use and Spatial Dynamics in Armoede

Land-Use / Spatial Trend	Typical Features Observed	Causes and Contributing Factors	Consequences / Risk Implications
Stand densification	Additional backyard rooms, zinc structures, rental rooms emerging	Household growth; return migration; lack of serviced land; multi-family plots	Increased fire spread risk; stormwater blockage; overloading of pit latrines and yards
Peripheral expansion	Informal extensions beyond original planned boundary	Population growth; delayed formal land allocation; lack of municipal planning enforcement	Pressure on grazing/open land; settlement sprawl; emergency access challenges
Encroachment into drainage lines	Structures and fences crossing natural water paths	No stormwater plan; lack of land-use control	Yard flooding, erosion, property damage
Erosion-prone access routes	Deep ruts forming along paths and internal gravel roads	Unmanaged runoff; unplanned tracks	Seasonal road inaccessibility; delays for clinics/emergency services
Mining proximity impact	Settlement creeping closer to mine boundary	Land shortage; perceived employment advantage	Exposure to blasting vibrations, dust, noise
Tenure confusion	Family disputes over yard boundaries and extensions	Overlapping roles between traditional and municipal land allocation	Social conflict risk and informal land occupation
Limited	Few structured	Planning gap during	Lack of safe assembly

public facilities	community spaces, no formal economic node	relocation; no structured development follow-up	points; weak economic base
Loss of agricultural land	No arable land for ploughing or grazing camp	Relocation from Motlhotlo removed land economy	Livelihood loss; dependency on cash economy
Unplanned business activity	Spaza shops, home-based businesses along roads	Organic economy; no zoning framework	Traffic conflict; no commercial waste system

Traditional land allocation continues to influence the shape and direction of settlement expansion. While relocation introduced a surveyed layout, traditional leadership structures still perform a recognised social role in facilitating land access. In some cases, this has resulted in overlapping authority between customary land processes and municipal spatial planning systems. As stands fill up, community pressure for land has increasingly been directed towards traditional structures, leading to informal allocation of peripheral plots without coordinated layout planning. This has introduced new risks, including uncertainty about future land ownership, boundary disputes between families, and the growth of settlement zones outside municipal planning oversight.

Another dimension of land pressure is the loss of agricultural and grazing land following relocation. Households that once relied on ploughing fields and communal land for cattle and goats now face restricted land availability. Livestock ownership has not stopped, but grazing land has been reduced and pushed to the outer edges of the village. As a result, livestock paths cut through residential sections, exposing children to road accidents and accelerating surface erosion. Livestock also compete for space with expanding residential zones, and during the dry season they move closer to homesteads, increasing hygiene risks and crop damage in home gardens.

These land-use patterns reflect a settlement system responding to real population and livelihood demands without the support of a guiding spatial strategy. They are visualised in Figure 2.2 below, which illustrates both the original planned layout and the pattern of outward and inward expansion that has occurred since relocation.

Figure 2.2: Armoede Settlement Layout and Expansion Pattern
(Insert current satellite/aerial layout with original grid and expansion annotations)



The proximity of Armoede to the mining belt has also shaped spatial development. While the mine boundary is physically separated from the village, the economic pull of mining employment has drawn settlement growth closer to mining-linked access roads. This growth pattern follows movement corridors rather than planning logic. At the same time, mining haul

routes and altered drainage patterns have shifted surface water flows, contributing to wash-away sections where new informal structures are located. Dust fallout from haul roads has also influenced land-use behaviour, as some households avoid certain areas for gardening and instead build rental rooms to compensate for lost land-based income.

The lack of open space planning has also become more evident over time. Areas originally intended for public use or future development have gradually been absorbed by residential expansion, leaving limited communal land. Without designated community spaces, people informally use road reserves, unoccupied plots or church grounds for gatherings. However, the absence of managed open space has also contributed to illegal dumping, environmental degradation, and the loss of safe assembly points that are essential during emergency events or evacuations.

The current settlement form of Armoede is the result of relocation legacy, household growth, structural poverty and limited land governance rather than formal spatial planning. It is predominantly residential with weak service alignment and no economic structure, reflecting a development model driven by survival rather than planning. While this form is typical of resettled rural communities in mining zones, it has created predictable pressures on land, services and environmental conditions, which later translate into disaster risk drivers assessed in Chapter 3.

2.5 Demographic and Socio-Economic Profile

This socio-economic profile provides an overview of the human development conditions of Armoede (Ga-Sekhaolelo) to understand the demographic pressures, economic realities and social vulnerabilities that shape local development capacity and disaster risk. The information draws from community demographic insights, household profiling inputs and municipal development records.

2.5.1 Population and Household Characteristics

Armoede has a youthful and growing population, shaped by household expansion since relocation from Motlhotlo. The community consists mainly of extended families, with 4–6 persons per household being the most common structure. Women make up a slightly higher proportion of the population and play a major caregiving and household management role. The age structure is young, with nearly half of the population under 35 years, creating high dependency ratios as few wage earners support many dependants. Labour migration continues to influence household dynamics: some working-age adults relocate temporarily for employment in mining towns and return during holidays, while others remain unemployed and dependent on family support.

Table 2.5: Summary of Key Socio-Economic Indicators – Armoede (Ga-Sekhaolelo)

Indicator	Estimate / Observation	Comparison / Note
Average household size	4–6 persons	Above Limpopo average (4.2)
Youth (<35 years)	67%	High economic dependency
Female population	51%	Slight female majority
Unemployment rate	55–56%	Above Mogalakwena average (48%)
Stable employment	38–44%	Mostly informal/unstable
Households earning < R5	72%	Widespread low income

000/month		
Households below poverty line	49%	High poverty incidence
Households receiving social grants	28%	Core income source
Households receiving remittances	26%	Labour migration-linked
Post-school qualification	12%	Weak skills base
Disability prevalence	2.8%	Chronic disease-related
Crime exposure	11%	Mostly localised petty crime

2.5.2 Education and Human Development Context

Education levels in Armoede reflect a constrained skills environment. School attendance is generally high at primary level, but progression declines sharply after Grade 10 due to financial challenges, teenage pregnancy, family responsibilities and limited local career options. Only about one-fifth of adults have completed Grade 12, and only 12% report any form of post-secondary education. There is no local access to TVET or tertiary institutions, and transport costs to Mokopane or Polokwane limit skills advancement. The result is a workforce with limited formal qualifications, reinforcing long-term unemployment and dependence on low-paying work.

Health vulnerability is present and linked to both physical access to services and chronic conditions. The nearest clinic serves a large catchment and offers limited services, operating mainly on weekdays. Elderly residents and persons with disabilities face transport barriers when seeking care. Common health challenges include hypertension, asthma and tuberculosis, with dust exposure from unpaved roads and mining-related activity aggravating respiratory problems. Health shocks increase household vulnerability by reducing earning potential and increasing dependency.

2.5.3 Economic Conditions and Income Sources

The local economy is small and underdeveloped. A minority of residents obtain formal employment, mainly through contract work in nearby mining operations or seasonal jobs in towns. Unemployment levels remain high, particularly among youth, and informal work such as piece jobs, transport services and basic construction makes up a large share of livelihood activity. The main sources of household income are social grants, remittances from family members working elsewhere, informal trading and occasional wage labour. Social grants, including child support, old-age and disability grants, anchor the household survival economy. However, most income is absorbed by basic needs such as food, transport, education fees and burial society payments.

Inequality is increasing within the village. Households with access to mining-related employment or remittance flows are able to invest in home extensions and trade activities, while others remain locked in chronic poverty. Relocation benefits were distributed unequally over time, and some families continue to dispute land compensation and development support. This historical imbalance contributes to uneven economic recovery and pockets of social tension.

2.5.4 Livelihoods and Social Vulnerability

Livelihood strategies are largely survival-oriented and dependent on low and irregular income. With no agricultural land and limited grazing space, households cannot rely on farming for food security as they did prior to relocation. Instead, many depend on purchased food, making them vulnerable to price increases and seasonal shortages. Female-headed households face significant constraints due to limited access to productive assets and income opportunities. Youth are particularly affected by unemployment, leading to frustration, idleness and engagement in risky behaviour.

Social vulnerability is visible in several forms. Alcohol abuse is a concern among unemployed youth and some adults, often used as a coping mechanism for economic stress. Although serious crime rates remain low, incidents of petty theft and social conflict are periodically reported. Household vulnerability is worsened by limited access to clinics, financial instability, overcrowded living conditions and growing psychological stress linked to unemployment.

2.5.5 Community Resilience and Social Strengths

Despite socio-economic hardship, Armoede has strong community resilience. Residents rely on collective support through burial societies, stokvels, church groups, street committees and sports teams. Traditional leadership and community structures such as the Armoede Community Development Trust (ACDT) continue to provide governance, dispute management and local coordination. Informal mutual aid systems ensure that households receive support during funerals, medical crises and emergencies. Community cohesion remains strong, and social organisation capacity is a major foundation for future disaster risk management structures.

These socio-economic conditions shape local development priorities and directly influence service demand, household vulnerability and institutional capacity. These dynamics inform the analysis of infrastructure, service access and development conditions in Section 2.6.

2.6 Infrastructure and Basic Services Profile

Infrastructure access and service performance in Armoede reflect the legacy of relocation planning, incremental upgrading, and population growth that has outpaced maintenance and capacity. Basic networks exist, but reliability, coverage and upkeep are uneven across the village, producing a pattern of partial service where infrastructure is present yet frequently underperforms. To frame the discussion that follows, Table 2.6 summarises the current status of key services, the principal challenges, and their implications for households and public facilities.

Table 2.6: Summary of Infrastructure and Service Conditions in Armoede

Service Type	Current Status	Challenges	Implications
Water Supply	Piped supply to yard taps and communal points	Intermittent supply, ageing pipes, pressure loss	Water insecurity and hygiene risks
Sanitation	Predominantly VIP latrines	Full pits, poor maintenance, no desludging plan	Odour, groundwater pollution risks
Electricity	Eskom grid-connected	Load shedding, illegal connections, low voltage	Appliance damage, safety risks
Roads & Stormwater	Gravel roads, informal stormwater paths	Erosion, no formal drainage, limited access	Road wash-aways and isolation during rains
Waste Management	Communal disposal points, irregular collection	Illegal dumping, wind-blown waste	Environmental health impacts
Housing	Formal RDP-type structures, backyard	Structural cracks, overcrowding	Fire spread and shelter risk

	rooms		
Community Facilities	Schools, churches, clinic, community hall	Limited capacity, uneven distribution	Pressure on public services

The summary indicates that Armoede's challenge is less about the absolute absence of services and more about **service quality, reliability and coverage equity**. Intermittent supply, maintenance lag and informal work-arounds shift costs and risks onto households, that is especially women, older persons, persons with disabilities and child-headed households, who bear the time and health penalties when systems underperform. The subsections below unpack each service in detail and explain how technical constraints, settlement form and daily usage patterns interact to produce the current service outcomes.

2.6.1 Water Supply

Water is accessed through a piped reticulation system. The main challenge is reliability and pressure stability rather than total absence of supply. The village is supplied through a network that is fed by boreholes and storage tanks. Most households have yard taps while others still rely on shared street taps. Under normal conditions the system provides sufficient water for basic household use. However, water pressure often drops during peak morning and evening usage. Streets located on slightly higher ground or further from the main supply line experience longer periods of low flow or no flow. This creates uneven access conditions within the same village and leads to frustration among residents.

The system faces several maintenance and operational challenges. Old asbestos and plastic pipes are slowly failing and are prone to leakage. Some leaking points go unrepaired for long periods which leads to water loss and further pressure reduction. Where residents extend pipes informally to backyard rooms or neighbouring stands, flow is reduced and faults become harder to detect. These informal connections also weaken pipe joints and increase the likelihood of bursts.

During interruptions households store water in containers for cooking, cleaning and drinking. This is a necessary coping strategy but increases health risks when containers are not cleaned regularly or when water is stored without being covered. Storage also increases the time burden for women, older persons and persons with disabilities who are responsible for most household water tasks. Schools also experience difficulties when water flow is too weak for toilets and handwashing during school hours.

Electricity load shedding indirectly affects the water supply because it disrupts pump operation and delays the refilling of storage tanks. When pumping resumes, it can take hours before water reaches the furthest households. The combined effect of ageing infrastructure, uncontrolled connections, slow repairs and power interruptions results in an unreliable water service that affects hygiene, school attendance and household wellbeing. Improving reliability does not require a new system but rather consistent maintenance, pressure management and protection of water infrastructure from informal alteration.

2.6.2 Sanitation

Sanitation in Armoede is mainly provided through Ventilated Improved Pit (VIP) toilets that were installed during relocation. While these toilets improved sanitation access at the time of resettlement, they have now reached a stage where many pits are full and require replacement or emptying. There is currently no structured desludging programme in the village and households are responsible for managing sanitation problems on their own. This results in informal solutions that create health and environmental risks.

Many households have responded to full VIP pits by constructing additional makeshift toilets on their stands using materials such as zinc sheets and timber. These structures are often built without proper lining or ventilation. They are usually located closer to dwellings than recommended, which creates unpleasant odour and increases the risk of disease transmission. In some cases, new toilets are dug in areas where groundwater is shallow, which increases the risk of contamination of underground water sources used by neighbouring households.

Sanitation challenges are worsened by the settlement layout. In the more compact parts of the village, toilets are positioned close to cooking areas, washing areas and child play spaces. When it rains, overflow from these sites may spread across yards and footpaths. During community discussions, residents also reported that children and elderly persons face safety problems when using distant or poorly maintained toilets at night. Lack of lighting increases this challenge for households without electricity or money for paraffin lamps.

Sanitation is also a matter of dignity and public health. Households that cannot build new pits sometimes share toilets with neighbours. These shared arrangements strain relationships inside the community and result in uneven responsibility for cleaning and maintenance. Without an organised sanitation support plan, households carry the health and financial burden alone. The sanitation situation also has a direct impact on school readiness and hygiene, especially during times of water interruption when handwashing becomes difficult. Addressing sanitation in Armoede requires a long-term approach that combines pit replacement, periodic desludging arrangements and basic community hygiene support.

2.6.3 Electricity

Most households in Armoede are connected to the national electricity grid through Eskom. Electricity access has brought important improvements to daily life, including lighting, telephone charging, small business activity and safer movement at night. However, electricity supply is not reliable. Households experience frequent interruptions caused by national load shedding, local load reduction and transformer overload. These power cuts force many households to continue using traditional fuels such as firewood for cooking and heating, as shown in **Figure 2.4**.

Figure 2.4: Continued Use of Firewood for Cooking in Armoede Due to Electricity Interruptions

(Insert photograph of a scotch cart transporting firewood here)



The photograph shows that even though electricity is available, many households collect firewood using scotch carts or wheelbarrows, especially during long periods of load shedding or when they do not have money to buy electricity units. Firewood is used as a substitute energy source and is also sold informally. This is a practical coping method, but it increases pressure on nearby vegetation, contributes to environmental degradation and exposes women and children to long distances of wood collection, often late in the afternoon.

Load shedding disrupts many aspects of community life. It affects homework and learning for school children, delays business operations and affects small food traders who depend on refrigeration. Some households have lost groceries due to power cuts because food spoils before it is used. Elderly residents and persons with chronic illness also experience difficulty in storing medication safely without power. Voltage fluctuations are another problem and often damage home appliances such as televisions, fridges and radios.

Informal electricity connections across yards and along street boundaries create safety risks. Overloaded transformers and damaged lines slow down restoration when faults occur. Electricity is a foundation service for the village, but it is not yet stable enough to support local economic development. Improving transformer capacity and reducing illegal connections would improve supply stability. Introducing basic household energy safety education would also reduce fire risk and electricity loss.

2.6.4 Roads and Stormwater

Road access to Armoede is provided through a tarred road that connects the village to the N11 regional route and surrounding communities. This access road is in reasonable condition and allows taxis, private vehicles and scholar transport to reach the village throughout the year. However, all internal roads within the village are gravel roads, and these are the roads that experience serious wear and frequent damage. Internal roads are essential for daily movement, but their condition limits reliable access to services and opportunities. Figure 2.3 below illustrates a typical internal road in Armoede, showing surface erosion and drainage wear.

Figure 2.3: Condition of Internal Gravel Roads in Armoede
(Insert photograph of internal road erosion here)



Internal roads deteriorate quickly during the rainy season because there is no engineered stormwater drainage system in the village. Water flows along the road surface instead of through side channels, which causes erosion and water pooling. Roads become muddy and slippery, and in some sections they become temporarily impassable after heavy rainfall. Households report that scholar transport sometimes refuses to enter deeper parts of the village, which affects punctuality and school attendance. Access to the clinic and emergency assistance is also delayed during wet periods.

During dry conditions, dust from gravel roads spreads across the settlement and affects household health. Dust causes eye and chest irritation and is especially difficult for people living with asthma, small children and elderly residents. Dust also settles on roofs and reduces water quality for residents who harvest rainwater. These seasonal challenges show that the road problem in Armoede is not only a transport issue, but a health and service access concern.

Road maintenance is the responsibility of the local municipality, but current maintenance efforts are limited to occasional grading. This approach does not last because it does not address the root cause of the problem, which is the absence of proper stormwater control. Without side drains, stormwater channels and culverts, road surfaces will continue to erode each year. Some residents attempt to fill eroded road sections with stones or soil, but these measures are temporary and may divert water into yards, increasing the risk of property damage.

Although traffic related to mine employees travelling to and from work passes near the village during shift changes, this does not significantly damage internal roads. The greatest sources of road wear are uncontrolled stormwater flow, increasing settlement density and the natural limits of gravel surfaces that carry regular transport. Addressing roads and stormwater in Armoede requires basic but well planned interventions that focus on drainage, durable surfacing of priority routes and regular maintenance that prevents seasonal road failure.

2.6.5 Waste Management

Solid waste management services are present in Armoede, but their performance is inconsistent and access to bins is uneven across the village. Most households received municipal wheeled bins and are scheduled for weekly waste collection. However, collection does not take place regularly, and waste often remains uncollected for long periods. When this happens, households begin to look for alternative disposal methods, and waste is placed in open spaces, along internal roads and in old pits that were not designed for waste disposal. Figure 2.4 presents a collage that shows the mixed waste environment in Armoede.

Figure 2.4: Waste Management Conditions in Armoede – Wheelie bins and informal dumping

(Insert collage showing wheelie bins in use and illegal dumping sites in the village)



The figure shows that while waste bins are available in many stands, illegal dumping still takes place because of irregular collection services. Animals such as goats and dogs tear open refuse bags left outside full bins, and wind scatters loose waste around the settlement. This creates unsanitary conditions near homes and along pedestrian routes. Households report that during long collection delays, especially after public holidays or service disruptions, dumping areas increase in size and become difficult to manage.

Waste service coverage is also uneven inside the village. Some households, especially in the RDP section, do not have wheeled bins. Community members report that the distribution of bins has been linked in some cases to possession of a traditional council green file, which records proof of settlement status. Some households who could not afford the R500 payment required to obtain the green file stated that they were not given bins and now struggle with household waste. As a result, they are forced to share bins with neighbours or dispose of waste informally.

Community members have raised complaints about inconsistent waste collection during village meetings and community forums. Residents have asked for improved waste collection schedules and for equal access to bins for all households. However, follow up has been slow, and there is no local waste management support plan. In the absence of regular collection, some households burn their waste, which creates smoke and releases harmful fumes from burning plastic and rubber. This affects air quality and increases the risk of fire in densely settled areas.

Informal disposal and burning practices are not a result of resident negligence but a response to gaps in the waste service system. Short term improvements are possible if regular collection is restored, missing bins are distributed fairly, and a simple community plan is introduced to manage temporary storage and safe disposal. Waste management in Armoede can be improved without major infrastructure investment, but it requires consistent service delivery and fair access to bins for all households.

2.6.6 Housing and Settlement Services

Housing in Armoede reflects a mixed settlement profile with clear variation in structure type, building quality and investment levels. The majority of formal houses were constructed as part of the relocation process between 2004 and 2010 and were built by **Valterra**. These houses are generally of good construction quality when compared to typical government subsidised housing and have serviced stands with space for future expansion. Alongside these relocation

houses are **RDP houses** that were built earlier for residents originally settled on Farm Armoede before the relocation project began. Housing development since then has expanded through self-built structures, backyard rooms and informal dwellings that have emerged in response to household growth and accommodation demand. Figure 2.5 below shows the three dominant housing types that now exist in Armoede: relocation houses built by Valterra, earlier RDP houses and newer informal structures and extensions.

Figure 2.5: Housing Types in Armoede – Valterra Relocation Units, RDP Houses and Informal Extensions
(Insert collage with images showing three housing types side by side)



The housing landscape is becoming more diverse. While Valterra-built relocation houses remain in good condition in most cases, many have been extended with additional rooms to accommodate growing households. These extensions are sometimes built using masonry, but in many stands they are made from zinc sheets, timber frames or mixed materials. These additions are often unregulated and constructed without building plans or safety standards. In some cases, poor foundation preparation and lightweight structures increase the risk of wind damage or structural collapse during storms.

A notable feature in the housing pattern is the growth of informal structures that are being used as rental rooms or as markers of land occupation. Some households have constructed shacks or single-room structures at the back of their stands as a source of rental income. Others build informal units to signal a claim to land or to prepare for future development. Over time, some of these temporary structures become permanent. Their spread increases density in parts of the village and places pressure on water access, sanitation and electricity supply points.

In addition to this, Armoede has begun to show early signs of housing inequality. There are households that have invested in higher-value properties, often built by families with stable incomes, especially from mine-related employment and business activity. These houses have better finishes, walling, paving, security features and sometimes solar backup systems. This reflects a growing divide in household circumstances and signals emerging social and spatial differentiation inside the village.

While housing development in Armoede shows progress and investment by residents, it also presents growing settlement risks. Increased density from backyard rooms reduces open space between dwellings, which increases the probability of fire spread. High settlement density in some sections also reduces emergency vehicle access due to narrow movement space. Informal structures often lack safe electrical wiring and stable roofing, which increases exposure to strong winds. Without basic planning guidance and controlled layout expansion, informal build-up could undermine long-term safety, movement routes and service delivery efficiency.

2.6.7 Community Facilities and Social Services

Armoede has a growing range of community facilities that support education, health, learning, administration and community life. However, these facilities vary in quality, accessibility and functionality. While progress has been made in recent years, particularly through community-led initiatives, service coverage remains uneven and some facilities are overstretched relative to population size.

The village hosts two primary schools, Jan Malebana Primary School and Armoede Primary School, which provide foundation and intermediate education. For secondary schooling, learners attend Mphunye Secondary School located in Ga-Puka, which is shared between

Armoede and neighbouring communities. As a result, many learners depend on daily transport and walking long distances, which affects attendance during rainy conditions or when road access is disrupted.

Health services are accessed through a clinic located between Armoede and Ga-Puka, which serves both communities. The clinic operates on weekdays, but residents report long queues, medicine shortages and limited staff capacity. For more serious medical treatment, residents travel to Mosesetjane Clinic or Mokopane Hospital, which increases transport costs and delays access to health care for elderly residents and people with chronic illness.

A major development milestone was the construction and handover of the Ga-Sekhaolelo Community Hall in 2024, which now serves as the main venue for public meetings, consultations and community functions. The facility is centrally located and improves local access to formal community gathering space, which was previously limited. The hall is shown in Figure 2.5.

Figure 2.5: Ga-Sekhaolelo Community Hall (Completed 2024)
(Insert photograph of newly completed Ga-Sekhaolelo Community Hall here)



Before the new hall was built, the community used a smaller multi-purpose building for meetings. This older structure has since been repurposed by the Armoede Community Development Trust (ACDT) and now functions as a Study Centre, providing a safe and supervised learning space for school learners and youth. This conversion reflects a positive community-led initiative to support education and personal development. The Study Centre is shown in Figure 2.6.

Figure 2.6: Ga-Sekhaolelo Study Centre (Converted Former Community Hall)
(Insert photograph of the study centre here)



Early childhood development services are also present in the village. Armoede has five Early Childhood Development Centres (ECDCs) that support early learning and school readiness. One of these, the Ga-Sekhaolelo ECDC, is uniquely community owned and community managed, while the others operate as independent centres. However, most of these ECD facilities have limited play infrastructure, inadequate sanitation and depend on parent contributions to remain functional.

In addition to formal public facilities, Armoede also has church buildings and burial society offices, which are important service spaces at household level, especially during times of crisis, funerals and religious gatherings. These buildings provide valuable community support functions even though they are not part of formal state infrastructure.

A former residential facility built as an old age home is now under consideration for conversion into a Sports and Wellness Centre. Community discussions on this transformation are ongoing. If completed, the facility could provide much needed recreational space for youth and organised community programmes.

The range of community facilities in Armoede can be summarised as follows:

- Educational facilities: two primary schools and access to a shared secondary school
- Health infrastructure: one shared clinic
- Community infrastructure: Ga-Sekhaolelo Community Hall and Ga-Sekhaolelo Study Centre
- Early learning services: five Early Childhood Development Centres
- Faith-based and burial support facilities: church buildings and burial society offices
- Repurposed development facility: old age home proposed for community sports and wellness
- Limited recreational spaces: informal sports fields and open grounds

While these facilities play an essential role in community life, many are undersupplied relative to population size. Distance, transport cost and service pressure limit access for many households. Social service departments do not maintain a permanent presence in the village, which leaves gaps in youth development, social worker support and family services. These weaknesses in community services shape local vulnerability and reduce opportunities for community growth. They also highlight the need for stronger institutional coordination, which is discussed in Section 2.7.

2.7 Governance, Community Structures and Institutional Systems

Governance in Armoede (Ga-Sekhaolelo) operates through a combination of customary leadership, elected local government structures and community-based organisations. These institutions shape decision making, development coordination and local accountability. While governance structures exist, they do not always work in a coordinated manner and some operate with limited resources and authority. Governance in the village is therefore active but fragmented, and institutional effectiveness depends heavily on community initiative.

2.7.1 Traditional Leadership

Armoede falls under the Mapela Traditional Authority, with local governance exercised at village level through the Ga-Sekhaolelo Traditional Council under the authority of the Senior Traditional Leader. The traditional leadership structure plays a central role in community land allocation, cultural affairs and dispute resolution. It remains the primary authority for settlement administration, allocation of sites, confirmation of local residence and issuing of settlement records, including the document known locally as the green file.

Traditional leadership is also involved in community affairs beyond cultural and land matters. The Traditional Council acts as a key point of engagement for external stakeholders, including government entities and development partners. It facilitates community meetings, known as makgotla, and sets protocols for negotiations affecting the community. While traditional authority remains respected, some challenges affect its governance reach. These include disputes over land allocation fairness, community concerns regarding the cost of administrative documents such as the green file and delays in decision making when matters require escalation to higher structures. Despite these limitations, traditional leadership remains the most consistent governance presence in the village.

2.7.2 Ward Representation and Municipal Governance

Armoede falls within Ward 6 of Mogalakwena Local Municipality. Representation of local government is mainly visible through the Ward Councillor and the Ward Committee, rather than direct municipal service delivery. The Ward Committee is responsible for communicating development needs from the community to the municipality and for reporting back decisions. In practice, however, this feedback loop is inconsistent. Community consultation processes are often limited to public announcements and do not result in clear development outcomes.
Mogalakwena Local Municipality is constitutionally responsible for basic service provision, but municipal involvement in Armoede is minimal and service follow-up is generally slow. Community records indicate that municipal officials seldom attend village meetings, and residents report that service-related requests often go unresolved. The Waterberg District Municipality, which is responsible for coordination and disaster management oversight, is largely distant in practice with limited proactive engagement in the village. Government departments such as Social Development, Home Affairs, Education and Health only appear ad hoc in response to specific cases or invitation, rather than through structured service programmes.

This gap in sustained municipal and sector department presence has created a reliance on informal communication channels. The Ward Committee plays a bridging role, but without municipal capacity support or regular departmental outreach, community expectations of government remain largely unmet.

2.7.3 Community Organisations and Development Structures

Unlike many villages, Armoede has a strong community-led development structure in the form of the Armoede Community Development Trust (ACDT). The Trust plays a major role in coordinating community development projects, managing agreements linked to relocation commitments and acting as an implementing partner for some initiatives. The Trust is recognised by both the Traditional Authority and external stakeholders and serves as a negotiation platform on community development issues.

In addition to the ACDT, project committees are formed when needed, especially around service delivery concerns such as waste management, housing and land disputes. However, these committees often lack formal recognition and are not integrated into municipal development forums. This reduces their influence in decision making processes. Community-based groups such as youth development initiatives, women support groups and sports committees exist, but they operate independently without shared planning structures.

ACDT plays a valuable coordination role, but it also carries high developmental expectations from the community. Without formal, legally recognised funding and support mechanisms, its capacity is stretched, and it is sometimes expected to take on responsibilities that belong to government. While it has demonstrated leadership in community-driven initiatives, it requires stronger institutional backing to sustain long term impact.

2.7.4 Coordination, Participation and Institutional Gaps

There is no formal village-wide development coordination platform in Armoede that brings together all leadership structures on a regular and structured basis. Coordination happens, but it is issue-based and depends on who calls the meeting. When matters relate to land or settlement administration, meetings are led by the Traditional Council. When matters relate to municipal services or development complaints, meetings are facilitated through the Ward Councillor and Ward Committee. When matters relate to development agreements or community projects, the Armoede Community Development Trust (ACDT) normally takes the lead. This results in parallel decision spaces rather than a single coordinated development forum.

Because coordination is not structured, meetings are often reactive and occur only when disputes arise or services fail. There is no standing development agenda that tracks priorities over time. Community participation levels are high, and meetings are generally well attended, but decisions made in meetings are not always followed by implementation because there is no shared system for responsibility tracking and follow up. This contributes to slow response to development issues, repeated discussions of the same problems and community frustration when progress is not seen.

Municipal and sector department participation is irregular. Government departments engage only when invited on specific matters, for example housing verification, water outages, social grant enquiries or school infrastructure complaints. However, Mogalakwena Local Municipality and Waterberg District Municipality rarely participate directly in Armoede development meetings, and feedback from municipal offices is often delayed. This creates gaps in accountability and service follow up, and shifts the burden of coordination to community leadership structures that do not have resources or legal authority to enforce implementation.

Although Armoede does not have a formal coordination forum, it has strong potential to establish one. There is active leadership, high public participation and existing spaces of organisation such as school committees, church forums, burial society networks and ACDT development meetings. These could form the base for a structured village development forum in future, which would reduce duplication, improve planning and support integrated disaster risk management later in the plan.

2.8 Local Economy and Livelihood Systems

The local economy of Armoede is characterised by limited formal employment, a small and thin local business base, and strong dependence on survivalist and informal economic activities. The economic environment reflects both the legacy of relocation, which reduced access to land-based livelihoods, and the village's partial connection to the mine-driven economy of the broader Mapela region. While there are indicators of improvement through household investment, the majority of residents remain economically vulnerable and rely on low and irregular income.

The largest share of household income comes from three main sources: wage labour linked to the mining sector and secondary services in Mokopane, small-scale trading and informal services within the village, and transfer income through social grants and remittances. Many residents seek employment at nearby mines or mining contractors, but most are absorbed into short-term and unstable contract jobs rather than permanent work. Daily and weekly transport to work locations outside the village connects Armoede to the wider regional economy, but economic benefits do not circulate strongly inside the village because most wages are used for basic survival rather than investment.

The informal economy plays a central role in household survival and local circulation of money. There are several tuckshops and spaza stores that operate from home yards, along with

cooked food vendors, street sellers, plastering and house building teams, small welding services, informal motor mechanics, taxi operators, local transport services using vans, backyard rental rooms and small livestock farming. This informal economy is unstructured but active and offers limited income stability. It operates without business infrastructure, working capital or formal support. However, it represents a real economic base that can be strengthened through local enterprise development support.

Armoede has begun to experience housing-led economic development through room rental. Families with access to some capital invest in backyard rental units to generate monthly income. These rentals accommodate local single men working in nearby mines and contractors, unemployed youth who want independence but stay in the village, and small families who have not received land allocations. Rentals now form a growing part of the village economy and increase settlement density.

Despite the presence of economic activity, local economic growth is constrained by clear structural factors. There is no formal business area, no marketplace infrastructure and no small enterprise support centre. Most trade happens from home yards or along roads, which limits economic development. There is no banking access point in the village, and residents travel to Mapela or Mokopane for most purchases. This means local money flows out of the village rather than circulating inside, and the local economy remains shallow.

Relocation also removed access to productive agricultural land. Households previously used fields and grazing areas for subsistence food production. In Armoede, households do not have arable land and very few home gardens exist due to water scarcity and poor soils. This has forced a full dependence on the cash economy, which increases economic vulnerability during income shocks such as illness, job loss or delayed grant payments.

Even with these constraints, the community shows economic resilience through adaptation. Households combine multiple livelihood strategies, such as trading, casual work, transport services, construction skills, sewing and catering for social events. Burial societies and savings clubs provide basic financial protection, and families share income within extended household networks. Armoede has a strong culture of economic cooperation which limits extreme poverty, but without formal economic support and opportunity creation, economic progress will remain slow.The structure of the local economy is directly linked to risk and disaster vulnerability. Limited income reduces the ability of households to repair damaged houses after storms, maintain safe electrical connections, buy water storage drums or improve sanitation. Economic dependency and lack of opportunity for youth also increase social pressure and frustration, which contributes to conflict risk and social instability. Building economic resilience is therefore a strategic part of community safety and risk reduction.

2.9 Social Dynamics, Cultural Norms and Community Values

Social life in Armoede is rooted in a strong sense of community identity, shaped by shared history, cultural heritage and collective memory of relocation. Despite economic and service delivery challenges, the village maintains a culture of mutual assistance, respect for traditional leadership and participation in community gatherings. The use of Sepedi as the dominant language reinforces cultural cohesion, and traditional practices such as initiation, customary marriage and ancestral ceremonies continue to hold social significance.

The extended family remains the foundation of social organisation. Households are multi-generational, and care responsibilities are shared among family members. Even with rising unemployment, families combine resources to support education, funerals and major life events. Burial societies and savings groups play an important role in household stability by providing financial support during crises. These informal safety networks reduce the burden on individual families and prevent total destitution.

Community cohesion, however, is not uniform. There are **emerging social tensions** linked to land allocation disputes, competition for opportunities, unequal access to services and the legacy of relocation. Youth frustration is increasing due to limited employment and recreation opportunities. This contributes to substance abuse, crime risk and conflict between generations, especially when young people feel excluded from decision making structures. Despite this, many youths express an interest in development but lack platforms and mentorship to engage actively.

Religion and cultural practices coexist and reinforce social order. Churches play a unifying role and are involved in counselling, welfare support and youth moral guidance. Cultural values emphasise respect, community solidarity and collective responsibility. Social norms still discourage behaviour that threatens community safety, such as theft or disrespect towards elders, but social control is weakening as economic pressure increases and informal settlement density rises.

Social resilience is one of Armoede's strongest assets. Households recover from setbacks through social networks rather than formal assistance. Community members mobilise quickly for funerals, family crises, housing repairs and community projects. However, without better youth inclusion, structured social programmes and conflict resolution mechanisms, the risk of social fragmentation may grow over time. Strengthening social cohesion is therefore essential for long term community stability and successful disaster risk management.

2.10 Development Challenges and Opportunities

Armoede (Ga-Sekhaolelo) has a strong social foundation and active community leadership, but development progress remains constrained by structural challenges that limit service delivery, economic growth and social stability. These conditions increase household vulnerability and weaken the ability of the community to cope with shocks such as storms, economic disruption, health crises and fire incidents. At the same time, the village has development strengths that can support growth if they are harnessed effectively. The key development challenges and opportunities are summarised in Table 2.6 before being discussed in more detail in the subsections that follow.

Table 2.6: Summary of Key Development Challenges and Opportunities

Development Area	Key Challenges	Opportunities
Basic services and infrastructure	Water interruptions, weak internal roads, electricity instability, poor stormwater control, irregular waste collection	Infrastructure upgrades through joint planning, use of community reporting systems and maintenance partnerships
Local economy	Limited formal jobs, shallow local market, low-income levels, no formal business infrastructure	Growth in informal trade, rental rooms, construction services and potential enterprise support
Land and housing	Settlement pressure, informal extensions, densification, land allocation disputes	Structured settlement planning, formalised extensions and incremental housing improvement
Governance	Fragmented development coordination, low municipal	Strong local leadership base and existing community

	visibility, limited departmental engagement	mobilisation capacity
Social development	Youth unemployment, substance abuse, leadership contestation, limited recreation facilities	Active social networks, study centre, early learning facilities and community development initiatives
Environmental and disaster risk	Soil erosion, localised flooding, household fire risk, climate variability	Community readiness for disaster risk reduction and potential for resilience programmes

2.10.1 Key Development Challenges

Armoede faces persistent service delivery and development backlogs. As discussed in Section 2.6, water reliability remains a problem, sanitation infrastructure is ageing, internal gravel roads are heavily eroded during the rainy season and stormwater flows through informal channels that damage yards and roads. Electricity interruptions and low voltage damage household appliances and reduce productivity. Waste collection is irregular and waste dumping is spreading across open spaces. These service challenges reduce quality of life and increase development pressure on households.

The local economy remains weak and unable to absorb local labour. As set out in Section 2.8, most employment is found outside the village through contract work linked to mines or services in Mokopane. Economic activity inside the village is mostly informal and survival based, with small margins and low growth potential. There is no local economic infrastructure such as a market space, workshop zone, business hub or cooperative centre to support economic activity and income diversification.

Land and housing pressures are increasing. Section 2.6.6 showed that informal structures are expanding as households increase backyard rooms or build informal dwellings to claim space for future development. Without basic planning guidance or layout control, settlement density is increasing, which reduces access for emergency vehicles and increases fire risk. Land allocation disputes also weaken community cohesion and delay service delivery projects where land ownership is contested.

Governance coordination remains limited. As discussed in Section 2.7, community meetings are held regularly, but decision making is fragmented and often reactive. There is no unified planning platform that brings together traditional leadership, the Ward Committee and community development structures. Mogalakwena Local Municipality and Waterberg District Municipality have limited local presence, and sector departments engage only when specific issues arise. This slows down service delivery responses and development implementation.
Social pressures are increasing. Section 2.9 highlighted growing youth frustration due to lack of employment and recreation options, rising substance abuse and conflict linked to leadership contestation and inequality. Crime is not widespread, but theft and social conflict are increasing in some areas. These pressures undermine social cohesion and reduce participation in development initiatives.

Environmental risk is increasing due to erosion, blocked drainage paths, soil disturbance and increased settlement density. Households rely on self-built structures that may not withstand strong winds or prolonged rainfall. Many houses use informal electrical connections that increase fire risk. These environmental risks, combined with social and economic pressures, increase disaster vulnerability in the village.

2.10.2 Development Opportunities

Despite these constraints, Armoede has clear development strengths. The community has strong internal organisation and high participation in public meetings and community initiatives. The presence of the Armoede Community Development Trust provides a central structure for community-driven development, negotiation and planning. Traditional leadership remains active and influential, providing continuity and cultural stability in governance matters. There are also economic opportunities. The village has local construction teams, transport operators, food vendors, spaza shops and service trades that demonstrate economic initiative. The growth of rental accommodation and local aggregation points for goods shows economic adaptation. With structured enterprise support, business training and access to microfinance, local livelihoods can become more sustainable.

The village has social development assets that can be expanded. The study centre, community hall and five early childhood centres provide foundations for education, training and youth development. Churches and burial societies support vulnerable households and strengthen social stability. There is land available for managed development, and community labour contributions can support local projects if they are properly coordinated.

Armoede also has strategic value in regional development planning. Its location near the N11 and within the Mapela mining region positions it well for development partnerships. Opportunities exist in skills development, local procurement agreements, agricultural renewal through home gardens and communal food projects, and community-based disaster risk reduction. These development challenges do not exist in isolation. They shape exposure to risk and limit the ability of households and institutions to prevent, withstand and recover from shocks. The development profile in this chapter provides the foundation for understanding risk in Armoede. The next chapter examines the **hazards, vulnerabilities and risks** that affect the village and identifies priority risk drivers that must be addressed in the Disaster Risk Management Plan.

3. Hazard, Risk and Vulnerability Profile

3.1 Introduction

Disaster risk in Armoede (Ga-Sekhaolelo) is shaped by localised but persistent hazards that affect households and community systems every year. Unlike large-scale disaster events, the risks in Armoede are daily and seasonal, linked to recurring conditions such as flooding from poor stormwater drainage, strong winds causing roof damage, household fires from unsafe energy use, health outbreaks linked to poor sanitation and waste disposal, and access disruptions due to damaged internal roads. These risks accumulate over time and slowly erode household assets, livelihoods and social wellbeing.

This risk profile aligns with the Disaster Management Act (57 of 2002), the National Disaster Management Framework (NDMF) and the Sendai Framework for Disaster Risk Reduction (2015–2030), which emphasise risk understanding as a foundation for integrated disaster risk reduction. The profile integrates community knowledge, local development realities and technical assessment to provide an accurate analysis of current and emerging risks in the village.

As one elder noted during consultations held in July 2024:

> ***"Our problems are not once-off disasters. They repeat themselves every year and no one really cares about what happens in the villages. The focus is on towns."***

These recurring events are not random "natural" problems. They reveal the interaction between hazards and underlying vulnerabilities caused by development failures. Without addressing root causes, Armoede will continue to experience avoidable risk, and losses will accumulate silently at household level.

3.2 Risk Analysis Methodology

The standard disaster risk logic promoted by South Africa's National Disaster Management Framework (NDMF, 2005) and aligned with international best practice (Sendai Framework for Disaster Risk Reduction 2015–2030, UNDRR, IFRC, World Bank GFDRR) follows a risk-based, prevention-centred approach. This logic emphasises understanding risk, reducing vulnerability, building resilience, and strengthening preparedness and response capacity.

At its foundation, disaster risk is understood as a **function of hazard, vulnerability, and capacity**, moderated by exposure:

Disaster Risk = Hazard × Vulnerability × Exposure ÷ Capacity

Where:

Hazard	:	Potential damaging physical event (e.g. floods, drought, disease outbreaks, hazardous materials)
Vulnerability	:	Conditions that increase susceptibility to harm (e.g. poverty, weak infrastructure, poor planning)
Exposure	:	People, assets, services or systems located in hazard-prone areas
Capacity	:	Capabilities and resources to prevent, withstand, and recover (e.g. early warning, emergency services, governance)

This logical sequence drives both the NDMF and global DRR practice:

1. **Know Your Risk:** Hazard, exposure, and vulnerability profiling
2. **Prevent and Mitigate:** Reduce risk drivers, build resilience, apply land use control, enforce standards
3. **Prepare:** Plans, systems, early warning, simulations, readiness
4. **Respond:** Coordinated, rapid interventions to save lives and protect livelihoods
5. **Recover and Build Back Better:** Restore and improve systems to reduce future risk

This mirrors the risk management cycle and supports development continuity rather than crisis reaction. The standard disaster risk logic promoted by the NDMF and global best practice is proactive, risk-informed, development-supportive and resilience-focused. It avoids viewing disasters as acts of fate and instead frames them as risk governance challenges that can be anticipated, reduced, and managed. Given this background, the methodology combined field evidence, community risk perceptions, spatial observation and institutional review. Risks were ranked using a likelihood and impact scoring method, shown in Table 3.1.

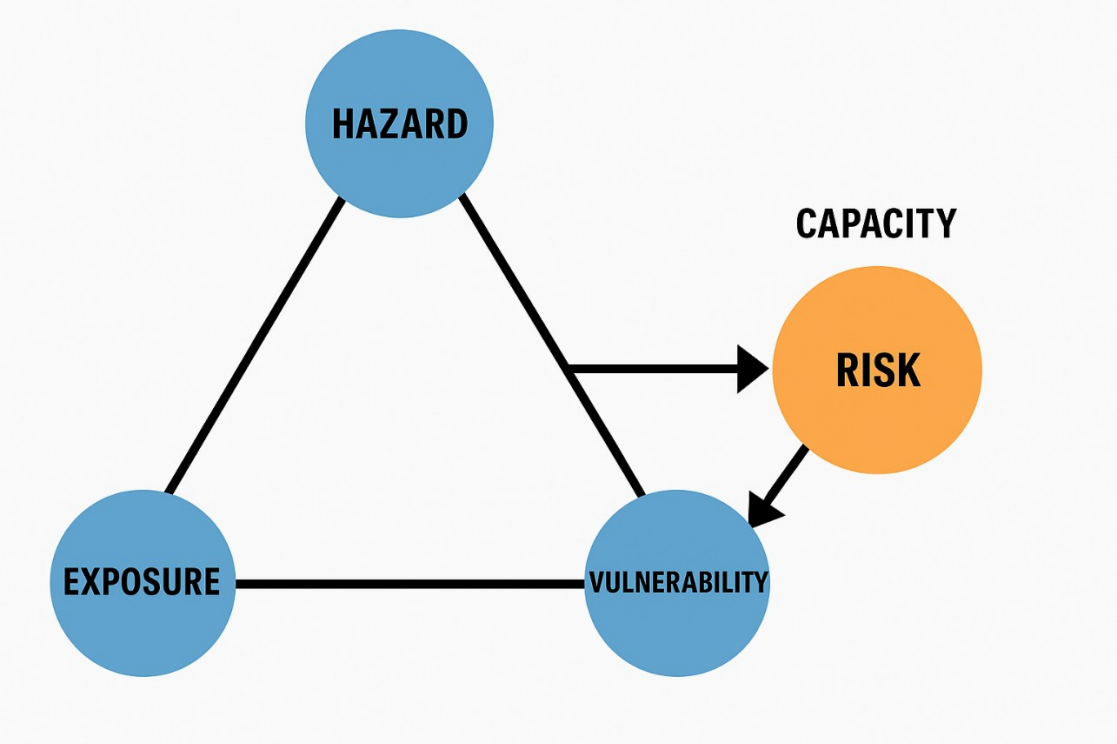
Table 3.1: Risk Rating Framework

Likelihood Score	Description	Impact Score	Description
1	Rare event	1	Negligible
2	Possible but infrequent	2	Minor household-level impact
3	Occurs annually	3	Moderate community disruption
4	Seasonally recurrent	4	Significant local impact
5	Very frequent or ongoing	5	Serious and widespread impact

Figure 3.1 illustrates the disaster risk logic used to frame the assessment.

Figure 3.1: Disaster Risk Logic Model

(Insert triangular or circular model linking Hazard + Exposure + Vulnerability → Risk, moderated by Capacity)



This method ensures that risk is not treated as a product of hazard alone. It exposes the development and governance weaknesses that create vulnerability. In Armoede, risk is driven more by vulnerability conditions than by hazard intensity, which shows that disaster resilience cannot be achieved through response alone but must be built through corrective development planning.

3.3 Hazard Identification

Disaster risk in Armoede is driven by a combination of recurring local hazards and systemic vulnerability conditions that expose households, livelihoods and community infrastructure to loss and disruption. Hazard identification is therefore a critical entry point in understanding the risk environment and forms the basis for prioritised risk reduction planning. This process does not treat disasters as isolated events but as the outcome of predictable hazard patterns interacting with conditions of vulnerability, consistent with the National Disaster Management Framework (NDMF) Key Performance Area 2 and the Sendai Framework Priority 1: Understanding Disaster Risk.

Hazards affecting Armoede were identified through multiple evidence sources, including field engagement, community risk mapping exercises, seasonal impact recall, observations of hazardous locations and technical input from local service providers. Particular weight was given to lived community experience, as recurring low-intensity but high-frequency events have a cumulative effect on poverty, household resilience and local development. As one community member noted during consultations held in July 2024, ***"Every year we know what will happen. The rain will damage the roads, the wind will take roofs, and fire will burn rooms. We live with risk."*** This statement reflects the chronic nature of risk in Armoede—local hazards repeat seasonally, often without formal disaster classification, but steadily erode development gains.

Hazards were classified into seven standard categories aligned with recognised disaster risk taxonomy: hydro-meteorological, climatological, environmental degradation, technological/ infrastructure, biological/public health, human-induced socio-economic, and mining-related hazards. These hazard categories were further validated through spatial observation, historical accounts of events and risk pattern verification. Table 3.2 summarises the hazard

inventory affecting Armoede, including local triggers, recurrence behaviour and source of evidence.

Table 3.2: Summary of Hazards Identified in Armoede

Hazard Category	Specific Hazards Identified	Local Triggers/Drivers	Recurrence Pattern	Source of Evidence
Hydro-meteorological	Flooding and stormwater runoff	No formal stormwater drainage, runoff from nearby slopes, blocked channels	Seasonal (rainy season)	Community consultations (2024), field observation
Hydro-meteorological	Lightning	Open terrain, metal roof structures, lack of lightning protection systems	Seasonal (summer storms)	Community accounts, school risk reports
Hydro-meteorological	Strong winds	Poorly anchored roof structures, informal construction	Seasonal (summer & spring)	Incident recall, builder testimony
Climatological	Heat stress	Rising temperatures, outdoor labour exposure	Increasing trend	Seasonal observations
Climatological	Drought and water scarcity	Borehole dependency, groundwater stress	Multi-year cycle	Local water committee input
Environmental Degradation	Soil erosion	Overgrazing, informal footpaths, stormwater scouring	Seasonal	Field inspection
Environmental Degradation	Veld and grass fires	Dry vegetation, uncontrolled burning	Seasonal (dry periods)	Household interviews
Environmental Degradation	Land degradation	Overuse of communal land, settlement expansion	Continuous	Spatial observation
Technological / Infrastructure	Electricity disruptions	Overloaded transformer, aged network	Frequent	Service provider communication
Technological / Infrastructure	Unsafe electrical wiring	Informal household connections	Continuous	Visual inspection
Technological / Infrastructure	Water supply interruptions	Borehole failure, pump breakdown, vandalism	Frequent	Water committee
Technological / Infrastructure	Road access failures	Poor road surfaces, rainfall washaways	Seasonal	Transport user reports
Public Health	Waterborne diseases	Contaminated water storage practices	Seasonal	Clinic data
Public Health	Sanitation-related disease	Pit latrine contamination, poor hygiene	Continuous	Health worker input

Public Health	Tuberculosis risk	Overcrowding, indoor smoke exposure	Continuous	Clinic staff
Biological (Livestock)	Animal disease outbreaks	Communal grazing, limited veterinary access	Annual	Farmers and herders
Human-induced	Household/structural fires	Paraffin and gas use, illegal connections	Frequent	Community reports
Human-induced	Alcohol-related violence	Social strain, youth unemployment	High frequency	SAPS records
Human-induced	Road transport accidents	Poor visibility at junctions, speed	Frequent	Community transport forum
Mining-related	Dust and air pollution	Mine haul road traffic	Continuous	Field observation
Mining-related	Blasting vibration effects	Nearby mining activity	Weekly/bi-weekly	Community risk mapping
Mining-related	Hazardous material spills	Fuel transport routes	Possible	Road observation
Mining-related	Tailings dam failure risk	Mining presence, i.e., proximity to Mogalakwena Platinum mine.	Very rare but catastrophic	External risk review
Socio-economic	Food insecurity and loss of income	Climate shocks, unemployment	Continuous	Household assessments

These hazards do not exist in isolation; many are interconnected and create cascading risk effects. For example, flooding accelerates soil erosion, which weakens road access, resulting in delayed medical care and increased exposure to secondary health risks. Water interruptions intensify sanitation-related disease exposure, while electricity instability increases dependence on unsafe energy sources, driving household fire risk. These hazard interactions reflect a risk system rather than discrete incidents.

While Table 3.2 captures the full range of hazards affecting Armoede, a hazard only translates into disaster risk when it interacts with exposed households, vulnerable livelihoods or fragile infrastructure systems. Section 3.4 therefore builds on this inventory by analysing how each hazard behaves in Armoede, the vulnerability conditions that intensify its impact and the systems most at risk. This ensures that subsequent sections on exposure profiling (3.5) and vulnerability analysis (3.6) are grounded in a clear and evidence-based hazard logic.

3.4 Hazard Analysis

Hazards in Armoede do not present as isolated events but operate as part of an interconnected risk system shaped by environmental exposure, fragile infrastructure, poverty conditions and limited local response capacity. Following hazard identification in Section 3.3, this section provides an analytical interpretation of the behaviour of each hazard, the triggers that drive its occurrence, the population and systems exposed, the vulnerability factors that amplify damage and the consequences observed in the community. This aligns with the standard disaster risk logic applied by the National Disaster Management Framework (NDMF) and the Sendai Framework Priority 1: Understanding Disaster Risk.

The analysis reveals that most hazards affecting Armoede are chronic and repetitive rather than sudden and catastrophic. While many are low-intensity in nature, they occur frequently, causing steady erosion of household assets, strain on local infrastructure and socio-economic instability. A resident summarised this reality clearly during consultations: ***"We are not surprised by disasters here. We expect them, rain, fires, no electricity. these things always come back."*** The hazard analysis in Table 3.3 below categorises and ranks hazards according to their local drivers, exposure conditions, consequences and risk tier.

The analysis shows that hazard frequency in Armoede is high, but severity is intensified by vulnerability conditions linked to underdevelopment—limited drainage, unsafe energy use, dependency on a single borehole and constrained local response capacity. Hazards interact and produce cascading effects; for example, flooding accelerates soil erosion, damages roads, isolates households and triggers water contamination, which then drives health outbreaks. Similarly, electricity disruptions lead to higher fire risk as households switch to candles and paraffin.

Hydro-meteorological and technological hazards dominate the local risk profile and are compounded by settlement design and service delivery gaps. Mining-related hazards contribute to chronic health and structural impacts, while socio-economic hazards heighten social instability and safety risks. Public health vulnerabilities remain persistent, creating a silent but severe hazard burden.

Table 3.3: Hazard Analysis Summary for Armoede

Hazard Category	No.	Hazard	Local Causes / Drivers	Exposure Profile	Vulnerability Factors	Consequences
Hydro-Meteorological Hazards						
1	Flooding and stormwater runoff	No formal drainage; blocked channels; run-off from slopes; compacted soils	Low-lying homes; internal access roads; school walking routes	Weak foundations; soil erosion; no diversion systems	Property damage, road washaways, displacement, water contamination	**Tier 1 – Life-threatening priority risk**
2	Lightning	No lightning rods on buildings; exposed metal roofs; outdoor activities	Schools, churches, sports fields, yard areas	Lack of lightning protection; no awareness training	Injuries, deaths, damaged appliances	**Tier 2 – Serious recurring risk**
3	Strong winds	Seasonal thunderstorms; poor roof reinforcement; unsecured structures	Informal rooms, RDP houses, schools	Lightweight roofing material; nail-only fixing	Roof uplift, home damage, injuries	**Tier 1 – Life-threatening priority risk**
Climatological Hazards						
4	Heat stress and extreme temperatures	High summer heat exposure; no cooling infrastructure	Outdoor workers, school learners, elderly	Low water access; no shade structures	Heat exhaustion, dehydration	**Tier 2 – Serious recurring risk**
5	Drought and water scarcity	Borehole reliance; low groundwater recharge; climate variability	Entire village	No alternative supply; household water rationing	Reduced hygiene, water conflict, livestock loss	**Tier 1 – Life-threatening priority risk**
Environmental Degradation Hazards						
6	Soil erosion	Concentrated runoff; overgrazed land; no soil stabilisation	Footpaths, internal roads, home plots	Sloped terrain; lack of erosion control	Loss of productive land, road damage	**Tier 2 – Serious recurring risk**
7	Veld and grass fires	Dry season burning; accidental ignition; unmanaged land	Grazing fields, homesteads, boundary fences	Lack of firebreaks; no firefighting equipment	Loss of grazing, property damage	**Tier 2 – Serious recurring risk**
8	Land degradation	Overgrazing; deforestation; uncontrolled expansion	Communal land, farmland	Poor land-use control	Declining land productivity	**Tier 3 – Emerging manageable risk**
Technological and Infrastructure Hazards						
9	Electricity disruptions	Overloaded transformers; poor maintenance; lightning strikes	Entire village	No backup energy systems	Loss of perishable food, security threats	**Tier 2 – Serious recurring risk**
10	Unsafe electrical wiring	Illegal connections; overloaded multiplugs; damaged cables	Informal households, backyard rooms	Poor electrical safety	Fire outbreaks, electrocution	**Tier 1 – Life-threatening priority risk**
11	Water supply interruptions	Borehole breakdown; diesel shortages; vandalism	Entire village water system	Single supply dependency	Water crises, school closures	**Tier 1 – Life-threatening priority risk**
12	Road access failure	Storm damage to gravel roads; no culverts	Clinic routes, school transport	Weak road base, erosion-prone soil	Isolation during rain, delayed emergencies	**Tier 2 – Serious recurring risk**
Biological / Public Health Hazards						
13	Waterborne disease	Dirty water storage; runoff contamination	Children, families using communal taps	Limited sanitation	Diarrhoea, dehydration, illness	**Tier 1 – Life-threatening priority risk**
14	Sanitation-related disease	Overloaded pit toilets; greywater stagnation	Densely populated yards	No drainage	Illness outbreaks, unsafe environment	**Tier 2 – Serious recurring risk**
15	Tuberculosis (TB)	Poor indoor ventilation; overcrowding	Multi-family households	Indoor smoke exposure	Long-term respiratory illness	**Tier 2 – Serious recurring risk**
16	Animal disease outbreaks	Communal grazing; poor animal vaccination	Cattle, goats, livestock keepers	Shared grazing; no vet support	Livestock death, food insecurity	**Tier 2 – Serious recurring risk**
Human-Induced / Socio-Economic Hazards						
17	Household and structural fires	Paraffin use; candles; unsafe wiring	Informal homes, young children	Flammable structures	Deaths, asset loss	**Tier 1 – Life-threatening priority risk**
18	Alcohol-related violence and crime	High unemployment; tavern concentration; poor policing	Youth, women	Weak law enforcement	Injuries, trauma	**Tier 2 – Serious recurring risk**
19	Road transport accidents	No road markings; narrow roads; speeding taxis	Pedestrians, learners	No sidewalks or road safety measures	Injuries, deaths	**Tier 2 – Serious recurring risk**
Mining-Related Hazards						

20	Dust and air pollution	Mine haul trucks using gravel roads	Households near roadside	Chronic inhalation	Asthma, eye irritation	**Tier 2 – Serious recurring risk**
21	Blasting vibration	Mine blasting shockwaves	Houses near mine	Cracked walls, old RDP units	Structural collapse risk	**Tier 2 – Serious recurring risk**
22	Hazardous material spills	Fuel tankers passing village roads	Households and livestock along route	No hazmat response	Toxic exposure risk	**Tier 3 – Emerging manageable risk**
23	Tailings dam failure	Proximity to mining region	Downstream areas	No evacuation plan	Catastrophic flooding	**Tier 4 – Low-probability escalation risk**

Top 5 Priority Hazards for Armoede

Based on likelihood of occurrence, severity of impact, population exposure and vulnerability conditions, the top five priority hazards are:

1. Water supply interruptions: Due to single-source dependency and life-threatening consequences
2. Flooding and stormwater runoff: Seasonal recurrence and cumulative infrastructure damage
3. Household/structural fires: High frequency and severe social displacement impacts
4. Unsafe electrical wiring: Compounding fire risk and child safety threats
5. Drought and water scarcity: Growing climate pressure on water security and livelihoods

These priority hazards will guide risk reduction planning and mitigation strategy design in Chapter 4. The next section (3.5) examines exposure patterns, identifying who and what is at risk from these hazards in Armoede.

3.5 Exposure Analysis

Exposure refers to the people, physical assets, resources, services and systems that are located in hazard prone conditions and are therefore likely to experience damage, disruption or loss when hazard events occur. While hazards exist independently, exposure determines who and what is placed in harm's way. In Armoede, exposure is shaped by settlement geography, weak infrastructure layout, service delivery limitations and dependency on single point systems such as the borehole and electricity transformer. Understanding exposure is essential to identifying disaster risk because it reveals the scale of potential impact and where risk is concentrated.

Exposure in Armoede reflects development challenges that position people and assets in hazardous locations without appropriate protection or resilience measures. Many households are located along natural runoff shapes where seasonal flooding affects homes and properties. Informal housing structures are concentrated in high density yards with limited spacing, which increases exposure to fires and disease transmission. Critical services such as water and electricity have centralised supply points, which means a single system failure exposes the entire village. A community member explained during consultations that, "Water is life. When the borehole fails, it feels like life stops in the village." This reflects how significant exposure is in this community because one hazard can affect many parts of daily life.

The patterns of exposure for people, infrastructure, basic services, livelihoods and natural resources in Armoede are summarised in Table 3.4 below. Exposure in Armoede follows a spatial pattern in which some locations and groups experience higher risk because they are repeatedly exposed to harmful conditions. Flood exposure is concentrated in low level settlement areas where drainage pathways have formed naturally due to a lack of stormwater systems. Road exposure is concentrated in areas where gravel roads pass through soft soils that erode quickly during rainfall. Livelihood exposure is concentrated in grazing zones and household livestock kraals that are located close to roads and face drought and disease exposure.

Critical service exposure is one of the highest risk factors in the village. Because the entire community depends on one borehole, one electricity transformer and one main internal road network, a single hazard event can disrupt water access, energy supply and mobility for every household. This creates systemic exposure, which multiplies the effect of hazards. When the water system fails, sanitation exposure increases. When roads fail, emergency access exposure increases. When power fails, fire exposure increases because families turn to open flame alternatives.

Table 3.4: Exposure Analysis Summary for Armoede

Exposure Category	Exposed Elements	Geographic Concentration	Scale of Exposure	Risk Implication
Population exposure	Children, elderly people, people with disabilities and chronic illnesses	High density yards and low-lying areas	High	Direct exposure to hazard impact and emergency delays
Housing and settlement exposure	RDP units, informal rooms and mud brick extensions	Along stormwater paths and erosion zones	Very high	Structural damage from flooding, fire and wind
Water supply exposure	Single borehole, storage tanks and communal taps	Central borehole zone	Very high	Total community exposure to water failures
Energy exposure	Electricity transformer and informal household wiring	Central supply point and informal yards	High	Exposure to electrical fires and outages
Transport exposure	Gravel roads, footpaths and school transport routes	Southern and central internal routes	High	Exposure to access delays during storms
Livelihood exposure	Livestock, subsistence gardens and small businesses	Western grazing belts and village centre	High	Loss of livestock and income during disasters
Social service exposure	Primary school, community hall and churches	Central settlement cluster	Medium to high	Exposure to closure or disruption during hazard events
Public health exposure	Pit latrines, waste disposal points and washing sites	High density residential zones	High	Exposure to disease outbreak after flooding
Environmental exposure	Soil, trees, grassland and river channels	Settlement outer edges and grazing land	Medium	Exposure to erosion and environmental decline
Institutional exposure	Local committees and community governance	Meeting points and informal offices	Medium	Exposure to coordination breakdown during emergencies

Exposure also has a strong social dimension. Households with fewer economic resources often live in more exposed areas such as riverbanks, erosion lines and informal extensions built close together. These households are more exposed to multiple hazards at the same time and have limited ability to strengthen their homes or relocate. Exposure also affects communal life. Schools, churches and community halls are exposed to hazard disruption because they do not have protective infrastructure such as stormwater control, rapid evacuation routes or physical safety barriers.

Exposure defines who and what is at risk from hazard impact. However, exposure alone does not determine the severity of disaster loss. That depends on the vulnerability of people and systems. Section 3.6 therefore examines vulnerability conditions in Armoede and explains why exposed households, services and livelihoods experience repeated setbacks during hazard events.

3.6 Vulnerability Analysis

Vulnerability refers to the conditions that increase the susceptibility of people, livelihoods, infrastructure and systems to the harmful effects of hazards. Unlike exposure, which identifies who or what is at risk, vulnerability explains why some groups in Armoede experience greater hardship and loss when hazard events occur. Vulnerability results from physical weakness, economic limitation, social inequality, environmental deterioration and institutional shortfalls. These factors influence the ability of households and systems to anticipate, absorb, respond to and recover from hazard impact.

The National Disaster Management Framework explains that disaster risk in South Africa is fundamentally a development issue, driven less by extreme hazard events and more by the conditions that make communities vulnerable. This is evident in Armoede, where risk is intensified by poverty, dependence on informal livelihoods, inadequate planning, limited infrastructure protection and weak access to essential services. Vulnerability is therefore not caused by nature but by social and development conditions that place people at risk and restrict their ability to cope.

In Armoede, certain groups carry a higher vulnerability burden, including children, elderly residents, people with disabilities, female headed households and residents without reliable income. They often lack strong housing, savings, transport, safe sanitation and access to formal support. This reduces their ability to prepare for hazard events or recover from damage. As one resident stated during consultations, “Small problems become big disasters for poor families because they have nothing to fall back on.” This reflects how vulnerability deepens disaster impact through social and economic pressure.

The key vulnerability conditions affecting Armoede are summarised in Table 3.5, which consolidates the primary dimensions of vulnerability and their implications for disaster risk in the village. Vulnerability in Armoede is strongly linked to development conditions that increase the severity of hazard impact. The analysis in Table 3.5 shows that vulnerability is not only a result of individual household challenges but also of structural conditions such as inadequate planning, inequality in service access and limited livelihood security. This means that vulnerability is patterned and predictable, not random, and certain groups and areas experience repeated hardship from even small hazard events.

Table 3.5: Vulnerability Analysis Summary for Armoede

Vulnerability Category	Underlying Causes	Who or What is Most Affected	Evidence from Armoede	Risk Implication
Social vulnerability	Poverty, social exclusion, household dependency, limited safety networks	Children, elderly people, people with disabilities, female headed households	Households rely on neighbours for support and lack emergency resources	Increased human impact from hazards and slow recovery
Economic vulnerability	High unemployment, grant dependence, low household income, fragile informal economy	Households without stable income, spaza traders, livestock owners	Loss of work during hazards leads to stress and food insecurity	Hazard impact triggers long term livelihood disruption
Physical vulnerability	Weak building materials, informal extensions, poor construction quality	Informal backyard rooms, unreinforced walls and roofs	Wall cracks, collapsed structures and frequent wind damage observed	High risk of housing failure during storms and floods
Environmental vulnerability	Overgrazing, soil exposure, lack of land management practices	Communal grazing land, farmland and settlement edge	Visible erosion channels and degraded grazing belts	Environmental decline increases flood and drought impact
Institutional vulnerability	Limited access to disaster services, weak governance resources, slow response	Local committees, ward structures and community organisations	No emergency equipment or contingency resources available	Disaster coordination and support capacity is limited
Health vulnerability	Poor sanitation, high dust exposure, shared water storage, indoor smoke	Households with respiratory illness, children, elderly	Clinic reports show frequent cases of asthma and water related illness	Disease risk increases disaster impact and slows recovery
Livelihood vulnerability	Dependence on livestock and small informal trading, climate sensitive activities	Livestock keepers, food garden producers and informal workers	Livestock deaths reported during drought and disease outbreaks	Loss of income and erosion of coping capacity

A major contributor to vulnerability in Armoede is weak physical resilience at household level. Many homes are built using low-cost materials without proper foundations or reinforced roofing, which increases vulnerability to strong winds, floods and storms. Informal extensions and backyard rooms are especially at risk. Economic vulnerability is also significant, as many households rely on social grants and informal work. This limits their ability to save money, invest in safer structures or recover quickly from damage or loss. As a result, disaster events have a long-lasting effect on their wellbeing.

Vulnerability is also intensified by fragile essential services and limited institutional support. The village is dependent on a single borehole and electricity transformer, which increases systemic vulnerability because a single failure affects the entire community. Vulnerability is worsened by limited access to emergency services, no local disaster response equipment and delays in municipal response time. Health vulnerability is also evident in Armoede due to dust exposure from mine haul routes, poor sanitation in dense yard clusters and limited access to clean water storage options. Livelihoods are vulnerable due to climate stress, repeated drought and livestock disease, which reduce household resilience over time.

These interlinked vulnerability conditions show that disaster risk in Armoede cannot be reduced through emergency response alone. Risk reduction must focus on addressing the root causes of vulnerability including poor housing conditions, weak planning control, limited protection of livelihoods and lack of basic resilience infrastructure. Vulnerability reduction is therefore a central requirement for building long term community resilience. Section 3.7 will build on this analysis by examining existing capacity in Armoede and identifying the strengths that can be used to reduce vulnerability and manage disaster risk.

Vulnerability in Armoede is not only a technical or physical condition. It is also a lived social reality shaped by inequality and limited choices. During the community consultations, one resident expressed this clearly by saying, “When disaster comes, the poor are always the ones who feel it the most because we have nothing to protect ourselves.” This experience is common in the village and demonstrates that vulnerability is a barrier to recovery and long-term development. Unless vulnerability is reduced through deliberate planning and investment, hazards will continue to push households deeper into poverty and risk.

Section 3.7 assesses the existing capacities and coping mechanisms in Armoede. While vulnerability highlights risk and weakness, capacity identifies the strengths, resources and local systems that can reduce disaster impact. Understanding both vulnerability and capacity is essential to building resilience and informing practical disaster risk reduction strategies for the village.

3.7 Capacity and Coping Mechanisms

While Section 3.6 highlighted the vulnerability conditions that increase disaster risk in Armoede, this section focuses on the strengths, resources and coping strategies that support resilience at household and community level. Capacity in disaster risk management refers to the abilities, knowledge, systems and networks that people and organisations use to prevent, withstand, respond to and recover from hazard impacts. In Armoede, capacity exists across social, institutional and community structures even though it often operates informally and with limited resources. These structures provide a platform for coordinated disaster risk reduction planning. Capacity in Armoede is summarised in Table 3.6.

Table 3.6: Capacity and Coping Mechanisms in Armoede

Capacity Category	Existing Capacities	Evidence from Armoede	Limitations	Resilience Contribution
Community based capacity	Mutual aid, neighbour cooperation, informal disaster support networks	Households assist one another during storms, fires and funerals	Informal and dependent on voluntary effort	Strengthens social recovery after hazard events
Social support systems	Burial societies, churches, savings groups and community associations	Community support during death, poverty or crisis	No formal emergency fund or relief system	Reduces social strain and builds unity
Indigenous and local knowledge	Experience of seasonal patterns and safe grazing areas	Elders warn of approaching heavy rain and drought conditions	Not linked to formal planning systems	Enhances early preparedness behaviour
Local leadership and governance	Ward committee and traditional leadership authority	Community meetings coordinated to resolve local risks	Limited mandate on development and infrastructure	Enables community mobilisation
Institutional capacity	Water committee and local school board coordination	Community manages borehole queues and water access disputes	Committees lack resources and formal support	Provides problem solving structure
Early warning and communication	Word of mouth alerts, WhatsApp groups, community messengers	Community relays warnings rapidly during events	No official early warning system	Promotes fast risk communication
Emergency response capacity	Community rescue during fires, storms and road incidents	Residents respond quickly before services arrive	No first aid kits, no protective gear	Provides immediate household level support
Community based disaster management	Disaster Management Volunteer Force established in 2025	Twenty volunteers trained in a twelve week DRM programme	Limited equipment and no formal recognition	Strengthens organised emergency response
Livelihood adaptation capacity	Small gardens, livestock sharing and flexible income	Families produce food or share livestock during	Climate stress reduces production yield	Supports survival and food security

	strategies	hardships		

Local capacity in Armoede is driven by strong community cooperation, shared responsibility and practical problem solving. Residents have developed their own protective actions against recurring hazards such as flooding, fires and service disruptions. These include digging household water diversion channels, reinforcing roofs to reduce wind damage, establishing household emergency contacts and forming informal neighbourhood support networks. Capacity also exists through long standing social structures such as churches, burial societies, street committees and women led savings groups that provide support during hardship. These community-based systems play a crucial role in disaster recovery and social stability.

A significant development in local disaster readiness took place in 2025 when the Armoede Community Development Trust established the Disaster Management Volunteer Force (DMVF). This group consists of twenty trained local volunteers who completed a twelve-week Basic Disaster Risk Management First Responder Programme. The DMVF provides immediate response to emergencies such as household fires, missing persons, minor search and rescue, severe weather events and community safety alerts. They also support public awareness programmes and basic early warning dissemination. This represents an important step towards building structured local disaster capacity, even though the team currently operates with limited equipment and without formal municipal integration.

In addition to this emerging capacity, there are institutional strengths that can be developed further, including ward committee structures, traditional leadership authority, school governance and health outreach programmes.

Capacity in Armoede is rooted in social solidarity, practical action and community organisation. Although there are no formal emergency services based in the village, residents routinely assist each other during hazard events. The presence of the Disaster Management Volunteer Force demonstrates that the community is willing and able to organise around disaster readiness. This is a major capacity strength because local volunteers are the first to respond before external services arrive. The DMVF is already playing an important role in risk awareness, emergency coordination and basic incident response. Ideally, this local capacity should be formally recognised and integrated into the municipal disaster management system to improve coordination, training and access to essential equipment.

However, current capacity remains limited by resource constraints. There is no community disaster kit, no firefighting tools, no safe evacuation assembly points, no emergency transport and no formal risk reduction budget. Despite these limitations, local commitment to problem solving and mutual assistance remains strong. During field consultation, one community member said, “When something happens here, we do not wait for anyone. We stand together.” This demonstrates a high level of social cohesion, which is a core driver of resilience.

Section 3.8 builds on the hazard, exposure, vulnerability and capacity findings to prioritise disaster risks for Armoede. This prioritisation will guide the risk reduction focus areas presented in Chapter 4.

3.8 Risk Prioritisation

Risk prioritisation combines the results of the hazard, exposure, vulnerability and capacity assessments to determine which disaster risks in Armoede require the most urgent attention. Not all hazards present the same level of risk. Some occur frequently and cause repeated damage, while others happen less often but may cause severe losses. Prioritisation enables effective decision making by identifying where risk reduction investment and preparedness actions must be focused first.

This prioritisation follows the risk logic used in the National Disaster Management Framework and the Sendai Framework, where disaster risk is determined as a function of hazard likelihood, exposure, vulnerability and capacity. The following formula was applied:

Risk = Hazard × Vulnerability × Exposure ÷ Capacity

Hazards were scored on a scale of one to five for each factor, with capacity scored in reverse because stronger capacity lowers risk. Risk scores were then classified using the rating framework in Table 3.7.

Table 3.7: Risk Rating Classification Framework

Risk Score Range	Risk Tier	Priority Level
80 to 125	Tier 1	Priority risk requiring urgent action
50 to 79	Tier 2	Serious recurring risk
25 to 49	Tier 3	Manageable emerging risk
5 to 24	Tier 4	Low escalation risk

The risk score for each hazard identified in Section 3.4 was calculated and ranked using the risk rating methodology described earlier. Each hazard was assessed based on its likelihood of occurring, its potential impact on people and assets, the level of vulnerability in affected areas and the degree of exposure within the community. Capacity levels were also factored in, since stronger capacity reduces overall risk. This produced a comparative risk profile that makes it possible to distinguish priority risks from lower level or emerging risks. The full risk ranking is presented in Table 3.8.

The risk prioritisation confirms that Armoede faces a combination of systemic service risks and high frequency community level hazards. Tier 1 hazards represent severe and urgent risks to life, health and community stability. These include water supply interruption, flooding, household fires, unsafe electrical wiring and drought. These hazards affect the entire community and have the potential to escalate into humanitarian crises if not addressed.

Tier 2 hazards such as electricity disruptions, strong winds, road access failure and veld fires are serious recurring risks that intensify vulnerability and disrupt livelihoods and essential services. Tier 3 hazards are emerging risks linked to environmental decline, climate impacts and social pressures. Tier 4 hazards such as lightning and tailings dam failure have lower probability but high consequence potential. These must be monitored through early warning and emergency preparedness.

Table 3.8: Risk Prioritisation Matrix for Armoede

No	Hazard	Likelihood (1 to 5)	Impact (1 to 5)	Vulnerability (1 to 5)	Exposure (1 to 5)	Capacity (1 to 5 reverse)	Risk Score	Risk Tier
1	Water supply interruption	5	5	5	5	2	125	Tier 1
2	Flooding and stormwater runoff	4	5	5	5	2	100	Tier 1
3	Household and structural fires	5	4	5	4	3	93	Tier 1
4	Unsafe electrical wiring	5	4	5	4	3	93	Tier 1
5	Drought and water scarcity	4	4	5	5	3	87	Tier 1
6	Electricity disruptions	4	4	4	4	3	75	Tier 2
7	Strong winds	4	4	4	4	3	75	Tier 2
8	Road access failure	4	4	4	4	3	75	Tier 2
9	Waterborne disease	4	4	5	4	4	72	Tier 2
10	Veld and grass fires	4	4	4	4	3	75	Tier 2
11	Animal disease outbreaks	3	4	4	4	3	64	Tier 2
12	Heat stress	3	3	3	4	3	45	Tier 3
13	Sanitation related disease	3	3	4	3	3	39	Tier 3
14	Soil erosion	3	3	3	3	4	33	Tier 3
15	Dust and air pollution	4	3	3	4	4	45	Tier 3
16	Road transport accidents	3	3	4	4	4	45	Tier 3
17	Blasting vibration effects	3	3	3	3	3	33	Tier 3

18	Hazardous material spills	2	4	3	3	3	28	Tier 3
19	Land degradation	3	2	3	3	3	26	Tier 3
20	Lightning	3	2	3	2	3	20	Tier 4
21	Tailings dam failure	1	5	4	4	2	20	Tier 4
22	Public health disease outbreaks	3	3	4	3	4	36	Tier 3
23	Social instability and crime	3	3	4	4	4	45	Tier 3

The risk profile for Armoede confirms that risk reduction efforts must focus on water security, settlement protection, energy safety, emergency response capacity and infrastructure resilience. These findings directly inform the risk reduction strategies outlined in Chapter 4.

3.9 Implications for Disaster Risk Reduction

The risk profile of Armoede confirms that disaster risk in the village is not driven by extreme natural events but by development failures that place people, livelihoods and essential services in conditions of chronic risk. High risk hazards such as water supply failure, household fires, unsafe electrical wiring and flooding are direct results of inadequate basic service provision, unplanned settlement growth, weak enforcement of building standards and lack of protective infrastructure. Risk is therefore systematic and built into the daily functioning of the village rather than being caused by occasional hazard events.

The analysis in this chapter shows that risk is concentrated in predictable locations and among specific groups. High density residential clusters with informal housing face repeated flood, fire and electrical hazards. Households located along informal stormwater paths and eroded access roads experience isolation during rainfall. Livestock owners face repeated drought and animal disease shocks due to declining grazing land and lack of veterinary services. Critical services such as water and electricity have single point failure systems that expose the entire community to disruption. Risk is therefore structural and social in nature, shaped by poverty, inequality, low resilience and spatial marginalisation.

The implications of this risk profile are clear. Disaster risk in Armoede cannot be reduced through emergency response alone, because the main causes of risk are embedded in settlement layout, infrastructure design, service delivery gaps and environmental degradation. This means that disaster risk reduction must be integrated into local development planning, service improvement programmes and infrastructure investment. Risk in Armoede is a development planning issue and must be treated as such by the Local Municipality, the District Municipality and all development partners, including the private sector operating in the surrounding mining area.

Risk governance is currently weak due to a lack of local risk management planning, no formal early warning system, limited technical support from municipal disaster management services and resource constrained community level capacity. Although Armoede has social resilience through informal support networks and the Disaster Management Volunteer Force, these capacities operate without sustainable tools, funding or institutional backing. This increases long term vulnerability and limits recovery after hazard events. Without planning intervention, the village faces escalating risk and repeated local disasters that will continue to erode safety, dignity and development gains.

The risk profile confirms five priority areas that must form the foundation of risk reduction in Armoede:

1. Strengthening water security and drought management
2. Reducing flood risk through drainage, land use planning and settlement protection
3. Improving energy safety and reducing fire risk
4. Protecting critical services and safe mobility
5. Building organised community-based disaster capacity

These implications provide the direction for Chapter 4, which presents the disaster risk reduction strategy for Armoede. The strategy will identify specific risk reduction interventions, roles and responsibilities and a phased implementation framework linked to local development priorities.

4. Disaster Risk Reduction Strategy

4.1 Introduction and Purpose

Disaster risk in Armoede is not an occasional disruption. It is a persistent development challenge that affects safety, livelihoods and service delivery on a daily and seasonal basis. The risk profile presented in Chapter 3 confirmed that the major risks facing Armoede are driven not by extreme events but by underlying conditions such as service fragility, weak infrastructure planning, unsafe settlement expansion and environmental pressure. These conditions expose households to repeated hardship and prevent progress towards sustainable development.

This Disaster Risk Reduction Strategy provides a long-term strategic direction for reducing disaster risk in Armoede. It moves beyond emergency response towards a proactive system of prevention, risk mitigation and resilience building. It defines the shift from managing disasters after they occur to systematically reducing the conditions that cause them. The strategy recognises disaster risk reduction as a development function that must be integrated into planning, service delivery and local decision making.

This strategy has been developed in alignment with Key Performance Area 3 of the National Disaster Management Framework, which requires the implementation of disaster risk reduction through prevention and mitigation planning. It fulfils the requirements of Section 53 of the Disaster Management Act (Act 57 of 2002), which mandates that municipal disaster management plans prioritise risk reduction over disaster response. The strategy also supports the principles of the Sendai Framework for Disaster Risk Reduction, which calls for strengthening local resilience and governance as the foundation of risk reduction.

This strategy is intended to guide decision making by:

- Setting a clear disaster risk reduction direction for the community
- Informing development planning and investment in risk sensitive ways
- Building local risk governance and disaster management capability
- Reducing vulnerability and strengthening resilience across the community
- Directing collaborative action between households, local structures, government and partners

It provides the strategic foundation for the Disaster Risk Management Plan, which follows in Chapter 5 and outlines implementation measures.

4.2 Strategic Framework

4.2.1 Strategic Posture

The Strategic Framework defines the direction and intent of disaster risk reduction in Armoede. It sets out the core components that guide how risk will be reduced over time and how resilience will be built. It reflects the realities of disaster risk in the community, the need for coordinated action and the commitment to long term risk governance. The framework recognises that disaster risk reduction must build on both local capability and institutional support to be effective.

Vision

A safe, prepared and resilient Armoede where lives, livelihoods and community assets are protected from disaster risk.

Mission

To reduce disaster risk through proactive prevention, targeted risk mitigation, strengthened local capacity and coordinated risk governance that supports sustainable community development.

4.2.2 Strategic Objectives

This strategy is guided by five strategic objectives that provide a clear direction for long term risk reduction and resilience building in Armoede:

1. Reduce priority disaster risks that threaten lives, livelihoods and community assets through targeted prevention and mitigation.
2. Protect critical services including water supply, energy provision, road access and public health systems from disruption.
3. Promote safer settlement development, infrastructure resilience and environmental protection to address physical vulnerability.
4. Strengthen community and institutional capacity for disaster risk management through coordination, training, awareness and preparedness.
5. Mobilise resources and partnerships to support sustainable implementation of disaster risk reduction.

4.2.3 Strategic Pillars

These five pillars provide the structure through which the strategy will be delivered:

- Risk reduction and safer settlements
- Protection of essential and productive services
- Community preparedness and resilience
- Risk governance and accountability
- Sustainability and partnerships

4.2.4 Priority Risk Focus Areas

This strategy prioritises urgent attention to the top five risks identified in Chapter 3:

- Water supply interruption
- Flooding and stormwater runoff
- Household and structural fires
- Unsafe electrical wiring
- Drought and water scarcity

4.2.5 Strategic Principles

The strategy is governed by the following principles:

- Risk reduction before response
- Community centred resilience
- Prevention through development
- Shared responsibility
- Local capability strengthening
- Transparency and accountability
- Sustainability and continuity

4.2.6 Risk Reduction Enablers

This strategy will be enabled by:

- Strengthening the local institutional base through structured community disaster management capacity
- Supporting and utilising the Disaster Management Volunteer Force established in 2025

- Leveraging the role of the Armoede Community Development Trust in local development coordination
- Strengthening local risk information and early warning
- Establishing strategic partnerships with government, mining and social partners
- Embedding disaster risk reduction in local development planning and service delivery decision making

4.3 Strategic Alignment

This Disaster Risk Reduction Strategy is aligned with the legislative, policy and planning frameworks that guide disaster management and development in South Africa and the region. Strategic alignment ensures institutional legitimacy, policy coherence and coordinated implementation across local, district, national, regional and global systems. It also enables collaborative governance by linking community risk reduction efforts in Armoede with formal disaster management structures and development planning processes.

Disaster risk reduction in Armoede is not an isolated initiative. It forms part of a broader resilience agenda that connects local action to national disaster management priorities, regional cooperation frameworks and global disaster risk governance principles. Table 4.1 summarises the alignment of this strategy with the key frameworks that inform and support disaster risk reduction.

Table 4.1: Alignment of the Armoede Disaster Risk Reduction Strategy with Policy Frameworks

Policy or Framework	Requirement	Alignment with this Strategy
Disaster Management Act (Act 57 of 2002)	Section 53 requires disaster prevention and mitigation within DRM plans	Strategy prioritises proactive risk reduction
National Disaster Management Framework (NDMF, 2005)	KPA 3 focuses on integrated disaster risk reduction	Strategy structured to deliver KPA 3 outcomes
Sendai Framework for Disaster Risk Reduction (2015–2030)	Strengthen local resilience and reduce disaster risk	Strategy aligns with the four priorities of the framework.
SADC Disaster Risk Reduction Strategy (2020–2030)	Regional cooperation and resilience building	Strategy supports community-based resilience action
African Union Programme of Action for DRR (2015–2030)	Implement DRR and resilience in Africa	Strategy advances local implementation of DRR priorities
Mogalakwena Local Municipality Integrated Development Plan (IDP)	Integrate DRR in development planning	Strategy supports risk informed development planning
Mogalakwena Spatial Development Framework (SDF)	Promote safe land use and settlement planning	Strategy addresses risk sensitive land development
Waterberg District Disaster Management Plan	Local implementation linked to district DRM systems	Strategy aligns with district coordination mechanisms

This alignment ensures that disaster risk reduction in Armoede is integrated into governance and development planning processes at all levels. It confirms compliance with national legislation, supports local planning coherence and positions the strategy within recognised resilience frameworks. This strengthens opportunities for institutional support and resource mobilisation while reinforcing the strategic legitimacy of disaster risk reduction as a development priority. The next section outlines the strategic approach that guides how this strategy will be implemented.

4.4 Strategic Approach

The strategic approach defines how disaster risk reduction will be achieved in Armoede. It provides the practical logic for shifting from fragmented and reactive crisis response towards coordinated risk reduction and resilience building. This approach recognises that disaster risk is not caused by natural events alone but is produced by vulnerability and development conditions which can be reduced through deliberate and sustained action. The strategic logic guiding this approach is illustrated in **Figure 4.1**, which shows how community resilience, service protection, governance strengthening and prevention measures combine to reduce disaster risk over time.

Figure 4.1: Armoede Disaster Risk Reduction Strategic Approach Model



Community Centred Resilience

The foundation of this strategy is community resilience. Risk reduction will only be effective when local capability is strengthened, because emergency services are distant and external support is delayed during hazard impact. This approach builds local agency by supporting collective action, risk awareness, first responder capability and local preparedness planning. Community structures such as the Disaster Management Volunteer Force and civil society partners play a central role in risk detection, early warning, preparedness and local response. Risk reduction measures will be designed to be appropriate for local conditions, locally owned and sustained over time.

Integrated Risk Governance
Disaster risk is a governance challenge. To reduce systemic risk, local coordination must improve, decision making must be risk informed and risk reduction must be integrated into service delivery and planning. The strategic approach therefore promotes shared responsibility and coordinated implementation between the community, Mogalakwena Local Municipality and the Waterberg District Disaster Management Centre. Risk governance will be strengthened through structured communication, reporting, accountability and monitoring arrangements. This approach ensures that disaster risk reduction is institutionalised rather than treated as a once off project.

Protection of Essential Services
Service fragility is a major driver of disaster risk in Armoede. Interruptions to water supply, electricity and road access quickly escalate into disaster conditions and threaten lives and local livelihoods. The strategic approach therefore prioritises the protection of essential services and critical infrastructure. This includes strengthening water security, improving electrical safety, maintaining safe access routes and reducing flood risk through drainage interventions. Service protection is central to preserving community stability, supporting economic activity and protecting vulnerable households.

Prevention, Mitigation and Preparedness
This strategy is built on the prevention first principle. It intentionally shifts Armoede from a reactive disaster response posture to a proactive risk reduction system. Prevention will be achieved through risk aware planning and development control. Mitigation will reduce hazard and exposure conditions through targeted measures such as drainage improvements, fire safety upgrades and infrastructure strengthening. Preparedness will ensure readiness through training, early warning systems and contingency planning.

This strategic approach strengthens the conditions for resilience by reducing risk drivers, increasing capacity and improving local coordination. It provides the logic for operationalising disaster risk reduction in practice. The next section concludes the strategic direction before it is translated into action through the Disaster Risk Management Plan in Chapter 5.

4.5 Strategic Direction Summary
This Disaster Risk Reduction Strategy establishes a clear direction for reducing disaster risk in Armoede and safeguarding the wellbeing of the community. It responds directly to the local risk conditions identified in Chapter 3 and provides a strategic framework to guide coordinated and sustained action. The strategy recognises that disaster risk in Armoede is primarily a development challenge driven by service fragility, unsafe settlement patterns and limited risk governance. Addressing these root causes requires a shift from reactive disaster response towards proactive and resilience driven development.

The strategy provides long term guidance anchored in risk reduction, community capability, service protection and institutional alignment. It positions disaster risk reduction as a shared responsibility between the community, local government, district disaster management structures and development partners. The strategic approach is transformational in nature. It commits to preventing risk creation, reducing existing risks and building the resilience required to withstand future shocks. The next step is to translate this strategy into practical implementation. Chapter 5 presents the Disaster Risk Management Plan for Armoede, which converts this strategy into priority risk reduction measures, implementation responsibilities and timelines.

5. Disaster Risk Management Plan

5.1 Introduction and Purpose

This chapter translates the Disaster Risk Reduction Strategy outlined in Chapter 4 into a practical Disaster Risk Management Plan for Armoede. It provides clear operational measures to prevent, reduce and manage disaster risk in the community. The plan is grounded in the understanding that disaster risk in Armoede is a daily and seasonal reality driven by development conditions, service disruptions and recurring hazards. Effective disaster risk management must therefore combine risk reduction with preparedness, response and recovery measures.

The purpose of this chapter is to set out a structured plan that defines the actions required to reduce risk, protect lives and livelihoods and strengthen local resilience. It provides an implementation framework that assigns responsibilities, establishes timeframes and guides coordinated action between community structures, the Mogalakwena Local Municipality, the Waterberg District Disaster Management Centre and relevant public and private partners. The plan covers all hazard groups identified in the risk profile and takes into account the local service and infrastructure conditions that influence vulnerability.This plan is aligned with the Disaster Management Act (Act 57 of 2002) and the National Disaster Management Framework, which require disaster management plans to prioritise prevention, mitigation and preparedness, supported by coordinated response and recovery mechanisms. It is designed to be practical and action focused, ensuring that risk management responsibilities are clear and that implementation can be monitored over time.

5.2 Operational Plan Context

This Disaster Risk Management Plan is based on the risk conditions identified in Chapter 3 and informed by the strategic direction outlined in Chapter 4. Armoede faces a complex risk environment dominated by recurring hazards rather than large scale catastrophic events. The drivers of risk are rooted in development challenges such as fragile service delivery systems, unsafe settlement expansion, erosion and land degradation, limited emergency response capacity and growing climate pressure. Disaster risk is therefore a continuous condition that affects daily life and service access in the community. This operational plan is therefore designed to reduce risk at household, service and community level.

5.2.1 Link to Strategic Objectives

This plan is guided by and implements the five Strategic Objectives defined in Chapter 4:

1. Reduce priority disaster risks that threaten lives, livelihoods and community assets.
2. Protect critical services including water, energy and access routes from disruption.
3. Promote safer settlements, resilient infrastructure and environmental rehabilitation.
4. Strengthen local capacity for disaster risk management, preparedness and response.
5. Mobilise and coordinate partnerships and resources for disaster risk reduction.

All prevention, preparedness, response and recovery measures in this chapter are aligned with these strategic objectives to ensure continuity between the strategy and implementation.

5.2.2 Hazard Coverage Scope

This plan covers all twenty-three (23) hazards identified in the risk profile. These hazards are grouped as follows:

i) Hydro meteorological hazards: Flooding and stormwater runoff, lightning, strong winds
ii) Climatological hazards: Heat stress and extreme temperatures, drought and water scarcity
iii) Environmental degradation hazards: Soil erosion, veld and grass fires, land degradation

iv) Technological and infrastructure hazards: Electricity disruption, unsafe electrical wiring, water supply interruption, road access failure
v) Biological and public health hazards: Waterborne disease, sanitation related disease, tuberculosis (TB), animal disease outbreaks
vi) Human induced and socio-economic hazards: Household and structural fires, alcohol related violence and crime, road transport accidents
vii) Mining related hazards: Dust and air pollution, blasting vibration, hazardous material spills, tailings dam failure

These hazards are addressed across the prevention, preparedness, response and recovery plans that follow. A detailed Hazard Coverage Matrix will be presented at the end of Chapter 5 to demonstrate full coverage of all 23 hazards.

5.2.3 Planning Assumptions

This plan has been developed based on the following assumptions:

i) Disaster risk will increase if no deliberate risk reduction action is taken.
ii) Local emergency response capacity is limited and community based first response remains essential.
iii) Climate variability will continue to increase water stress, heat risk and storm intensity over time.
iv) Water supply dependency on a single borehole system is a major risk until redundancy is established.
v) The municipality and district disaster management centre will provide technical and advisory support when required.
vi) Community structures will continue to play a central role in preparedness, response and early warning dissemination.
vii) Resources are limited and risk reduction actions must be prioritised based on urgency and feasibility.

5.2.4 Dependencies and Conditions for Success

The success of this plan depends on:

i) Functional cooperation between the community, Mogalakwena Local Municipality and Waterberg District Disaster Management Centre.
ii) Continuing role of the Disaster Management Volunteer Force and Armoede Community Development Trust in local risk management.
iii) Access to basic equipment for first response and early warning communication.
iv) Community participation in prevention and preparedness actions.
v) Regular review of risk conditions and adjustment of actions where required.
vi) Support from government, development partners and the private sector where specialised services are required.

These assumptions and dependencies must be monitored as part of plan implementation to ensure effectiveness and continuity

5.3 Prevention and Mitigation Plan

The prevention and mitigation plan focuses on reducing the root causes of disaster risk in Armoede by addressing chronic vulnerabilities, unsafe conditions, and service-related risk drivers identified in the risk profile (Section 3.4). The plan prioritises actions that prevent the creation of new risk, reduce existing risk, and protect essential services. It aligns with the National Disaster Management Framework Key Performance Area 3: Disaster Risk Reduction (NDMF, 2005) and supports Sendai Priority 1: Understanding Disaster Risk and Priority 2: Strengthening Disaster Risk Governance (Sendai Framework, 2015–2030). Actions are prioritised to ensure a realistic, community-led and actionable plan.

Objective P1: To reduce flood and stormwater risk in high-risk areas of Armoede

5.3.1 Objective P1: To reduce flood and stormwater risk and protect internal village access routes

Implementation Priority: High

Risk Justification: *Seasonal flooding in Armoede causes road washaways, blocks access to schools and the clinic, erodes footpaths, and damages low-lying homes. Without drainage control, these impacts worsen each rainy season and increase vulnerability among children, the elderly and people with mobility challenges.*

Hazards Addressed: Flooding, stormwater run-off, erosion, access disruption, water contamination risk.

Priority Actions – Immediate Risk Reduction

These actions must be implemented urgently before seasonal rains to prevent repeated infrastructure damage and protect households.

Action	Expected Result	Timeframe	Lead Responsibility	Support Required
Clear and reopen blocked stormwater paths in high water flow areas	Reduced surface flooding and improved water flow	0–6 months	DMVF	Mogalakwena LM (tools and technical guidance)
Create basic diversion furrows to protect low-lying households	Runoff redirected away from homes	0–6 months	Households and DMVF	ACDT (tools and sandbags)
Identify and map high-risk flood points for planning upgrades	Priority areas identified for future intervention	0–6 months	DMVF and Ward Committee	Waterberg DMC

Supporting Actions – Reduction of Secondary Flood Impacts
These actions reinforce protection of infrastructure and reduce long-term erosion effects.

Action	Expected Result	Timeframe	Lead Responsibility	Support Required
Construct sandbag protection barriers along	Protection of key	6–12	DMVF and Ward	Mogalakwena LM

vulnerable internal road edges	internal access routes	months	Committee	(guidance)
Promote household-level stormwater control using drainage channels and contouring	Reduced property damage during storms	6–12 months	Households	DMVF (awareness support)

5.3.2 Objective P2: To improve water security and reduce the impact of drought and water service interruptions

Implementation Priority: High

Risk Justification: *Armoede is highly water insecure due to dependence on a **single village water source**, inconsistent water availability and limited household storage capacity. Water shortages reduce hygiene, increase disease risk, and lead to conflict over limited water access.*
Hazards Addressed: Drought, water scarcity, water service interruptions, dehydration risk, livestock water stress.

Priority Actions – Protect Essential Water Availability

These urgent actions aim to secure community water access and reduce interruptions at household level.

Action	Expected Result	Timeframe	Lead Responsibility	Support Required
Secure the main village water source through fencing and controlled access to prevent tampering and contamination	Protection of the main community water supply	0–6 months	Ward Committee and ACDT	Mogalakwena LM – Water Services
Install community emergency water storage tanks (10,000L each) in central access areas	Improved access to water during supply disruption	6–12 months	Ward Committee and ACDT	Waterberg DMC, NGOs
Promote safe water handling and storage at household level	Reduced risk of waterborne disease	0–6 months	DMVF and Ward Committee	Local Clinic (Health Education)

Supporting Actions – Strengthen Long-Term Water Resilience

These actions build community resilience to ongoing water shortages.

Action	Expected Result	Timeframe	Lead Responsibility	Support Required
Promote village-level water conservation awareness campaigns	Reduced household water wastage	6–12 months	DMVF	Local Schools and Faith Leaders
Identify safe alternative water collection points	Improved emergency	6–12	Ward Committee	Mogalakwena LM

in case of breakdown of the main system	preparedness	months	and DMVF	and ACDT

5.3.3 Objective P3: To prevent household and community fires and reduce electrical safety risks

Implementation Priority: High

Risk Justification: *Household fire risk in Armoede is mainly caused by unsafe and illegal electrical connections, overloaded plug points and damaged wiring insulation. These unsafe conditions increase the risk of fire outbreaks and pose a threat especially to children, elderly residents and households living in poorly insulated structures.*

Hazards Addressed: Electrical fire risk, household fire outbreaks, loss of property, injury and safety risk.

Priority Actions – Immediate Fire Risk Reduction

These urgent actions aim to prevent avoidable house fires and protect families from electrical hazards.

Action	Expected Result	Timeframe	Lead Responsibility	Support Required
Identify and remove high-risk illegal electrical connections in identified hotspots	Reduced fire ignition risk from electrical faults	0–6 months	DMVF	Mogalakwena LM – Electrical Unit
Conduct household electrical safety awareness through door-to-door visits	Improved safe electricity use practices	0–6 months	DMVF	Local Schools and Ward Committee
Promote safe use of extension leads and multi-plugs	Reduced overload-related fires	0–6 months	DMVF	SAPS (crime prevention talks)

Supporting Actions – Community Fire Protection Measures

Action	Expected Result	Timeframe	Lead Responsibility	Support Required
Establish basic emergency fire control points with sand buckets and beaters at community gathering areas	Improved ability to control small fires quickly	6–12 months	Ward Committee and ACDT	District Fire Services
Promote proper disposal of burning materials and	Fewer open fire	6–12	Ward Committee	Local Clinic (health

waste to prevent fire starts	hazards near homes	months		messaging support)

5.3.4 Objective P4: To reduce environmental degradation, soil erosion and land-based disaster risk

Implementation Priority: Moderate

Risk Justification: *Environmental degradation in Armoede is driven by uncontrolled stormwater run-off, overgrazing, deforestation, illegal dumping and poor land use control. These conditions increase erosion, expose households to flood damage, weaken internal roads and reduce grazing and agricultural productivity over time. Environmental damage directly increases disaster vulnerability and threatens long-term community sustainability.*

Hazards Addressed: Soil erosion, land degradation, blocked drainage, environmental health risk, increased flood exposure.

Priority Actions – Immediate Environmental Protection Measures

These actions focus on restoring natural land stability and controlling erosion in risk-prone areas to prevent further damage.

Action	Expected Result	Timeframe	Lead Responsibility	Support Required
Stabilise erosion hotspots using soil bunds and stone packing along uncontrolled run-off paths *(community labour-based)*	Reduced soil loss and protection of household areas	0–6 months	Traditional Council	DMVF (technical support), Mogalakwena LM (material guidance)
Close unsafe and expanding erosion gullies using brush-packing and basic re-vegetation *(using local materials)*	Prevent further land collapse and property risk	6–12 months	Traditional Council	ACDT (tools support), Waterberg DMC
Protect vulnerable sections of internal roads from washaway using hand-built side drains *(low-cost priority action)*	Improved road resilience to stormwater flow	0–6 months	DMVF	Mogalakwena LM – Roads Unit

Supporting Actions – Sustainable Environmental Rehabilitation

These actions establish longer-term measures to protect communal land and prevent recurring erosion.

Action	Expected Result	Timeframe	Lead Responsibility	Support Required
Implement community awareness on controlled grazing and land protection *(public education)*	Reduced land stripping from livestock	6–12 months	Traditional Council	Local Schools and Ward Committee
Establish designated waste points and enforce no-dumping areas to protect waterways *(behavioural change)*	Cleaner environment and reduced flood blockages	0–6 months	ACDT	Mogalakwena LM – Waste Services

Policy Alignment: Supports the National Environmental Management Act (1998) and NDMF KPA 3: Disaster Risk Reduction (2005).

5.3.5 Objective P5: To reduce public health risks caused by poor sanitation, waste mismanagement and unsafe water handling

Implementation Priority: High

Risk Justification: *Public health risks in Armoede are strongly linked to poor household sanitation, unsafe disposal of greywater, open dumping of waste and unsafe water storage practices. These conditions (might)lead to recurring diarrhoeal cases, skin infections, mosquito breeding and contamination of the village water supply. Public health risk also increases during flooding when waste and stagnant water mix, exposing children and elderly people to higher infection risk.*

Hazards Addressed: Waterborne disease, sanitation-related illness, poor hygiene, waste pollution.

Priority Actions – Immediate Community Health Protection

These actions must be implemented urgently to prevent avoidable health risks and disease outbreaks.

Action	Expected Result	Timeframe	Lead Responsibility	Support Required
Introduce safe greywater disposal using household soak pits *(simple low-cost health protection)*	Reduced stagnant water and mosquito breeding	0–6 months	Local Clinic	DMVF (community mobilisation), Department of Health
Conduct village waste cleanup campaigns focusing on illegal dumping hotspots *(community mobilisation action)*	Reduction in water contamination and rodent risk	0–6 months	ACDT	Ward Committee, Mogalakwena LM – Waste Unit
Promote safe water storage and handling at households to prevent contamination *(health risk reduction)*	Reduction in diarrhoeal disease risk	0–6 months	Local Clinic	DMVF

Supporting Actions – Sustainable Community Health Measures

These actions create long-term improvements in hygiene and sanitation.

Action	Expected Result	Timeframe	Lead Responsibility	Support Required
Construct drainage channels around pit latrines to prevent flooding and overflow *(environmental health safety)*	Improved sanitation safety during rainfall	6–12 months	ACDT	Mogalakwena LM – Sanitation
Provide health and hygiene education through schools and churches *(prevention education)*	Improved community hygiene practices	6–12 months	Local Clinic	Local Schools and Faith Leaders

Policy Alignment: Supports the National Health Act (2003) and Sendai Framework Priority 3: Investing in Disaster Risk Reduction (2015–2030).

5.3.5 Objective P6: To reduce mining-related risks affecting household safety, village infrastructure and environmental health

Implementation Priority: Moderate

Risk Justification: *Armoede is exposed to risks linked to mining activity in the surrounding region, including dust pollution from mine haulage trucks, ground vibration from blasting operations and potential spills of hazardous materials transported on regional roads. These risks affect household safety, cause cracks and structural damage to RDP houses, reduce air quality and increase long-term respiratory illness risk. Mining risk must be reduced while maintaining constructive engagement to protect community rights and safety.*

Hazards Addressed: Dust exposure, blasting vibration, hazardous material transport, road safety risk, structural damage to houses.

Priority Actions – Immediate Community Protection from Mining Risk

These actions address the most direct safety and environmental impacts on the village.

Action	Expected Result	Timeframe	Lead Responsibility	Support Required
Record and report structural damage to homes linked to blasting vibration *(evidence-based protection action)*	Community able to claim remediation from responsible operators	0–6 months	ACDT	Ward Committee
Engage local transport authorities to enforce speed control for mine haulage trucks near settlement *(safety-first action)*	Reduced pedestrian and road accident risk	0–6 months	Ward Committee	SAPS and Traffic Police
Request scheduled blasting notifications and safety distance compliance from mining companies *(community protection action)*	Improved household safety during blasting	0–6 months	ACDT	Traditional Council

Supporting Actions – Strengthen Mining Accountability and Risk Communication

These measures ensure long-term protection, transparency and community involvement.

Action	Expected Result	Timeframe	Lead Responsibilit	Support Required

			y	
Establish a quarterly mining risk monitoring and engagement platform *(risk dialogue action)*	Improved mining risk accountability and reporting	6–12 months	ACDT	Waterberg DMC and Mogalakwena LM
Advocate for roadside dust suppression measures such as gravel wetting near settlement edges *(air quality action)*	Reduced community exposure to dust emissions	6–12 months	ACDT	Mining Environmental Officers

Conflict Sensitivity Note: All mining-related engagement must follow transparent reporting and community consultation procedures led by the Trust to prevent conflict and ensure community unity.

Policy Alignment: Supports NDMF KPA 3 and Sendai Framework Priority 2: Strengthening Risk Governance.

5.4 Preparedness and Early Warning Plan

While Preparedness enables early action before hazards cause harm. While prevention and mitigation reduce root risk conditions, preparedness strengthens readiness to act when hazard conditions begin to escalate. This section establishes a practical, community-based system for early warning and activation in Armoede, linked to priority hazards including flooding, strong winds, household fires, heat extremes, water supply failure, public health risks and mining-related vibration impacts.

Preparedness in Armoede is built on realistic, local capacity, using existing community structures and simple, low-cost systems to improve emergency readiness. The Trust provides leadership and coordination while the Disaster Management Volunteer Force (DMVF) executes operational readiness actions. Other local stakeholders such as the Traditional Council, Ward Committee, Local Clinic and schools provide support according to function.

5.4.1 Purpose and Approach

The purpose of this Preparedness and Early Warning Plan is to:

- Ensure a structured and reliable system for alerting the community about developing risks
- Strengthen readiness to respond safely and quickly to emergencies
- Define activation triggers and procedures for common hazards
- Establish local emergency roles, coordination and communication pathways
- Build incident readiness using local capacity before external help arrives

The approach to preparedness is based on the following principles:

- **Community command clarity** – the Trust provides direction and decision-making
- **Operational readiness** – DMVF executes emergency preparedness and activation tasks
- **Local first response** – immediate community action before escalation
- **Progressive activation** – actions increase with hazard level
- **Integrated communication** – consistent messaging across all structures
- **Support escalation** – municipal and external services are called only when required

This approach is aligned with Key Performance Area 4 of the National Disaster Management Framework (NDMF, 2005) and Sendai Framework Priority 4: Enhancing Disaster Preparedness for Effective Response (2015–2030).

5.4.2 Early Warning System

An Early Warning System allows timely protective action by providing alerts before and during hazard events. Armoede will use a **multi-channel early warning system** that works even when mobile network or electricity is unreliable.

Official early warning channels in Armoede:

- WhatsApp Broadcast Alerts managed by the Trust for verified emergency messaging;
- Loudhailer Announcements used in areas with limited mobile access;
- School Notification System risk messages relayed to parents via learners;
- Church Network Alerts extended coverage through faith groups;
- Community Information Points at shops, taxi stops and community halls; and
- Trigger Monitoring rainfall, wind, fire risk and mining blasting notifications.

Rules for early warning messages:

- Must be accurate, verified and short;
- Must include: WHAT is happening, WHERE, WHO is affected and WHAT to do;
- Must be authorised by the Trust before broadcast; and
- Must use standard wording to avoid confusion during emergencies.

5.4.3 Activation Protocol

A progressive activation system will be used to manage early warning levels and response activation. This prevents confusion and ensures readiness actions increase as hazards escalate.

Activation Level	Meaning	Example Triggers	Action Required
Level 1: Advisory	Potential risk developing	Weather notice issued; mine blasting notice	Monitor situation
Level 2: Watch	Risk increasing	Heavy rainfall forecast; strong wind alert	Prepare protective actions
Level 3: Warning	Hazard occurring	Local flooding; fire outbreak; water outage	Activate local response
Level 4: Emergency	Severe impact	Evacuation required; injury; risk to life	Call external support services

Emergency Reporting Pathway

All emergencies must follow the official escalation pathway for coordination and support:

> ***Household → Trust (lead authority) → DMVF (operational activation) → Traditional Council (community mobilisation) → External Support (SAPS, EMS, Fire Services, Mogalakwena LM, Waterberg DMC)***

This pathway prevents disorder during emergencies and establishes a clear command system for community safety.

5.4.4 Preparedness and Early Warning Implementation Plan

The following implementation plan outlines the practical actions required to strengthen preparedness, early warning and readiness in Armoede. Actions are grouped by preparedness category and specify responsibility, support requirements and verification methods to ensure implementation. The Trust provides leadership and coordination, while the Disaster Management Volunteer Force (DMVF) executes operational tasks.

Preparedness Category	Action	Expected Result	Timeframe	Lead Responsibility	Support Required	Resources Needed	Verification
Coordination	Establish a Village Emergency Coordination Team (VECT) to oversee activation and reporting	Local command structure operational	Immediate (0–3 months)	Trust (leads)	Ward Committee, Traditional Council	Meeting venue, duty roster	VECT appointment record
Coordination	Develop a Community Emergency Activation Protocol (CEAP) with clear steps	Standard action sequence established	Immediate (0–3 months)	Trust	DMVF	CEAP template, printed protocol	CEAP approved
Communication	Establish official emergency WhatsApp Broadcast Groups	Rapid and reliable alerts	Immediate (0–3 months)	Trust	DMVF (technical support)	Data, admin phones	Broadcast group active
Communication	Deploy loudhailer-based announcement method	Warning coverage for all zones	Short Term (3–6 months)	Trust	Traditional Council	Loudhailer, batteries	Test log
Readiness	Identify and prepare village safe assembly points for evacuation	Known safe gathering points	Short Term (3–6 months)	DMVF (operational)	Trust	Flags/signs, basic shelter	Safe points map
Readiness	Conduct seasonal preparedness briefings before rainy and fire seasons	Improved readiness	Medium Term (6–12 months)	DMVF	Trust	Venue + materials	Attendance register
Alert Monitoring	Appoint rainfall, fire risk and wind monitors per zone	Local triggers tracked	Short Term (3–6 months)	DMVF	Ward Committee	Reporting sheets	Weekly hazard log
Public Awareness	Train households in basic emergency actions for floods, windstorms and fires	Safer household response	Immediate (0–3 months)	DMVF	Local Clinic, Schools	Posters + flyers	Training records
Resource Readiness	Issue basic emergency kits to DMVF teams	Faster emergency	Medium Term (6–12	Trust	Donors, NGOs	Torches, vests, radios	Distribution list

		action	months)				
Activation	Conduct emergency response drills (flood/fire scenario practice)	Improved activation performance	Medium Term (6–12 months)	DMVF	Trust	Drill checklist	Drill report

5.4.5 Emergency Communication Structure

A clear communication structure is essential to manage emergency information flow, avoid confusion and control escalation. All emergency information must follow the approved reporting pathway.

Level	Role	Function	Communication Action
Level 1	Households	First to detect incidents	Report immediately
Level 2	Trust (ACDT) – Command Point	Confirms incident, authorises alerts	Activates DMVF
Level 3	DMVF	Deploys initial response	Coordinates on-site
Level 4	Traditional Council	Mobilises community support	Supports Trust
Level 5	External Support	Municipal and district support	Activated only when required

This system ensures that emergencies are managed locally first, escalation is controlled and communication remains credible and trusted.

5.5 Emergency Response Plan

Preparedness enables early action before hazards escalate, but emergency response is the coordinated action taken during and immediately after a hazardous event to protect life, reduce losses and stabilise the situation. This section defines how emergency response will be activated, led and managed in Armoede under a clear community-controlled command structure.

5.5.1 Purpose

The purpose of the Emergency Response Plan is to establish a structured, community-led system for managing emergency incidents in Armoede. It provides a clear command approach, rapid activation procedures and coordinated response actions to ensure that emergencies are handled efficiently and safely. The plan is designed to:

- Protect life and prevent injuries
- Reduce damage to homes, communal assets and infrastructure
- Maintain community order during emergencies
- Enable quick, controlled activation of emergency teams
- Use local capacity as the first line of response
- Escalate to municipal or district support only when necessary

This plan is aligned with Section 53 of the Disaster Management Act (Act 57 of 2002) and Key Performance Area 4 of the National Disaster Management Framework (NDMF, 2005).

5.5.2 Response Strategy

Emergency response in Armoede is guided by the following response principles:

- Life Safety First – Protection of human life is the highest priority
- Local Response First – Immediate action starts inside the community
- Clear Command – The Trust provides leadership and incident direction
- Operational Execution – The DMVF carries out response actions
- Controlled Escalation – External services activated only when needed
- Communication Discipline – Only verified information is shared
- Protection of Critical Services – Water, access roads and electricity restored early
- Order and Stability – Prevent panic, maintain safety

Command and Control for Response

Command Function	Role Description
Incident Leadership	Trust provides leadership, coordination and authorisation
Operational Response	DMVF executes on-site response actions
Community Authority Support	Traditional Council mobilises local support and order
Coordination Support	Ward Committee assists communication with municipality
Health Support	Local Clinic provides first-aid and health alerts
Specialised Intervention	SAPS, EMS, Fire Services, Mogalakwena LM and Waterberg DMC activated only if required

This model ensures discipline, accountability and community control during emergencies.

5.5.3 Response Levels

Emergency response actions in Armoede will follow a structured Response Severity Scale to determine the appropriate level of activation and support.

Response Level	Type of Incident	Example Scenario	Response Action
Level 1: Local Incident	Minor, managed locally	Small household fire, tree blocking path	DMVF response only
Level 2: Community Incident	Affects multiple households	Flooding in low-lying area	Trust activates coordinated response
Level 3: Village Incident	Significant damage, major public risk	Windstorm damage, multiple families displaced	External support contacted
Level 4: Disaster Incident	Severe, overwhelms local system	Large-scale destruction	Full district and provincial support required

Response levels ensure organised escalation and prevent over-reaction or confusion.

5.5.4 Emergency Response Implementation Plan

The Emergency Response Implementation Plan outlines the priority response actions for Armoede during emergencies. Actions are grouped by functional response areas to enable rapid coordination and timely intervention. The Trust provides leadership and authorisation, while the DMVF carries out operational response. The Traditional Council supports mobilisation, and external services are called only when needed.

Response Area	Action	Expected Result	Timeframe	Lead Responsibility	Support Required	Resources Needed	Verification
Incident Command	Activate Village Emergency Coordination Point (VECP) at designated site near Trust office	Central command established	Immediate	Trust	Ward Committee	Activation checklist	VECP activated
Incident Command	Assign Incident Log Keeper for all emergency events	Accurate records kept	Immediate	Trust	DMVF	Incident register	Log completed
Search and Rescue	Deploy DMVF search team to locate and assist trapped or missing people	Lives protected	Immediate	DMVF	Community volunteers	Ropes, headlamps, whistles	Response record
Fire Response	Suppress small fires using fire beaters and sand	Fire spread prevented	Immediate	DMVF	Community Fire Volunteers	Fire beaters, sand buckets	Fire report
Flood Response	Support evacuation from flooded homes and redirect water using sandbags	Lives and homes protected	Immediate	DMVF	Traditional Council	Sandbags, spades	Evacuation log
Medical Support	Provide basic first-aid while Clinic is contacted	Injuries stabilised	Immediate	DMVF First Aid Team	Local Clinic	First-aid kits	First-aid register
Evacuation	Move affected people to pre-identified safe points	Safe relocation	Immediate	DMVF	Traditional Council	Safe point signage	Evacuation list
Traffic and Access Control	Clear blocked access routes for responders	Route opened	Immediate	DMVF	Community volunteers	Hand tools, reflective vests	Route report
Relief Coordination	Coordinate emergency relief (water, blankets, temporary shelter)	Basic needs met	Within 24 hours	Trust	NGOs, CBOs, private donors	Relief stock	Relief distribution log
Communication	Issue verified emergency updates to community	Controlled communication	Immediate	Trust	Ward Committee	Loudhailer, phone data	Alert message record

Damage Assessment	Conduct rapid assessment of affected homes and services	Situation known	Within 24 hours	DMVF Assessment Team	Ward Committee	Assessment forms	Rapid assessment report

5.5.5 Support Escalation

Some incidents may exceed local capacity. In such cases, escalation must follow a controlled and formal process to ensure coordinated support without loss of community leadership.

Emergency Escalation Pathway

External assistance is only requested when response needs exceed local resources:

> ***Household → Trust (lead authority) → DMVF (operational response) → Traditional Council (community support) → Mogalakwena Local Municipality / Waterberg District Disaster Centre / SAPS / EMS / Fire Services***
> ***Conditions for Escalation***

External support will be requested when:

- Lives are in danger beyond local rescue capacity.
- Fire cannot be contained locally.
- Roads become impassable and isolation occurs.
- Multiple families require emergency shelter.
- Hazardous materials or injuries overwhelm local capability.

Emergency Communication Rules

- All emergency information must be confirmed by the Trust before release.
- Only incident facts must be communicated.
- No unofficial social media reporting is allowed.
- All reports must be documented in the Incident Log.
- DMVF must send situation updates every 2 hours during active incidents.

5.6 Recovery and Continuity Plan

While emergency response stabilises immediate danger, recovery focuses on restoring normal life after an emergency, ensuring that affected households, services and livelihoods return to functional conditions. Recovery is not a once-off activity but a gradual process that begins immediately after response operations and continues until the community is stabilised. Continuity ensures that essential services remain available during and after emergencies, preventing long-term disruption.

5.6.1 Purpose

The purpose of the Recovery and Continuity Plan is to guide how Armoede resumes normal life after emergency events in a structured and organised way. The plan provides a framework to:

- Restore essential services such as water, access roads, electricity and sanitation
- Support affected households to recover safely and with dignity
- Reduce long-term disaster impacts and prevent secondary risks
- Maintain livelihoods and local economic activity
- Coordinate community, institutional and external support for recovery
- Ensure early recovery actions link to long-term risk reduction

This plan ensures that recovery in Armoede is community-led, coordinated and development-linked, rather than reactive or dependent.

5.6.2 Recovery Strategy

The Recovery Strategy ensures that recovery is systematic, fair and aligned with priority needs. Recovery efforts will follow these recovery principles:

Recovery Principle	Application in Armoede
Safety First	Recovery actions must not expose people to further risk
Restore Quickly	Focus on rapid restoration of basic services
Support First the Most Affected	Vulnerable households prioritised for recovery
Protect Livelihoods	Assist families to resume income-generating activities
Build Back Better Where Possible	Repair in a way that reduces future risk (e.g. stronger drainage)
Community Ownership	Local leadership directs the recovery process
Partnership Approach	Coordinate support with NGOs, government and private sector

Scope of Recovery and Continuity in Armoede

Recovery actions in this plan cover:

- Household recovery: Repairs to homes and living spaces
- Infrastructure recovery: Restoring roads, drainage, electricity and water supply
- Social recovery: Support for affected families, safety of children, community support
- Livelihood continuity: Support for small businesses, farmers and income security
- Service continuity: Ensure schools, the clinic and essential services continue operating
- Institutional recovery: Maintain functioning local coordination through the Trust and DMVF

The Trust will lead recovery coordination through community engagement and prioritisation, while the DMVF will support by conducting assessments and mobilising recovery actions. In addition, the Ward Committee and Traditional Council will provide governance support, and partner organisations such as NGOs, private companies and government departments will be engaged for assistance where needed.

5.6.3 Recovery and Continuity Implementation Plan

The Recovery and Continuity Implementation Plan provides structured actions to restore households, services and livelihoods after emergencies. Actions are grouped by recovery priority areas. The Trust leads recovery coordination, and the DMVF conducts operational recovery support, working with affected households and local partners.

Recovery Area	Action	Expected Result	Timeframe	Lead Responsibility	Support Required	Resources Needed	Verification
Household Recovery	Conduct rapid household damage assessments after incidents	Damage information collected for recovery planning	Immediate (0–3 days)	DMVF Assessment Team	Ward Committee	Assessment forms	Assessment report
Household Recovery	Support basic emergency repairs for damaged homes (roof patching, doors, windows)	Families stabilised quickly	Short Term (0–2 weeks)	DMVF	Community volunteers	Tools, nails, plastic sheeting	Repair log
Service Continuity	Restore community access routes affected by flooding or erosion	Roads passable for emergency access	Immediate (0–3 days)	DMVF	Mogalakwena LM (Roads)	Gravel, hand tools	Road clearance record
Water Supply Recovery	Reconnect or secure temporary emergency water access where supply is disrupted	Emergency water access restored	Immediate (0–72 hours)	Trust	Mogalakwena LM Water Unit	Water tanks, pipes	Water access report
Electricity Continuity	Report and track electricity faults affecting recovery areas	Faster power restoration	Short Term (0–7 days)	Trust	Eskom Fault Unit	Reporting contacts	Fault closure records
Livelihood Continuity	Support affected informal traders and small farmers with recovery start-up help	Local income restored	Medium Term (1–4 weeks)	Trust	Local business partners	Temporary stalls, feed support	Livelihood recovery list
Social Recovery	Provide community debriefing and support for affected families	Psycho-social recovery initiated	Short Term (0–3 weeks)	Traditional Council	Local Churches, DSD	Meeting space	Debrief record
Displacement Support	Assist displaced families with temporary shelter and reintegration	Families safely accommodated	Immediate (first 48 hours)	Trust	Ward Committee, NGOs	Mats, blankets, tarps	Shelter register

Risk Reduction in Recovery	Integrate risk reduction into reconstruction (e.g. drainage added to repaired homes)	Safer recovery achieved	Medium Term (1–3 months)	Trust	DMVF	Sandbags, stone lining	Safer repair record

5.6.4 Transition from Response to Recovery

To ensure continuity and control, recovery must begin **before** emergency response is fully closed. Transitioning from response to recovery in Armoede will follow the system outlined below.

Transition Step	Action	Responsibility
1. Incident stabilisation	Confirm danger has passed and area is safe	Trust + DMVF
2. Impact verification	Complete rapid assessment of affected people and damage	DMVF Assessment Team
3. Recovery activation	Launch Recovery Coordination Meeting	Trust
4. Prioritisation	Identify most affected households and services	Trust + Traditional Council
5. Resource allocation	Assign tools, volunteers and support for recovery	Trust
6. Coordination	Link recovery with ward, municipal and district officials	Trust + Ward Committee
7. Handover	Log recovery actions and close incident formally	Trust

Recovery must be documented and coordinated, with recovery progress reviewed weekly by the Trust until normal living conditions and essential services are restored. Continuity actions must ensure critical services like water, access, health and schooling never fully collapse, even during ongoing recovery.

5.7 Hazard-Specific Contingency Planning

Contingency planning provides predefined emergency actions for priority hazards to ensure that Armoede can respond quickly, safely and in a coordinated way when an event occurs. These plans are short operational guides for use during high-risk situations. They align with Section 3.8 (Risk Prioritisation) and are informed by Armoede's local hazard conditions.

Each contingency plan defines early warning triggers, standby actions, immediate response actions, continuity measures and clear command logic. The Trust provides leadership and coordination, and the DMVF executes operational response. Traditional Council and Ward structures support mobilisation and order.

5.7.1 Flooding and Flash Floods

Risk Justification: Flooding is a Tier 1 priority hazard in Armoede due to repeated seasonal flooding of internal roads, erosion damage and water entering homes in low-lying areas.

Early Warning Triggers:

- Continuous rainfall for more than 24 hours
- Water accumulating in flood-prone zones
- Overflow of natural drainage paths

Contingency Objective: Prevent loss of life, protect homes and maintain safe access routes during flood conditions.

Standby Actions (Before Event):

- Clear stormwater channels in high-risk areas
- Pre-position sandbags near low-lying homes
- Identify and prepare evacuation safe points
- Warn flood-prone households via WhatsApp Alerts

Immediate Actions (During Event):

- Relocate people from flooded homes to safe points
- Redirect flow using sandbags to protect homes
- Keep main access routes open for response
- Activate emergency lighting if power is lost

Continuity Measures:

- Maintain access to school, clinic and water points
- Record affected households for recovery support

Contingency Command Logic: Trust leads flood activation and coordination. DMVF implements evacuation, road clearance and protection tasks. Traditional Council supports community mobilisation. Mogalakwena LM Roads Unit requested only if access routes fail.

5.7.2 Household and Structural Fires

Risk Justification: Fires are a major recurring risk due to overcrowded housing conditions, unsafe electricity connections and open-flame cooking.

Early Warning Triggers:

- Visible smoke or flame reports from homes
- Community members reporting sparks or electrical smell
- Wind conditions increasing fire spread risk

Contingency Objective: Stop fire spread quickly and prevent loss of life and property.

Standby Actions (Before Event):

- Train families on fire-safe behaviour
- Pre-position fire beaters and sand at strategic points
- Identify nearest water access points for fire control
- Map high-risk homes with illegal wiring

Immediate Actions (During Event):

- Suppress small fires with beaters and sand
- Evacuate family members and neighbours
- Disconnect power from burning structure if safe
- Call Waterberg Fire Services if uncontrolled

Continuity Measures:

- Provide emergency shelter for displaced families
- Begin safe debris removal to prevent re-ignition

Contingency Command Logic: Trust leads fire incident prioritisation. DMVF Fire Unit executes suppression and evacuation. Traditional Council supports perimeter control. External fire services called only if fire escalates.

5.7.3 Water Supply Failure (Emergency Water Crisis)

Risk Justification: Water supply failure is the **highest priority hazard** in Armoede due to reliance on a **single water source** and regular breakdowns.

Early Warning Triggers:
- Noticeable reduction in water pressure
- Borehole failure or pump malfunction
- Diesel supply interruption or vandalism

Contingency Objective: Ensure emergency access to safe water and prevent health and sanitation risks.

Standby Actions (Before Event):
- Identify emergency water collection points
- Pre-position 5 × 10,000L storage tanks
- Train households in water treatment (chlorine/boiling)
- Establish borehole security watch

Immediate Actions (During Event):
- Activate emergency water distribution schedule
- Deliver water to vulnerable households (elderly, disabled, childcare homes)
- Issue hygiene protection alerts to prevent illness

Continuity Measures:
- Maintain supply priority for clinic and schools
- Organise temporary water trucking if outage persists

Contingency Command Logic: Trust activates the Water Crisis Team. DMVF manages emergency storage and distribution. Water Committee supports supply coordination. Mogalakwena LM and Waterberg DMC contacted only for water trucking support.

5.7.4 Strong Winds and Storm Damage

Risk Justification: Strong seasonal winds in Armoede frequently damage roofs, expose households to cold and rain, and increase risk of flying debris injuries, especially in homes with weak roofing structures.

Early Warning Triggers:
- Severe wind alerts from SAWS or local observations
- Visible roof uplift or loose roofing sheets in windy conditions
- Trees or structures bending dangerously

Contingency Objective: Reduce risk to life and prevent roof and structure collapse during severe wind events.

Standby Actions (Before Event):
- Identify weak roofing in high-risk homes
- Pre-position roofing nails, rope and plastic sheeting
- Raise community awareness on safe in-house sheltering
- Clear loose materials that can become windborne debris

Immediate Actions (During Event):
- Evacuate unsafe structures temporarily
- Secure loose roofing sheets using emergency rope tie-down method
- Remove people from under powerlines and trees
- Report damaged electricity poles to Eskom fault line quickly

Continuity Measures:

- Restore shelter for affected families within 24–48 hours
- Prioritise temporary repairs to restore family safety

Contingency Command Logic: Trust leads activation. DMVF executes roof securing, safety evacuation and road clearance. Traditional Council supports community messaging. Mogalakwena Disaster Unit called only if families are displaced or infrastructure collapses.

5.7.5 Public Health Outbreak (Waterborne Disease)

Risk Justification: Poor waste disposal, pit latrine overflow and unsafe water storage increase the risk of diarrhoeal outbreaks, especially affecting children and the elderly.

Early Warning Triggers:

- Cluster of reported diarrhoea cases
- Clinic reports increase in gastrointestinal illness
- Contaminated or visibly dirty stored water

Contingency Objective: Prevent spread of infection and protect vulnerable groups during health outbreaks.

Standby Actions (Before Event):

- Train households in safe water storage and treatment
- Promote safe waste disposal and hygiene messaging
- Monitor sanitation hotspots in the village

Immediate Actions (During Event):

- Distribute chlorine tablets or boiling guidance
- Conduct hygiene education through loudhailer alerts
- Activate safe water points for drinking and cooking
- Refer severe diarrhoea cases immediately to clinic

Continuity Measures:

- Maintain clean water access points
- Track and report infection reduction weekly

Contingency Command Logic: Trust authorises outbreak alert. DMVF supports hygiene education and clean water distribution. Clinic leads case management. Department of Health Environmental Health is activated only if outbreaks escalate.

5.7.6 Drought and Prolonged Water Scarcity

Risk Justification: Armoede faces growing water stress due to climate variability, single-source dependency and overuse. Drought reduces drinking water, affects livestock and increases sanitation risks.

Early Warning Triggers:

- Decline in borehole recovery levels
- Longer queues at water collection points
- Livestock losing weight due to dry grazing

Contingency Objective: Protect household water security and support livestock survival during prolonged dry periods.

Standby Actions (Before Event):

- Implement water-saving awareness campaigns
- Activate borehole protection and maintenance checks
- Restore additional backup storage tanks

Immediate Actions (During Event):

- Introduce household water rationing schedule
- Prioritise vulnerable families for water supply
- Provide temporary animal water points at communal sites

Continuity Measures:

- Engage Waterberg District for long-term drought relief
- Support fodder banks for livestock keepers if needed

Contingency Command Logic: Trust activates drought measures. DMVF ensures rationing and backup distribution. Farmers' Committee supports livestock protection. Mogalakwena LM and Waterberg DMC engaged only for extended drought relief.

Closing Note on Contingency Planning

These hazard contingency plans ensure that Armoede does not respond to emergencies without preparation. Each plan defines who does what, when and how, using local capacity first before requesting outside support. These plans must be reviewed before each seasonal risk period and tested through simple simulation drills once a year.

5.8 Implementation Responsibilities Summary

Effective implementation of this Disaster Risk Management Plan requires a clear assignment of responsibilities to avoid duplication, confusion or response delays. Disaster risk management responsibilities in Armoede follow a structured local command hierarchy led by the Armoede Community Development Trust (ACDT), supported operationally by the Disaster Management Volunteer Force (DMVF) and strengthened by governance support structures such as the Traditional Council and Ward Committee, with external support partners engaged only when necessary. This structure protects local leadership, ensures accountability and aligns with Section 53 of the Disaster Management Act.

5.8.1 Purpose

The purpose of this section is to summarise the specific roles and responsibilities of key structures involved in the implementation of Chapter 5. It confirms responsibility ownership for prevention, preparedness, emergency response, recovery and contingency operations in Armoede. This framework promotes coordinated action, establishes who does what, and ensures that every function in the system has a responsible lead.

5.8.2 Roles and Responsibilities Framework

Implementation responsibilities in Armoede are organised into five levels of functional authority:

i) **Strategic Leadership (Primary Authority): Armoede Community Development Trust (ACDT)**

The Trust is the central authority for disaster risk management in Armoede. It provides leadership, approves activations, allocates community resources, chairs coordination meetings, communicates official messages, mobilises partnerships and ensures alignment with district and municipal structures. No activation or public communication occurs without Trust authorisation.

ii) **Operational Execution: Disaster Management Volunteer Force (DMVF)**

The DMVF is the formal operational arm for disaster risk management in Armoede. It carries out risk reduction actions (prevention and mitigation), preparedness measures, early warning activation, emergency response, assessments and recovery support. The DMVF operates under the direction of the Trust and is accountable to the local command system.

iii) **Governance Support and Community Mobilisation: Traditional Council and Ward Committee**

The Traditional Council provides cultural authority, land-use control support and mobilisation power during risk reduction and emergency operations. The Ward Committee links the community with Mogalakwena Local Municipality, facilitates formal reporting and advocates for municipal support during emergencies.

iv) **Functional Support Structures: Committees and Essential Services**

These include the Village Water Committee, Farmers Group, School Safety Teams and the Local Clinic. They ensure continuity of essential services like water access, education and public health during emergencies. They are activated through the Trust and support hazard-specific operations.

v) **External Support and Technical Assistance: Government and Partners**

External stakeholders such as Mogalakwena Local Municipality, Waterberg District Disaster Management Centre, SAPS, EMS, Fire Services, NGOs and private sector partners provide backup or specialised support only when local capacity is exceeded. Their role is supplementary, not directive.

5.8.3 Core Implementation Responsibilities

The implementation of this Disaster Risk Management Plan is structured around five functional pillars aligned with the disaster management continuum defined by the National Disaster Management Framework. These functional areas ensure clear accountability, unified command control, and coordinated execution across all phases of disaster risk management. Each pillar has a designated lead to prevent duplication and ensure operational discipline. Table 5.X provides a summary of the core implementation responsibilities across the five disaster management functions.

Table 5.X: Core Implementation Responsibilities by Disaster Management Function

Disaster Management Function	Lead Responsibility	Operational Execution	Support Structures	Core Responsibilities
Prevention & Mitigation	ACDT	DMVF (Risk Reduction)	Water Committee, Farmers, Traditional Council	Reduce hazard exposure; stormwater management; safe settlement measures; infrastructure protection
Preparedness & Early	ACDT	DMVF (Early Warning)	Ward Committee,	Community alerts; seasonal

Warning			Schools, Clinic	mobilisation; communication systems; readiness campaigns
Emergency Response	ACDT	DMVF (Incident Commander and Response Team)	SAPS, EMS, Fire Services, Traditional Council	Incident command; evacuation; first response actions; lifesaving
Recovery & Continuity	ACDT	DMVF (Recovery Team)	NGOs, Farmers, Ward Committee	Damage assessments; livelihood restoration; social recovery support
Hazard-Specific Contingency	ACDT	DMVF (Contingency Unit)	Clinic, Fire Services	Hazard standby plans; trigger monitoring; contingency action

This functional structure establishes a disciplined implementation system in Armoede where responsibilities are clear, centralised and enforceable. It reinforces the leadership role of the Armoede Community Development Trust as the entry point for disaster risk governance, while enabling the DMVF to perform structured operations under authorised activation. These implementation roles are translated into operational actions in Chapter 5 and coordinated through the institutional framework presented in Chapter 6.

5.8.4 Link to Chapter 6

This responsibilities framework provides the foundation for Chapter 6: Institutional and Coordination Arrangements, which will formalise how implementation is governed, sustained and supported. Chapter 6 will define the organisational structure, reporting lines, partnership mechanisms, resourcing strategies and accountability systems necessary to drive continuous implementation of the Disaster Risk Management Plan in Armoede.

6. Institutional and Community Coordination Arrangements

6.1 Purpose

This chapter establishes the governance and coordination system that will guide the implementation of the Armoede Community Disaster Risk Management Plan. It defines leadership roles, decision-making authority, operational coordination, partnerships, monitoring mechanisms and sustainability measures. Effective disaster risk management depends on strong institutional arrangements, and this chapter ensures that implementation is not fragmented or personality-dependent, but structured, accountable and sustainable.

Disaster risk management in Armoede is community-led, anchored by the Armoede Community Development Trust (ACDT) as the local coordination authority and official entry point for all disaster-related communication, coordination and support. The system is aligned with:

- The Disaster Management Act (57 of 2002);
- The National Disaster Management Framework (NDMF, 2005);
- Section 53 requirements for integrated disaster risk planning; and
- Cooperative Governance principles across local, district and provincial systems.

The institutional arrangements ensure that:

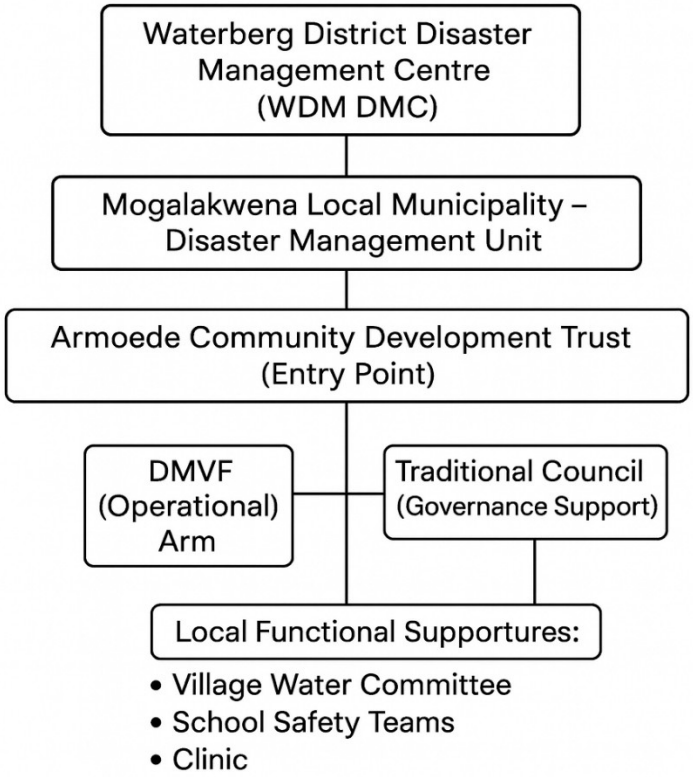
- Leadership remains inside the community;
- Roles and responsibilities are clearly distributed;
- Coordination is structured and disciplined;
- Volunteer management is safe and accountable;
- Partnerships and resources are controlled, not chaotic; and
- Reporting and compliance are maintained.

6.2 Community Leadership and Local Disaster Structure

Disaster risk governance in Armoede follows a local command model led by the Trust. It ensures that decision-making remains close to the community while being connected to government disaster management systems. The disaster risk governance structure for Armoede is illustrated in Figure 6.1 below, showing how local authority and external support organisations align under a single coordination pathway.

Figure 6.1: Armoede Disaster Risk Governance Structure





This structure ensures:

- **Clear authority** – Trust-led coordination prevents fragmented responses
- **Local operational capacity** – DMVF executes DRM actions
- **Governance legitimacy** – Traditional Council and Ward Committee support community mandate
- **Formal government link** – Mogalakwena Municipality and Waterberg District enable escalation
- **Participation** – Local committees ensure sectoral involvement

All disaster coordination and support must flow **through the Trust** to maintain a single, disciplined disaster management system.

6.3 Operational Coordination System

Armoede uses a three-level disaster coordination system that connects village decision-making with municipal and district disaster structures. This ensures local response remains primary, with external support activated only when required. Table 6.1 provides a summary of the coordination framework.

Table 6.1: Disaster Risk Management Coordination Levels for Armoede

Coordination Level	Coordination Platform	Function
Level 1: Local Command	Armoede Community Disaster Coordination Forum (ACDCF), chaired by Trust	Leads DRM implementation, approves activations, coordinates emergency actions
Level 2: Municipal Interface	Mogalakwena Local Municipality Disaster Unit	Provides technical DRM support, resources and reporting alignment

Level 3: District Support	Waterberg District Disaster Management Centre (DMC)	Disaster declarations, specialised resources, large-scale support

The Armoede Community Disaster Coordination Forum (ACDCF) meets:

- **Monthly** for implementation review
- **Seasonally** before rainy and fire seasons
- **Post-incident** for emergency reviews and recovery direction

Its core mandate includes:

- Approving local disaster readiness measures
- Monitoring high-risk periods and issuing local alerts
- Escalating emergency reports to the municipality/district
- Coordinating disaster recovery and rehabilitation

6.4 Volunteer and Community-Based Disaster Management

The Disaster Management Volunteer Force (DMVF) is the official operational unit supporting disaster management in Armoede. It functions as a single, disciplined volunteer force under the authority of the Trust, not as fragmented sub-teams. The DMVF is activated for community safety, prevention, response and recovery operations.

Core responsibilities of the DMVF include:

- Clearing stormwater drains, erosion control and hazard mitigation;
- Supporting early warning activation and alerts;
- Managing household evacuation and fire response support;
- Assisting vulnerable families during emergencies;
- Conducting rapid damage assessments; and
- Supporting relief distribution and recovery efforts.

The DMVF operates through:

- Authorised activation only (no informal deployments);
- Incident reporting procedures;
- Volunteer safety protocols; and
- Regular training and readiness sessions.

All deployments and external communications involving volunteers must go through the Trust to ensure safety, liability control and coordinated implementation.

6.5 Partnerships and Support Institutions

Effective disaster risk management requires structured partnerships that enhance community capability without weakening leadership control. All partnerships must be coordinated through the Trust as the official entry point. Table 6.2 provides a summary of partnership priorities and categories.

Table 6.2: Partnership Categories and Roles in Disaster Risk Management

Partnership Type	Stakeholders	Support Role
Government Disaster Support	Mogalakwena LM DRM Unit, Waterberg DMC	DRM technical support, training and escalation
Emergency Services	SAPS, EMS, Waterberg Fire Services	Life-saving response for fire, rescue and medical
Development and Relief Partners	NGOs, CBOs, FBOs, universities	Resilience programmes and humanitarian support

Economic and Sector Partners	Local mines, private sector, farmers	Logistics, water trucking, machinery support

Partnership Principles

Partnership principles, listed below, ensure that all external support strengthens community resilience rather than creating dependency, duplication or uncoordinated interventions:

- No organisation acts independently in Armoede
- All support aligns with this plan
- No political exploitation of disaster events
- Partnerships must be formalised wherever possible
- Community governance of aid is maintained

6.6 Monitoring, Evaluation and Learning (MEL)

Monitoring, Evaluation and Learning ensures this plan delivers results. It provides accountability, tracks progress and drives continuous improvement. Table 6.3 below outlines the MEL roles.

Table 6.3: MEL Roles and Responsibilities

MEL Function	Lead	Support
Overall, MEL coordination	Trust	Ward Committee
Implementation tracking	DMVF	Trust Secretariat
Incident data records	DMVF Operations	Traditional Council
Partner programme tracking	Trust	Ward Committee
Evaluation reviews	ACDCF	Waterberg DMC
Learning repository	Trust + DMVF	Local schools

Table 6.4 communicates the monitoring schedule.

Table 6.4: Monitoring Schedule

Activity	Frequency	Reporting To
DRMP action tracking	Monthly	Trust
Hazard watch	Continuous	DMVF
Seasonal readiness	Bi-annually	ACDCF
Emergency logs	Per incident	Trust
Partner activity	Quarterly	Trust

Table 6.5 shows the indicators for tracking progress.

Table 6.5: Community-Level DRM Indicators

Indicator Type	Indicator Example
Risk Reduction	Drainage cleared sites
Preparedness	Households reached by warnings
Response	Alert to activation time
Recovery	Affected households assisted
Institutional	DRM meetings held

Learning will be built through post-incident reviews, seasonal reflections and shared documentation.

6.7 Sustainability Framework

Sustainability ensures disaster risk management continues long term. This system is sustained through:

i) **Institutional Continuity:** DRM stays a permanent Trust function;
ii) **Leadership Continuity:** pipeline of future leaders via training;
iii) **Community Participation:** Ownership through involvement;
iv) **Resource Continuity:** Community-managed preparedness fund; and
v) **Partnership Continuity:** Long-term agreements with key partners.

The Trust commits to:

- Review this plan annually;
- Maintain DRM reporting;
- Protect community ownership of DRM; and
- Strengthen youth and women DRM leadership.

This chapter formally establishes disaster governance in Armoede, ensuring that risk management is coordinated, compliant and community-driven.

6.8 Conclusion

This Disaster Risk Management Plan establishes a structured, coordinated and community-led system for reducing risk, building resilience and protecting lives and livelihoods in Armoede. It moves the community beyond crisis-driven responses toward a disciplined approach to prevention, preparedness, response and recovery, ensuring that disaster risk management is not an event-based reaction, but a permanent function of local development governance. The institutional framework defined in this chapter makes disaster coordination clear, accountable and enforceable by establishing the Armoede Community Development Trust (ACDT) as the official entry point and coordination authority for disaster risk management in the village.

Successful implementation of this plan requires shared responsibility and cooperation between local leadership structures, the Disaster Management Volunteer Force (DMVF), traditional governance, community committees and external government and development partners. Implementation discipline will be secured through the monitoring and reporting mechanisms set out in Section 6.6, while long-term continuity will be ensured through the sustainability framework in Section 6.7. The plan reinforces the principle that disaster risk reduction is not a government task alone, but a community safety obligation tied to development progress, dignity and protection of vulnerable households.

This plan represents a collective commitment to proactive risk reduction, coordinated emergency management and resilience-building in Armoede. It provides a clear decision-making model, leadership structure, roles and procedures that enable decisive action before, during and after emergencies. With structured implementation, this plan will reduce hazard impacts, strengthen preparedness and enhance the village's ability to withstand and recover from disasters.

This Disaster Risk Management Plan is hereby adopted as the official disaster risk management framework for Armoede and will be implemented under the leadership of the Armoede Community Development Trust, with immediate effect.

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Annexure 1: Community Hazard and Risk Register

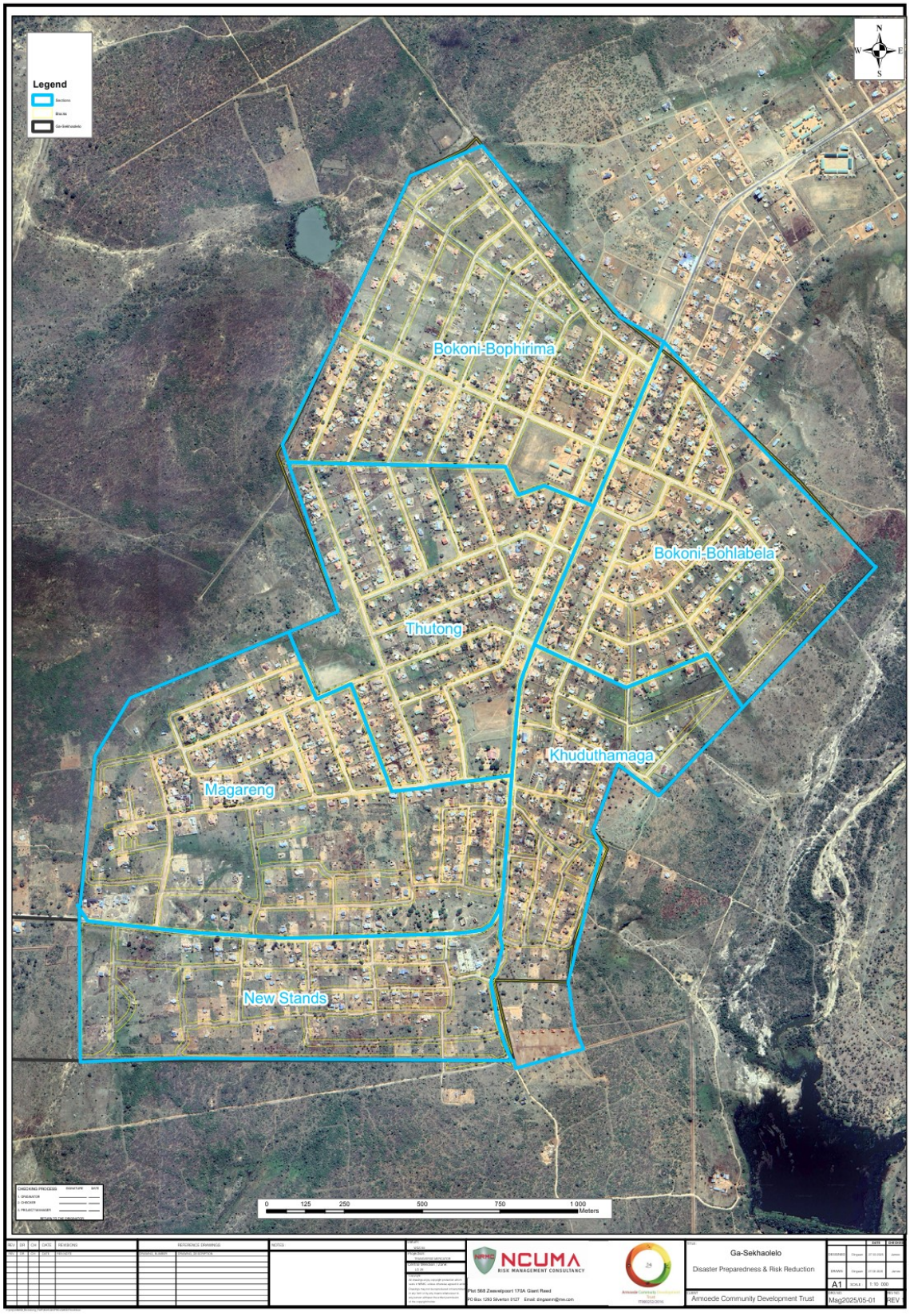
Referenced in Section 3.4 – Hazard Analysis and Section 3.8 – Risk Ranking

No.	Hazard Category	Hazard	Associated Vulnerability	Location / Area Affected	Potential Impact / Risk Description	Proposed Mitigation / Action
1	**Hydro-Meteorological**	Flooding and stormwater runoff	Infrastructure, property, health	Low-lying areas, road crossings	Flooding of homes, road washaways, water contamination, and isolation of households	Construct and maintain drainage channels and culverts; clear blockages; install diversion systems
2		Strong winds	Housing, infrastructure	Informal and RDP dwellings	Roof uplift, falling debris, injuries, and property loss	Reinforce roofing structures; secure lightweight materials; community awareness on wind hazards
3		Lightning	Human, livestock, property	Schools, churches, sports fields, open yards	Injury, death, damage to appliances and structures	Install lightning rods; provide awareness on lightning safety
4		Water supply interruption	Health, livelihoods, sanitation	Entire community (single borehole reliance)	Water shortages, hygiene decline, livestock loss, social conflict	Drill additional boreholes; install water storage tanks; promote conservation practices
5	**Climatological**	Drought and water scarcity	Agriculture, food security	Entire community and grazing lands	Crop failure, livestock deaths, household water rationing	Water harvesting; drought-tolerant crops; efficient irrigation and borehole maintenance
6		Heat stress and extreme temperatures	Human health, productivity	Outdoor workers, learners, elderly	Dehydration, fatigue, school absenteeism	Provide shaded areas; hydration programmes; awareness campaigns
7	**Environmental / Ecosystem Degradation**	Soil erosion	Environment, infrastructure	Slopes, unpaved roads, homesteads	Loss of fertile soil, road damage, structural instability	Re-vegetation, contouring, erosion barriers
8		Land degradation	Livelihoods, environment	Communal grazing and farmlands	Reduced agricultural productivity, dust generation	Controlled grazing; soil rehabilitation projects
9		Veld and grass fires	Livelihoods, property	Peri-urban areas, grazing fields, farm boundaries	Destruction of grazing land, property loss, smoke inhalation	Establish firebreaks; train community volunteers; provide firefighting tools
10	**Technological / Infrastructure**	Unsafe electrical wiring	Human, property, infrastructure	Informal settlements, small businesses	Electrical fires, electrocution, loss of property	Safety inspections; community rewiring support in partnership with Eskom
11		Electricity disruptions	Economic, social, and safety	Village-wide	Increased fire risk from candles and paraffin; disruption of livelihoods	Maintenance collaboration with Eskom; backup power (solar); community alert protocols
12		Road access failure	Emergency response, mobility	Internal and connecting routes	Delayed response to medical emergencies; access barriers	Routine road maintenance; culvert installation; drainage improvement
13		Hazardous material spills	Environment, human safety	Fuel transport routes	Soil and water contamination; health risk	Develop spill response plan; emergency coordination with transporters
14		Household and structural fires	Human safety, property	Residential clusters, high-density zones	Injury, loss of life, destruction of homes	Firefighting kits; community training; safer energy alternatives
15		Road transport accidents	Human safety	Main access roads, intersections	Injuries and fatalities from speeding or poor visibility	Road signage, speed control, pedestrian safety programmes
16	**Socio-Economic / Human-Induced**	Food insecurity and loss of income	Livelihoods, social stability	Vulnerable households	Malnutrition, poverty, theft	Community gardens; skills and microenterprise development
17		Alcohol-related violence	Social stability, health	Taverns, community gatherings	Domestic violence, crime, injury	Behavioural campaigns; social support and counselling
18		Animal disease outbreaks	Livelihoods, food security	Livestock zones	Loss of income, reduced nutrition, health risks	Veterinary outreach, regular vaccination campaigns
19		Dust and air pollution (mining-related)	Human health	Areas near haul roads and mining routes	Respiratory and eye ailments, reduced air quality	Dust suppression by mines; tree barriers; environmental monitoring

20		Tailings dam failure (external risk)	Human, environment	Downstream areas near Mogalakwena Mine	Large-scale flooding, toxic contamination	Emergency coordination with mining company; early warning system

Annexure 2: Community Hazard Maps

Referenced in Section 3.4 (Hazard Analysis) and Section 5.3 (Prevention and Mitigation Plan)



Annexure 3: Seasonal Hazard and Preparedness Calendar

Referenced in Section 5.4 (Preparedness and Early Warning)

Purpose

This annexure presents the seasonal timing of hazards affecting Armoede Village, linking each to early-warning triggers and preparedness measures. It supports coordinated readiness across the Armoede Community Development Trust (ACDT), Disaster Management Volunteer Force (DMVF), and partner institutions.

Table A3.1: Seasonal Hazard and Preparedness Calendar

Season (Period)	Hazard	Hazard Category	Indicative Triggers / Early Warning Signals	Preparedness and Mitigation Actions	Lead Responsibility	Supporting Partners
Summer (Dec – Feb)	Flooding and stormwater runoff	Hydro-Meteorological	Heavy rainfall; blocked drains; rising stream levels	Clear culverts; inspect bridges; activate evacuation routes	DMVF Early Warning Unit	ACDT, Municipality, Kgoro
	Lightning	Hydro-Meteorological	SAWS thunderstorm alerts; electrical activity	Maintain lightning rods; suspend outdoor events; safety education	DMVF Safety Desk	Schools, Churches, SAWS
	Strong winds / hailstorms	Hydro-Meteorological	Gusty conditions; storm warnings	Reinforce roofs; secure structures; prune trees	DMVF Response Team	Public Works, ACDT
	Heat stress / extreme temperatures	Climatological	Temperatures > 35 °C; heatwave alerts	Provide shaded areas; hydration campaigns	DMVF Health Desk	Clinics, Schools
	Road transport accidents	Technological / Infrastructure	Wet roads; festive traffic peaks	Maintain signage; patrols; first-aid readiness	DMVF Response Team	SAPS, Traffic Dept
Autumn (Mar – May)	Drought onset / water scarcity	Climatological	Declining rainfall; falling borehole yield	Promote water saving; service boreholes	ACDT Technical Team	DMVF, Farmers
	Water supply interruptions	Hydro-Meteorological	Equipment failure; low pressure	Preventive maintenance; standby power	ACDT Infrastructure Desk	Municipality
	Veld and grass fires (build-up phase)	Environmental / Ecosystem	Dry vegetation; open burning	Maintain firebreaks; awareness campaigns	DMVF Fire Sub-Unit	Fire Services, Farmers
	Soil erosion / land degradation	Environmental / Ecosystem	Early runoff; bare slopes	Vegetate bare soil; repair drains	ACDT Environmental Desk	Public Works
Winter (Jun – Aug)	Drought peak	Climatological	Extended dryness; low water tables	Enforce water-use controls; promote rainwater harvesting	ACDT Technical Team	Municipality, DMVF
	Veld and structural fires	Environmental / Ecosystem	Low humidity; windy conditions	Strengthen fire patrols; maintain equipment	DMVF Fire Sub-Unit	Fire Services, Kgoro
	Electricity disruptions	Technological / Infrastructure	Load-shedding; transformer faults	Maintain equipment; promote solar lighting	ACDT Technical Desk	Eskom
	Unsafe electrical wiring	Technological / Infrastructure	Illegal connections; exposed lines	Household inspections; rewiring campaigns	DMVF Safety Desk	Eskom, Kgoro
	Air pollution / respiratory illnesses	Environmental / Ecosystem	Smoke; dust; indoor fires	Health education; medical outreach	DMVF Health Desk	Clinics, Schools
Spring (Sep – Nov)	Early storms / flash floods	Hydro-Meteorological	First rains; storm alerts	Inspect culverts; reinforce embankments	DMVF Early Warning Unit	ACDT, Municipality
	Soil erosion (re-emergence)	Environmental / Ecosystem	Heavy rains on bare ground	Restore vegetation; clean stormwater channels	ACDT Environmental Desk	Public Works
	Animal disease outbreaks	Socio-Economic /	Veterinary alerts; livestock	Vaccination drives; isolate herds	DMVF Livelihoods	Veterinary Services

		Human-Induced	deaths		Desk	
	Dust and air pollution (mining-related)	Socio-Economic / Human-Induced	Windy conditions; mine operations	Engage mines; monitor air quality; plant trees	ACDT Environmental Desk	Mining Company
	Food insecurity / income loss	Socio-Economic / Human-Induced	Poor yields; low employment	Promote gardens; skills training; cash-for-work	ACDT Livelihoods Desk	DMVF, NGOs
	Alcohol-related violence	Socio-Economic / Human-Induced	Festive gatherings; tavern activity	Community dialogues; patrols; referral systems	DMVF Safety Desk	SAPS, Churches, Social Workers
	Tailings dam failure (external risk)	Socio-Economic / Human-Induced	Heavy rains near mine; seepage signs	Conduct drills; maintain mine early-warning link	ACDT Disaster Desk	Mine Management, DMVF
Year-Round (Continuous)	Electricity disruptions	Technological / Infrastructure	Load-shedding; theft; faults	Maintain transformers; adopt renewable lighting	ACDT Technical Desk	Eskom, Community Safety Forum
	Unsafe electrical wiring	Technological / Infrastructure	Informal connections; degraded lines	Rewiring campaigns; awareness sessions	DMVF Safety Desk	Eskom, Kgoro
	Food insecurity (low-income households)	Socio-Economic / Human-Induced	High unemployment; limited land	Livelihood support; youth skills training	ACDT Livelihoods Desk	NGOs, Farmers
	Alcohol-related violence and social conflict	Socio-Economic / Human-Induced	Substance abuse; household tensions	Counselling; peer-support; education	DMVF Safety Desk	SAPS, Churches
	Hazardous material spills / transport accidents	Technological / Infrastructure	Frequent road transport; fuel movement	Emergency spill-response readiness; coordination	DMVF Response Team	SAPS, Traffic Dept
	Abandoned structures / illegal dumping	Environmental / Ecosystem	Crime; waste accumulation	Secure structures; provide waste skips	ACDT Environmental Desk	Municipality, Kgoro

Summary

- **Seasonal profile:** Summer and Spring are dominated by hydro-meteorological hazards, while Winter heightens technological and environmental risks.
- **Year-round vigilance:** Chronic hazards such as unsafe wiring, social instability, and waste management require continuous attention.
- **Operational integration:** Quarterly reviews (Feb, May, Aug, Nov) ensure coordination across ACDT and DMVF portfolios.

Annexure 4: Armoede Disaster Management Volunteer Force (DMVF) Structure and Training Record

Referenced in Section 3.7 (Capacity and Preparedness) and Section 6.4 (Institutional Structure)

Purpose

This annexure provides a consolidated view of the organisational structure, composition, and training status of the Armoede DMVF. It records existing capacity, training achievements, and linkages within the Armoede Community Disaster Management System to support operational readiness, accountability, and continuous skills enhancement.

Overview of the DMVF

The Disaster Management Volunteer Force (DMVF) serves as the *operational arm* of the Armoede Community Development Trust (ACDT).

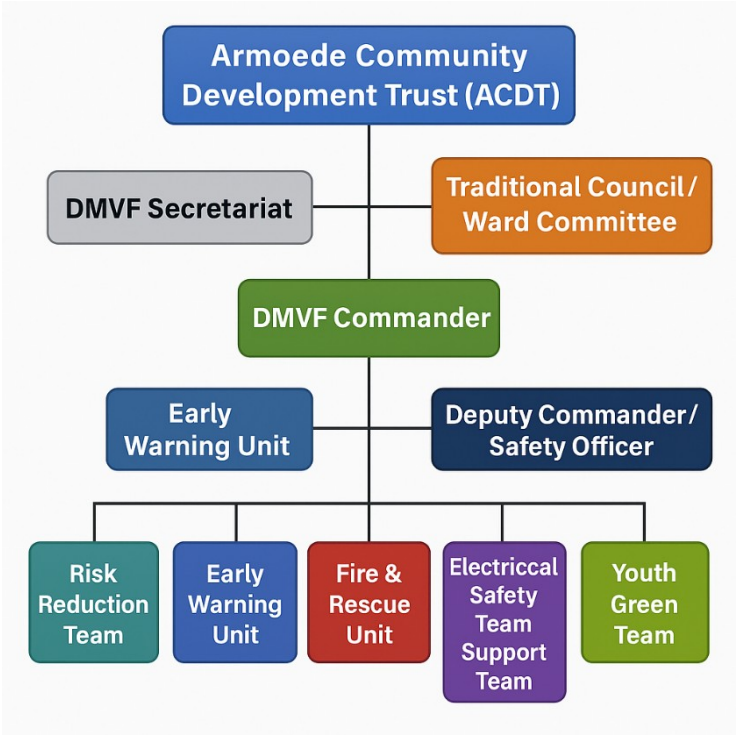
It consists of twenty trained local volunteers who completed a twelve-week Basic Disaster Risk Management First Responder Programme.

The DMVF executes community safety, early-warning, and first-response activities including:

- Clearing blocked stormwater drains and erosion control
- Supporting household evacuation and fire suppression
- Assisting vulnerable households during crises
- Conducting rapid damage and needs assessments
- Supporting relief distribution and post-incident recovery

All activations are authorised through the ACDT, ensuring safe, disciplined, and coordinated deployment.

Figure A4.1: Simplified DMVF Organisational Structure



DMVF Institutional Structure

Level / Unit	Designation / Function	Reporting Line	Key Responsibilities
ACDT Board of Trustees	Oversight Authority	Reports to Community & Regulators	Custodian of the DRM Plan; approves activations; liaises with municipal and district structures.
ACDT IOC/ DMVF Secretariat	Coordination and Administration	Reports to ACDT Board	Maintains incident registers, training records, and external communications.
DMVF Commander	Field Command & Control	Reports to ACDT Disaster Desk	Leads volunteer deployments; ensures safety and discipline.
Deputy Commander / Safety Officer	Safety and Incident Oversight	Reports to DMVF Commander	Oversees PPE use, first-aid readiness, and incident logs.
Risk Reduction Team	Drainage & Infrastructure Support	Reports to DMVF Commander	Drainage maintenance, erosion control, debris removal.
Early Warning & Communication Unit	Alerts & Information Flow	Reports to DMVF Secretariat	Disseminates warnings; maintains liaison with SAWS and schools.
Fire & Rescue Support Unit	Response & Rescue	Reports to DMVF Commander	Fire containment, household evacuation, light search and rescue.
Electrical Safety & Hazard Team	Infrastructure Safety	Reports to Deputy Commander	Detects faulty wiring; supports Eskom coordination.
Livelihoods & Health Support Team	Social and Health Relief	Reports to DMVF Secretariat	Assists vulnerable families, supports clinic referrals, manages relief distribution.
Youth Green Team	Environmental Rehabilitation	Reports to DMVF Commander	Implements gully stabilisation and revegetation activities.

This structure ensures:

- **Clear Authority:** ACDT leadership avoids fragmented responses.
- **Operational Capacity:** DMVF executes front-line DRM actions.
- **Governance Legitimacy:** Traditional Council and Ward Committee endorse and support the mandate.
- **Institutional Linkage:** Municipal and District DMCs provide technical support and escalation channels

D. Training and Capacity Development Record

Training Programme / Course	Duration	Provider / Facilitator	Participants (No.)	Core Competencies Covered	Status / Remarks
Basic Disaster Risk Management First Responder Programme	12 weeks	Waterberg DMC & Mogalakwena LM DRM Unit	20	First aid, fire response, evacuation techniques, incident reporting protocols	Completed (2025)
Early Warning and Simulation Exercise	2 days	ACDT + Schools + Clinic	20 + Community	Message relay, alert timing, communication chain testing	Annual drills completed
Electrical Safety Inspection Training	3 days	Mogalakwena LM Electrical Dept / DMVF Safety Unit	10	Household inspection procedures, risk identification and remediation	Ongoing rotational refreshers
Fire Prevention and Basic Firefighting	5 days	Waterberg Fire Services & DMVF Fire Sub-Unit	15	Use of extinguishers, safe evacuation, firebreak maintenance	Completed
Community Relief and Logistics Training	2 days	NGO Partner / ACDT Relief Desk	12	Needs assessment, distribution protocols, vulnerable group support	Completed
Digital DRM Monitoring Register Training	1 day	ACDT Secretariat + Ward Committee	8	Data capture and reporting for monitoring framework	Initiated; follow-up scheduled
Environmental Risk Mitigation and Gully Rehabilitation	1 week	ACDT & Farmers Forum	10	Brush-packing, grass planting, soil stabilisation	Completed with support from Youth Green Team

E. Sustainability and Next Steps

- Maintain a bi-annual training (or refresher) calendar and annual simulation schedule.
- Formalise recognition of the DMVF within municipal disaster systems for equipment and insurance support.
- Establish a Community Disaster Equipment Fund managed by the ACDT for PPE and response tools.
- Strengthen linkages with the Waterberg DMC for continuous skills development and joint exercises.

Annexure 5: Early Warning and Emergency Communication Protocol

Referenced in Section 5.4 - Preparedness and Early Warning and Section 6.4 - Coordination

Purpose
To provide a systematic and community-integrated protocol for detecting, validating, communicating, and acting on early warning information in Armoede. The protocol ensures that risk information translates into *timely, inclusive, and actionable communication* that saves lives, safeguards livelihoods, and protects local development gains.

This annexure operationalises the Sendai Framework Priority 4, *Enhancing Disaster Preparedness for Effective Response and to "Build Back Better" in Recovery, Rehabilitation and Reconstruction*, and aligns with Key Performance Area 4 of South Africa's National Disaster Management Framework (NDMF, 2005).

5.1 Conceptual Basis
Early warning is not an event but a systemic capability built upon science, communication, trust, and social mobilisation. Its effectiveness depends on four interdependent pillars:

i) Risk Knowledge: Understanding who and what is at risk;
ii) Monitoring and Forecasting: Continuous detection and interpretation of hazard data;
iii) Dissemination and Communication: Reliable, accessible, and trusted message delivery;
iv) Response Capability: The ability and willingness of people to act when warned.

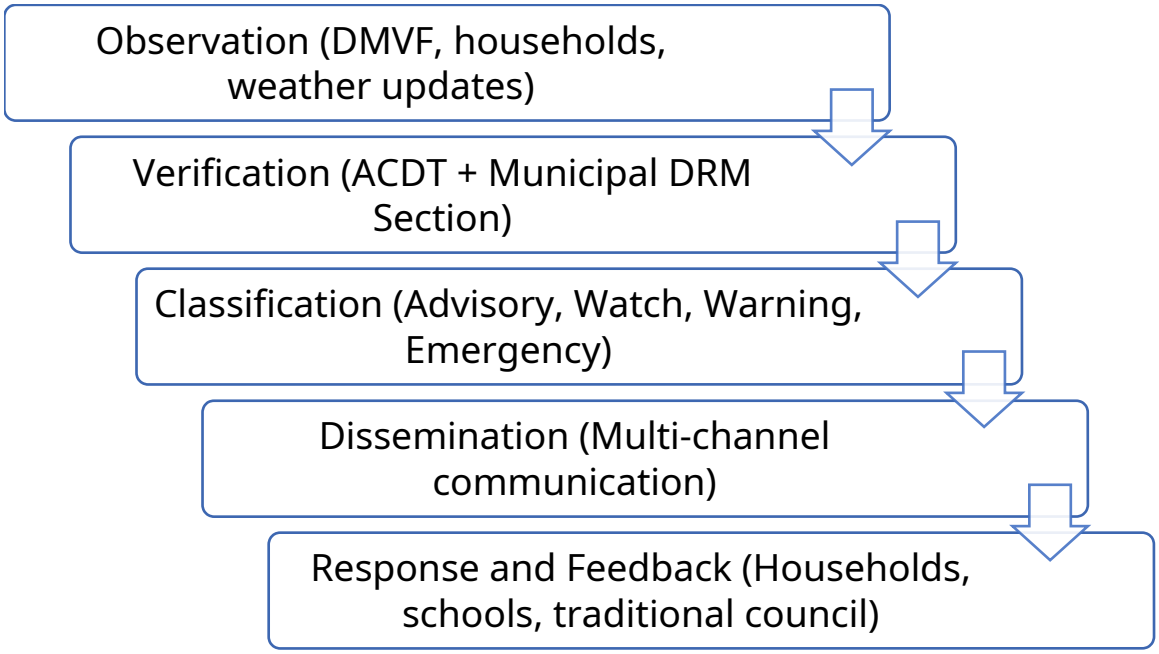
In the Armoede context, this framework integrates scientific observation (meteorological and hydrological), indigenous knowledge, and community reporting through a decentralised system led by the Armoede Community Development Trust (ACDT) and implemented by the Disaster Management Volunteer Force (DMVF).

5.2 Principles Guiding the Protocol
The protocol is anchored in the following principles, distilled from decades of field learning and global best practice:

- Speed and Accuracy: The first message must be the right message — timely, verified, and actionable.
- Trust and Credibility: Communities act when they trust the source; hence, all warnings are authenticated through established structures.
- Inclusivity and Reach: No household or vulnerable group should be left uninformed or unable to respond.
- Two-Way Communication: Early warning is as much about *receiving* local signals as it is about *sending* official alerts.
- Redundancy and Resilience: Multiple communication pathways ensure continuity during power or network failures.

5.3 Structure of the Early Warning System
Figure A5.1 (below) presents the Armoede Early Warning and Emergency Communication Flow, linking detection, validation, and dissemination.

Figure A5.1: Flow of Early Warning and Communication in Armoede



This model reinforces local agency within a structured command chain, ensuring that early action is triggered at the lowest competent level.

5.4 Warning Levels and Triggers

Early warnings are graded into four levels of escalating seriousness, harmonised with district and national protocols.

Level	Trigger / Condition	Examples of Hazard Indicators	Lead Activator	Expected Community Action
Level 1 – Advisory	Early signs of potential hazard	Continuous rainfall, rising water levels, reports of veld burning	DMVF Early Warning Unit	Observe, monitor, and alert leadership; prepare for update
Level 2 – Watch	Probability of hazard within 48–72 hours	Forecasted heavy rainfall, strong winds, power instability	ACDT Early Warning Lead	Ready equipment, check communication tools, brief households
Level 3 – Warning	Confirmed threat or hazard likely within 24 hours	Local flooding, veldfire moving toward settlement	ACDT + DMVF	Issue public alerts; activate partial standby and community mobilisation

Level 4 – Emergency	Hazard event confirmed or unfolding	Flash floods, uncontrolled fire, disease outbreak	Incident Commander (ACDT)	Execute emergency response; evacuate and protect lives

Each level corresponds with pre-defined action cards held by DMVF team leaders, ensuring discipline and uniformity across responses.

5.5 Communication Channels and Protocols
The Armoede system adopts a multi-channel, multi-institutional approach, drawing from field-tested communication redundancy principles:

Primary Channels:
- ACDT-verified WhatsApp broadcast lists (community-wide, bilingual).
- Loudhailer notifications through trained volunteers and vehicle-mounted systems.
- Direct liaison with Traditional Council, Ward Committee, School Principals, and Clinic Head.
- SMS or voice alerts for off-grid or remote households.

Backup Channels:
- Local faith-based and cultural institutions.
- Door-to-door messengers for elderly or disabled persons.
- Local radio (where possible) for wide-area alerts.

Each channel is tested quarterly, and activation roles are clearly mapped in the ACDT Emergency Communication Directory (updated annually).

5.6 Verification and Escalation Mechanism
All alerts follow a four-step verification process before dissemination to prevent false alarms or panic.

Step	**Verification Action**	**Responsible Entity**
1	Receive forecast, observation, or local report	DMVF Early Warning Unit
2	Validate data with the municipal or district DRM Centre	ACDT Early Warning Lead
3	Classify and authorise the warning level	ACDT Chairperson or delegated Incident Commander
4	Disseminate through designated communication channels	DMVF Communication Team
5	Notify the District DMC and maintain event log	ACDT Secretariat

This procedure ensures scientific accuracy, chain of command integrity, and accountability consistent with NDMF Enabler 2: Information Management and Communication Systems.

5.7 Warning Message Format
Messages must be **action-oriented** and avoid technical jargon. All warnings follow a unified format for clarity and consistency.

Component	Example
Header	"⚠ COMMUNITY FLOOD WARNING: ZONE 3 AT RISK"
Hazard Description	"Persistent rainfall and overflowing channels detected in lower Armoede."
Impact Summary	"Low-lying households and bridges may be affected within 6–12 hours."
Immediate Action Required	"Move valuables, avoid stormwater channels, and stand by for evacuation notice."
Issued by/Contact	"ACDT Early Warning Unit (Tel: 015 XXX XXXX). Valid until 18:00, 21 October."

This structure ensures the message is both technically sound and socially understandable, bridging science and community comprehension.

5.8 Inclusivity and Accessibility

The protocol recognises that warnings are only effective when accessible to all. Measures include:

- Bilingual messaging in English and Sepedi.
- Visual symbols (e.g., coloured flags, icons) for semi-literate recipients.
- Community focal points for elderly and persons with disabilities.
- School-based awareness so children act as secondary communicators within households.

The approach mirrors lessons from UNDRR's Global Platform (2023) on people-centred early warning systems, ensuring no one is left behind.

5.9 Review, Testing, and Continuous Improvement

The ACDT and DMVF conduct:

- Quarterly communication drills simulating real scenarios (fire, flood, drought).
- Annual multi-stakeholder review workshops involving Traditional Council, Ward Committee, schools, and local service departments.
- Post-event evaluations to assess timeliness, message clarity, and behavioural response.

Findings are documented in the M&E and Learning Framework (Annexure 8) to inform protocol refinement.

5.10 Institutional Roles

Function	Primary Responsibility	Supporting Role
System Oversight	ACDT (Early Warning Subcommittee)	Mogalakwena Local Municipality DRM Section
Monitoring & Observation	DMVF Early Warning Team	Community Watch Points
Communication Activation	DMVF Communication Officers	Ward Committee, School Heads
Escalation & Reporting	ACDT Secretariat	Waterberg District DMC
Feedback & Learning	ACDT + DMVF	Traditional Council, NGOs

5.11 Concluding Reflection

The Armoede Early Warning and Emergency Communication Protocol is more than a procedural tool. It is a *manifestation of institutional maturity and social trust*. It converts early warning into early action, embodying the highest principles of risk governance and community resilience.

As affirmed in over three decades of practice and research:

> ***"A warning not understood is a warning not issued."***

This annexure ensures that in Armoede, every signal becomes an instruction, and every message saves a life.

Annexure 6: Incident Response and Activation Flowchart

Referenced in Section 5.5 - Response Strategy and Section 5.7- Contingency Planning

Purpose
To present a clear and standardised process for activation, coordination, and escalation of incident response within the Armoede community. The flowchart and supporting framework ensure disciplined decision-making, unity of command, and timely deployment of available resources during any emergency. The system empowers the community to act swiftly while maintaining accountability and alignment with the Mogalakwena Local Municipality and Waterberg District Disaster Management Centres.

6.1 Conceptual Foundation
Incident response requires a balance between structure and flexibility. Drawing from more than three decades of operational experience in Africa, it is evident that community-based response functions best when:

i) Roles and lines of authority are clearly defined and widely understood.
ii) Decisions are taken as close as possible to the point of impact.
iii) Communication flows are short, direct, and verifiable.

The Armoede framework therefore follows a three-tier activation structure:

- Level 1 (Local Containment): Incidents manageable within the community using local resources.
- Level 2 (Supported Response): Incidents that exceed community capacity and require municipal or sectoral assistance.
- Level 3 (Escalated Disaster): Incidents beyond local and municipal capacity that require district or provincial coordination.

This model is consistent with the National Disaster Management Framework's emphasis on decentralised but coordinated response capability.

6.2 Activation Triggers
Incident response may be initiated through different types of triggers. The table below outlines how each trigger type originates and the corresponding verification process.

Table A6.1: Activation Triggers and Verification Pathways

Trigger Type	Source Example	Verification Pathway
Observation-Based Trigger	Household reports smoke, flooding, or structural collapse	DMVF verifies and informs ACDT Response Lead
Forecast-Based Trigger	Early warning from meteorological or health authority	ACDT validates and prepares response teams
Instruction-Based Trigger	Directive from Mogalakwena LM or District DMC	Immediate activation of coordination centre and DMVF

All triggers must be logged in the Incident Register, which is maintained by the ACDT Secretariat and linked to the Early Warning and Communication Protocol (Annexure 5).

6.3 Command Structure

The Armoede Incident Command System (ICS) adapts the national model to the local context. It ensures unity of command, operational clarity, and coordinated reporting.

Table A6.2: Local Incident Command Structure

Role	Designation / Responsibility
Incident Commander	ACDT Chairperson or delegated officer
Operations Lead	DMVF Team Leader for Response
Logistics Lead	Ward Committee representative supported by Traditional Council liaison
Planning and Intelligence Lead	DMVF Risk Information Officer
Communication and Liaison Lead	ACDT Secretariat

All functions report to the ACDT Board for overall oversight and compliance.

6.4 Incident Response and Activation Flow

The process below illustrates how an incident moves from detection to stabilisation and closure. It demonstrates the logical flow of decision-making and action.

Figure A6.1: Incident Response and Activation Flow

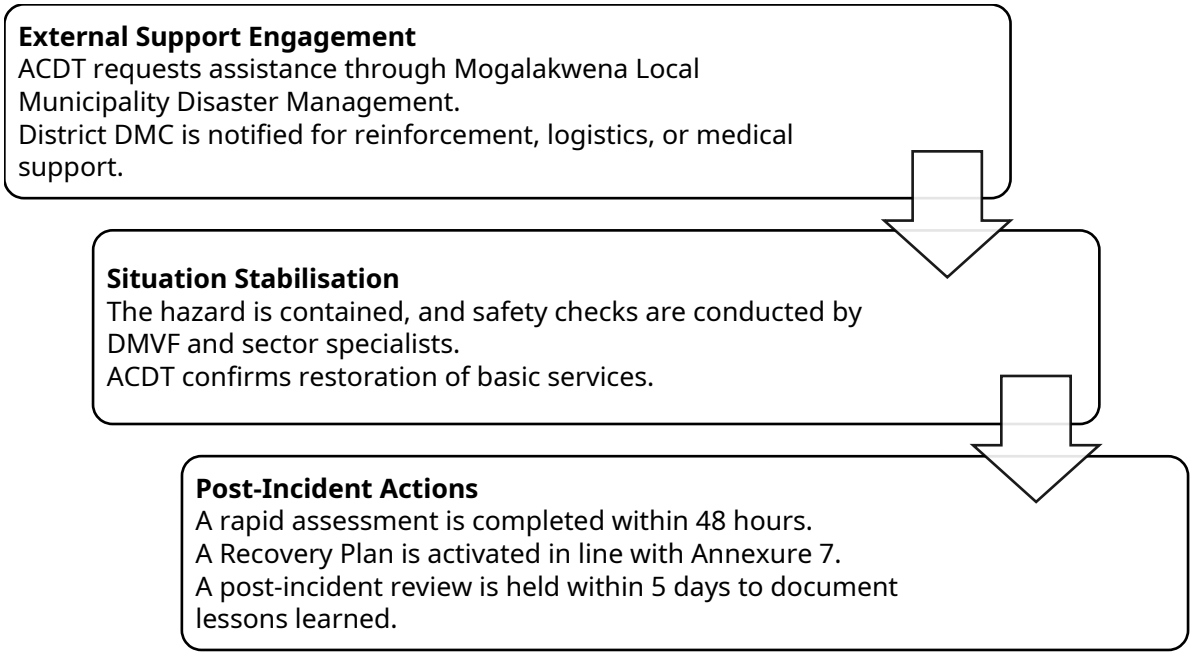
Incident Detection and Reporting
Hazard observed or reported by household, school, clinic, or DMVF volunteer.
The report is relayed immediately to the ACDT Response Lead.

Initial Verification
DMVF verifies the report on-site through direct observation, photography, or eyewitness accounts, and determines scale and severity.

Incident Classification
The event is classified as:
Level 1: Manageable locally by DMVF.
Level 2: Requires municipal or sectoral support.
Level 3: Major incident requiring district or provincial activation.

Activation and Mobilisation
The ACDT Incident Commander declares the activation level and mobilises response teams.
Community alerts are disseminated through the Early Warning Protocol (Annexure 5).

Response Coordination
On-site coordination posts are established.
Roles for evacuation, safety, documentation, and relief are assigned.
Communication with Traditional Council, Ward Committee, SAPS, and Fire Services is maintained.



6.5 Command Communication Matrix

Effective communication throughout the response cycle is crucial for maintaining operational control. The matrix below outlines the communication pathway, responsible parties, mediums, and expected outputs for each stage.

Table A6.3: Command Communication Matrix

Stage	Primary Communicator	Recipient(s)	Medium	Expected Output
Detection	Reporter (DMVF or Household)	ACDT Response Lead	Call or Message	Verified alert
Activation	ACDT Incident Commander	DMVF, Traditional Council, Ward Committee	Broadcast, Loudhailer	Mobilisation order
Coordination	ACDT Secretariat	LM DRM, Clinic, SAPS	Official Notification	Resource deployment
Situation Report	DMVF Operations Lead	ACDT Board, District DMC	Report Template	Situation Report
Closure	ACDT Chairperson	All Stakeholders	Meeting and Circular	Incident Closure Record

6.6 Operational Thresholds for Escalation

To ensure that incidents are managed proportionally and without confusion of authority, the thresholds below define when and how escalation occurs.

Table A6.4: Operational Thresholds for Escalation

Response Level	Indicative Criteria	Authority to Activate	External Notification
Level 1 – Local Containment	Limited to a single zone or fewer than five households	DMVF Response Lead	None
Level 2 – Supported Response	Multi-zone impact or infrastructure affected	ACDT Incident Commander	Mogalakwena LM
Level 3 – Declared Disaster	Widespread damage, fatalities, or displacement	ACDT Chairperson (through LM)	District and Provincial DMC

These thresholds preserve coordination discipline and align with disaster classification standards in the South African Disaster Management Framework.

6.7 Institutional Roles and Responsibilities

The following table clarifies the respective responsibilities of the main entities involved in incident response and coordination.

Table A6.5: Institutional Roles and Responsibilities

Entity	Core Responsibilities
Armoede Community Development Trust (ACDT)	Lead coordination, incident command, liaison, and reporting
DMVF	Field response, first aid, safety, and search operations
Traditional Council	Community mobilisation, access facilitation, and oversight of safety
Ward Committee	Communication support, logistics, and local reporting
Local Clinic	Emergency medical support and health reporting
Mogalakwena LM and Waterberg DMC	Technical reinforcement, transport, and specialist support

6.8 Post-Incident Learning and Reporting

Each incident, regardless of scale, must generate learning to strengthen preparedness and institutional maturity. This process consists of:

- Operational Debrief: Held within 24 hours of event stabilisation.
- After-Action Report (AAR): Compiled within 7 days by DMVF and validated by ACDT.
- Integration into Monitoring and Learning Systems: Findings incorporated into the M&E Framework (Annexure 8).

6.9 Concluding Reflection

The Armoede Incident Response and Activation Framework is not a theoretical construct but a practical and proven system rooted in community participation and institutional discipline. It transforms local capacity into a coordinated force that can prevent loss, reduce suffering, and accelerate recovery.

As years of DRM practice have shown:

“Communities that plan, communicate, and act together recover faster and stronger than those that wait for help to arrive.”

Annexure 7: Recovery and Continuity Action Framework

Section 5.6 (Recovery and Continuity Plan) and Section 6.6 (Monitoring, Evaluation and Learning)

Purpose
To provide a structured approach for restoring normalcy after an incident while embedding resilience-building measures that reduce future disaster risk. The framework ensures that post-disaster recovery in Armoede is coordinated, equitable, and development-oriented rather than reactive. It defines recovery phases, institutional roles, financing options, and monitoring parameters aligned with the principles of “Build Back Better” and “Leave No One Behind.”

7.1 Conceptual Foundation

Recovery is both a process and an opportunity. It aims not merely to return to pre-disaster conditions but to transform vulnerabilities into strengths. The Armoede approach to recovery is guided by four interrelated principles:

1. Restoration with Improvement: Every reconstruction activity should enhance resilience and reduce hazard exposure.
2. Community Ownership: Recovery must be locally led, supported by technical and financial partners.
3. Integrated Planning: Recovery must align with broader development and land-use planning instruments.
4. Equity and Inclusion: Vulnerable households must receive targeted support to ensure no one is left behind.

The recovery process in Armoede follows three interdependent phases (that is the Early Recovery, Intermediate Recovery, and Long-Term Rehabilitation) all supported by continuous learning and accountability mechanisms.

7.2 Recovery Phases and Objectives

The three phases of recovery are outlined in Table A7.1. Each phase builds on the previous one, ensuring smooth transition from emergency relief to sustainable development.

Table A7.1: Recovery Phases and Core Objectives

Recovery Phase	Timeframe	Primary Objective	Key Activities
Early Recovery	0–3 months	Stabilise affected households and re-establish essential services	Rapid needs assessment, emergency shelter, water and sanitation restoration, temporary livelihood support
Intermediate Recovery	3–12 months	Repair damaged infrastructure and restore community services	Rehabilitation of roads, schools, clinics, small-scale irrigation, livelihood diversification
Long-Term Rehabilitation	12–36 months	Achieve sustainable restoration and integrate risk reduction into development	Reconstruction of resilient housing, land rehabilitation, governance strengthening,

			institutional learning

7.3 Governance and Coordination for Recovery

Recovery governance is anchored in the Armoede Community Development Trust (ACDT) as the principal coordinating entity. Its oversight ensures that recovery activities remain transparent, inclusive, and well-aligned with local and district frameworks.

The table below outlines institutional roles for effective coordination and accountability.

Table A7.2: Institutional Roles in Recovery Governance

Entity	Core Responsibilities
Armoede Community Development Trust (ACDT)	Lead coordination, oversight of funding, and alignment with strategic recovery priorities
DMVF Recovery Unit	Field coordination, household recovery support, and damage verification
Traditional Council	Community mobilisation, access facilitation, and dispute resolution
Ward Committee	Local reporting, liaison with municipal services, and monitoring community needs
Mogalakwena Local Municipality	Technical assessment, infrastructure rehabilitation, and integration into municipal plans
Waterberg District DMC	Oversight, logistics, and access to district-level support mechanisms
Non-Governmental and Faith-Based Organisations	Support provision of relief materials, psychosocial care, and technical aid

7.4 Recovery Financing and Resource Mobilisation

Effective recovery depends on timely mobilisation of resources. The financing strategy combines community contributions, Trust resources, government allocations, and external donor support.

The resource mobilisation mechanism operates through the following pathways:

- Internal Financing: Utilisation of ACDT emergency and contingency funds.
- Public Sector Support: Requests to Mogalakwena LM and Waterberg DMC for infrastructure repair and technical services.
- Private Sector Partnerships: Engagement with local mines, businesses, and insurers for funding or materials.
- Community Contributions: Labour, local materials, and volunteerism for small-scale works.
- External Assistance: Formal requests to national or international humanitarian partners if recovery needs exceed local capacity.

Table A7.3: Recovery Resource Mobilisation Framework

Source	Instrument / Mechanism	Purpose of Funds	Accountability Channel
ACDT Contingency Reserve	Direct fund allocation	Immediate community-level recovery	ACDT Finance Committee
Municipal and District Budgets	Disaster relief and rehabilitation grants	Repair of public infrastructure	LM and DMC reporting

Corporate Social Investment	Partner agreements with local mines and companies	Livelihood recovery, training, and materials	Joint Recovery MoUs
NGOs and Donor Agencies	Project grants or in-kind support	Shelter, food security, psychosocial support	Donor reporting templates
Community Contributions	Voluntary labour and materials	Minor repairs and public clean-ups	DMVF documentation and oversight

7.5 Monitoring, Learning, and Accountability

Monitoring ensures transparency and helps track progress across recovery phases. ACDT, in collaboration with DMVF, uses a structured approach that integrates both qualitative and quantitative indicators. Learning and reflection are embedded into every stage of recovery.

The monitoring framework below presents performance indicators to be tracked through periodic progress reviews.

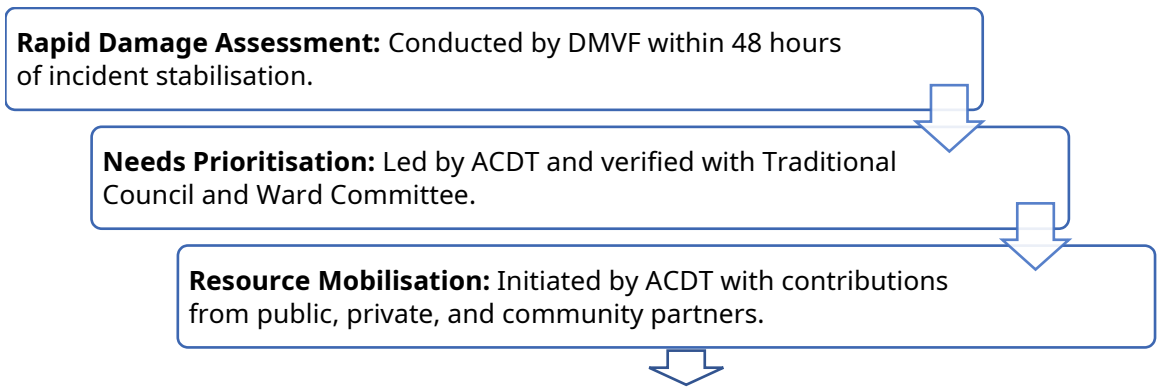
Table A7.4: Recovery Monitoring and Learning Framework

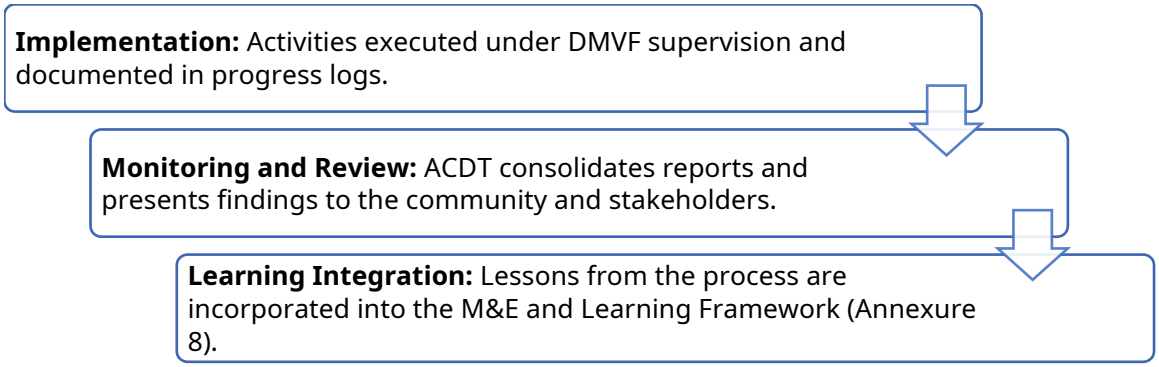
Objective	Key Indicators	Data Source	Reporting Frequency
Restoration of critical infrastructure	Number of facilities restored to operational level	Site inspection reports, LM records	Quarterly
Livelihood recovery	Number of households resuming productive activities	Household surveys, DMVF logs	Biannual
Social inclusion	Proportion of women and vulnerable households benefiting from recovery aid	Community verification and beneficiary lists	Quarterly
Institutional learning	Number of post-incident reviews and knowledge products produced	ACDT and DMVF reports	Annual
Community satisfaction	Community perception of fairness and transparency in recovery	Feedback sessions and focus groups	Annual

7.6 Post-Disaster Recovery Workflow

The following process flow ensures that recovery actions are systematic, inclusive, and progressively developmental.

Figure A7.1: Post-Disaster Recovery Workflow





7.7 Transition to Development

Recovery in Armoede is not a temporary process but a bridge to sustainable development. The transition from relief to rehabilitation and development is achieved through three reinforcing pathways:

- Institutional Integration: Recovery plans are absorbed into ACDT's annual workplan and Mogalakwena LM's Integrated Development Plan (IDP).
- Economic Revitalisation: Livelihoods are diversified through community enterprises and agricultural revitalisation.
- Resilient Infrastructure Development: Reconstruction integrates climate adaptation, local materials, and hazard-proof design.

These pathways transform recovery into a development opportunity rather than a return to vulnerability.

7.8 Concluding Reflection

The Recovery and Rehabilitation Framework affirms that the success of post-disaster action lies not in the speed of rebuilding but in the quality of renewal. True recovery strengthens governance, rebuilds trust, and restores dignity.

As experience from decades of disaster management practice teaches us:

"Recovery is not the end of a disaster; it is the beginning of resilience."

Annexure 8: Monitoring, Evaluation and Learning (MEL) Framework

Sections 6.6 (Monitoring, Evaluation and Learning) and 7.5 (Recovery Monitoring and Learning Framework)

Purpose
To provide a consolidated framework that guides systematic monitoring, periodic evaluation, and continuous learning in the implementation of the Armoede Disaster Risk Management (DRM) Plan. This annexure builds upon the operational tables in Chapters 5 and 6 and introduces an overarching mechanism for performance tracking, knowledge generation, and adaptive improvement aligned with the Sendai Framework Priority 4, NDMF KPA 4, and the Mogalakwena Local Municipality's performance management system.

8.1 Conceptual Basis
Monitoring and Evaluation (M&E) are not mere compliance tools but essential processes for accountability, institutional learning, and adaptive resilience. In the Armoede context, M&E ensures that:

- Activities under each DRM function are implemented as planned.
- Evidence informs decision-making, resource allocation, and partnership engagement.
- Lessons from implementation feed back into policy and practice improvement.

The Monitoring, Evaluation and Learning (MEL) system therefore integrates three complementary streams:

1. Performance Monitoring: Tracking implementation progress, resource use, and outputs against targets.
2. Evaluation: Periodic assessment of outcomes, effectiveness, and impact on resilience.
3. Learning: Capturing knowledge, best practices, and lessons to enhance future planning and response.

The overall process is illustrated in Figure A8.1, which depicts the continuous cycle of planning, implementation, monitoring, evaluation, reflection, and adaptation.

Figure A8.1: MEL System Flow



8.2 Institutional Responsibilities

Monitoring is coordinated through the Armoede Community Development Trust (ACDT) as the accountable institution, supported by the Disaster Management Volunteer Force (DMVF), Traditional Council, and Ward Committee. Oversight and technical guidance are provided by the Waterberg District Disaster Management Centre (DMC) and Mogalakwena Local Municipality.

Table A8.1: Institutional Roles for Monitoring and Learning

Entity	Key Responsibilities
ACDT	Overall coordination, data validation, consolidation of reports, and feedback to community and partners
DMVF	Field-level monitoring, data collection, documentation of incidents, and lesson sharing
Traditional Council	Oversight of compliance, community mobilisation for surveys, and inclusion assurance
Ward Committee	Verification of outputs and liaison with municipal structures
Waterberg DMC	External verification, evaluation support, and reporting alignment
Partner Organisations	Joint monitoring, technical input, and dissemination of findings

8.3 Performance Framework

The MEL framework uses a hierarchy of results indicators (inputs, outputs, outcomes, and impacts) consistent with both development evaluation and disaster risk management logic. The matrix below defines what is monitored, how it is measured, and how often reviews occur.

Table A8.2: Results-Based Monitoring Framework

Results Level	Focus of Measurement	Example Indicators	Data Sources	Reporting Frequency
Input	Resources allocated and used	• Budget utilisation • Staff time • Volunteer mobilisation	Financial records, activity logs	Quarterly
Output	Delivery of planned activities	• Number of trainings held • Length of firebreaks cleared • Infrastructure repaired	Project reports, DMVF logs	Quarterly
Outcome	Behavioural or system change achieved	• % of households with early warning access • Reduction in fire outbreaks	Household surveys, evaluation studies	Annual

Impact	Long-term resilience improvements	• Reduction in losses • Increased water security • Reduced response times	Disaster loss database, municipal reports	Biennial

8.4 Data Management and Quality Assurance

Monitoring quality depends on reliable data. Data management in Armoede will follow a Community–Trust–District data pathway ensuring consistency and integrity.

Table A8.3: Data Management Protocol

Process Step	Lead Actor	Tools / Mechanisms	Verification Approach
Data Collection	DMVF teams	Field forms, mobile data capture	Supervisor checks
Data Consolidation	ACDT Secretariat	Excel dashboard, community registers	Cross-verification with partner data
Data Validation	ACDT M&E Officer	Review meetings, sample audits	Triangulation with municipal data
Reporting	ACDT Board	Quarterly reports and presentations	Approval minutes
Archiving and Learning	ACDT & Ward Committee	Shared digital and physical repository	Annual review and update

8.5 Evaluation Approach

Evaluation assesses not only whether planned activities were implemented but also whether they were effective and relevant in reducing risk.

Evaluations will be guided by five key principles:

1. Relevance: Alignment of activities with local priorities and hazards.
2. Efficiency: Optimal use of resources and time.
3. Effectiveness: Achievement of intended outcomes.
4. Sustainability: Ability to maintain results beyond project cycles.
5. Learning and Adaptation: Use of evidence for continual improvement.

Table A8.4: Evaluation Plan

Evaluation Type	Purpose	Frequency	Lead Responsibility	Output
Process Evaluation	Assess implementation efficiency	Annually	ACDT	Annual performance report
Outcome Evaluation	Measure results and behavioural change	Every 2 years	ACDT with Waterberg DMC	Evaluation report
Impact Evaluation	Assess long-term resilience outcomes	Every 5 years	ACDT and partners	Impact report
After-Action Review	Capture lessons after major incidents	As needed	DMVF & ACDT	After-action summary

8.6 Learning, Reflection, and Knowledge Uptake

Learning is an intentional process embedded across all DRM functions. It turns data into wisdom by enabling reflection, innovation, and corrective action.

Learning will occur through structured quarterly learning circles, annual review workshops, and documentation of case studies on community response, recovery, and prevention.

Table A8.5: Learning and Knowledge Uptake Activities

Learning Mechanism	Frequency	Responsible Unit	Expected Output
Quarterly Reflection Sessions	Quarterly	ACDT Secretariat, DMVF	Learning briefs and practice notes
Annual Review Workshop	Annually	ACDT, Ward Committee, Traditional Council	Consolidated annual report and plan adjustments
Case Study Development	Annually	ACDT Research and Documentation Team	Knowledge products and shared best practices
Peer Exchange Visits	Every 2 years	ACDT with Waterberg DMC	Exchange reports and replication plans
Community Feedback Dialogues	Biannual	DMVF and Traditional Council	Community satisfaction reports

8.7 Reporting and Feedback Mechanisms

The reporting chain ensures accountability while maintaining two-way communication between the community and oversight bodies.

Table A8.6: Reporting and Feedback Framework

Level	Reporting To	Primary Output	Feedback Mechanism
Community / DMVF	ACDT	Activity updates, incident logs	Community meetings
ACDT	Ward Committee & LM DRM	Quarterly and annual reports	Review meetings
LM DRM	District DMC	Consolidated municipal report	Joint coordination sessions
District DMC	Provincial DRM	District-level summary report	Policy briefs and feedback loop

8.8 Indicator Alignment with Sendai Framework

To ensure international coherence, the community-level indicators are mapped against Sendai Framework Targets A–E. This ensures that local reporting contributes to national and global DRR tracking.

Table A8.7: Alignment of Local Indicators to Sendai Targets

Sendai Target	Global Focus	Local Indicator for Armoede
Target A	Reduce mortality	Number of lives saved through early warning
Target B	Reduce affected population	% of households with restored basic services within 7 days
Target C	Reduce economic loss	Value of community infrastructure restored

Target D	Reduce damage to critical infrastructure	% of functional facilities within 1 month post-event
Target E	Increase DRR strategies	Existence and annual review of the Armoede DRM Plan

8.9 Concluding Reflection

The Monitoring, Evaluation and Learning Framework transforms the DRM Plan from a static document into a living system that evolves through feedback, participation, and evidence. It ensures that accountability is upward (to authorities), horizontal (to partners), and downward (to the community).

> **"*Measurement without learning is bureaucracy; learning without measurement is guesswork. Together, measurement and learning, they create wisdom.*"**

Annexure 9: Stakeholder Coordination and Support Matrix

Referenced in Section 6.4 (Institutional Coordination)

Purpose

To clarify the relationships, mandates, and coordination pathways among all stakeholders engaged in disaster risk management within Armoede. The matrix ensures that operational and support actors understand their roles, avoid duplication, and adhere to the Trust-centred command, coordination, and communication system.

Coordination Architecture

The Armoede Disaster Risk Management architecture is designed around the principle that *local ownership drives effective coordination*. The Armoede Community Development Trust (ACDT) serves as the entry point and overall coordination authority, ensuring alignment with the Disaster Management Act (2002) and the district-provincial-national DRM system. The DMVF functions as the operational arm, while the Traditional Council safeguards customary governance, local legitimacy, and social cohesion. Support services, such as municipal, district, sectoral, and private partners, provide technical, logistical, and financial assistance in accordance with established protocols. The matrix below consolidates these linkages across the five functional pillars of disaster management.

Figure A9.1: Coordination Hierarchy Overview

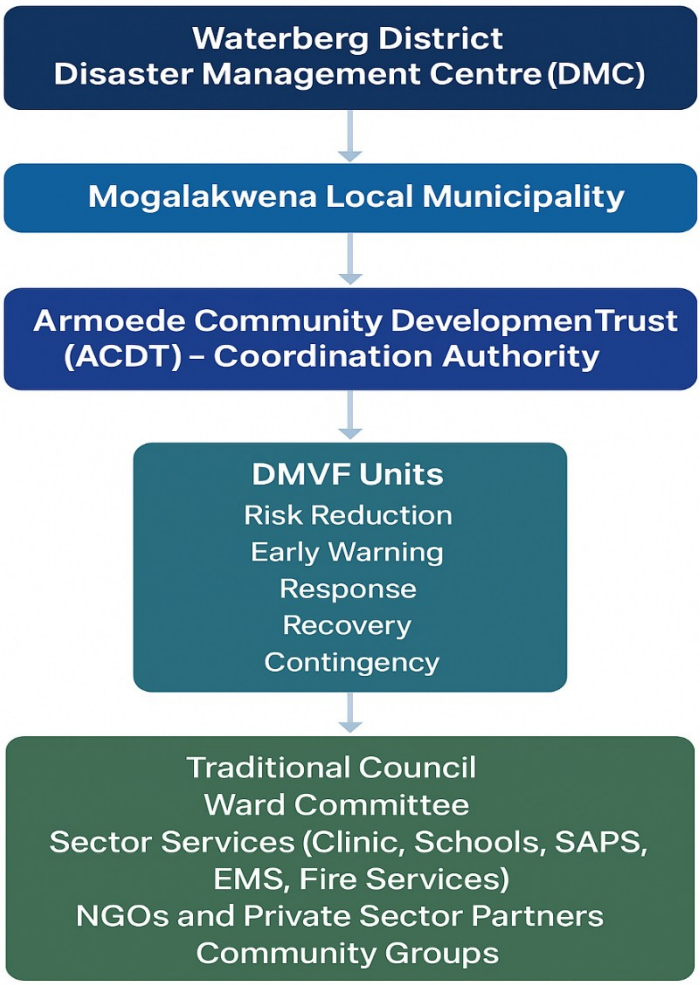

Waterberg District
Disaster Management Centre (DMC)
Mogalakwena Local Municipality
Armoede Community DevelopmenTrust
(ACDT) – Coordination Authority
DMVF Units
Risk Reduction
Early Warning
Response
Recovery
Contingency
Traditional Council
Ward Committee
Sector Services (Clinic, Schools, SAPS,
EMS, Fire Services)
NGOs and Private Sector Partners
Community Groups

Table A9.1: Stakeholder Coordination and Support Matrix

Disaster Management Function	Lead Responsibility	Operational Execution	Support Structures	Primary Interface / Escalation Pathway	Key Deliverables
Prevention & Mitigation	ACDT	DMVF – Risk Reduction Team	Traditional Council, Farmers Group, Ward Committee	Waterberg DMC → District Municipality	Annual risk-reduction programme implemented; erosion and flood hotspots mitigated
Preparedness & Early Warning	ACDT	DMVF – Early Warning Unit	Schools, Local Clinic, Ward Committee	Mogalakwena LM → Waterberg DMC	Functioning community early-warning and communication system; seasonal readiness campaigns completed
Emergency Response	ACDT (Incident Commander)	DMVF – Response Unit	SAPS, EMS, Fire Services, Traditional Council	Waterberg Disaster Operations Centre	Effective command and control; coordinated evacuation, rescue, and fire suppression
Recovery & Continuity	ACDT	DMVF – Recovery Team	NGOs, Farmers Group, Social Services, Ward Committee	Mogalakwena LM → Provincial Humanitarian Desk	Damage and needs assessment reports; restored access, services, and livelihoods
Hazard-Specific Contingency	ACDT	DMVF – Contingency Team	Clinic, Fire Services, Traditional Council	Waterberg DMC → Provincial Disaster Management Centre	Hazard-specific standby plans operationalised; triggers monitored and acted upon
Training & Awareness Support	ACDT	DMVF – Training Unit	Schools, NGOs, Faith Groups	Mogalakwena LM → District Training Forum	4 annual community training and awareness sessions delivered
Monitoring, Evaluation & Learning (MEL)	ACDT	Executive Administrator & DMVF Secretariat	Traditional Council Chair, Ward Committee	Waterberg DMC → SADC Learning Exchange Network	Quarterly MEL reports; lessons integrated into updated plans

Finance & Resource Mobilisation	ACDT (Trust Board)	Executive Administrator & Finance Desk	Private Sector Partners, NGOs, District Finance Unit	Waterberg DMC → Provincial Treasury	Secured funding and donor coordination; accountability for resource flows
Community Mobilisation & Social Cohesion	ACDT	Traditional Council & Ward Committee	Senior Citizens Forum, Youth Groups, Faith Organisations	Mogalakwena LM → Social Development Unit	Inclusive participation, verified community attendance, and equitable recovery assistance

The coordination hierarchy underpinning disaster risk management in Armoede is summarised in Figure A9.1 and elaborated through the matrix below. It delineates the vertical and horizontal linkages that connect community structures with municipal, district, and provincial systems, ensuring both accountability and coherence in the implementation of the Armoede Disaster Risk Management Plan.

Key coordination principles include:

i) **Hierarchy of Coordination:** The ACDT retains overall authority and convening power; DMVF acts as the technical and operational executor; Traditional Council ensures legitimacy and customary oversight.

ii) **Escalation Protocol:** All disaster-related matters escalate *upward* through ACDT → District DMC → Provincial Centre → NDMC, ensuring vertical coherence.

iii) **Horizontal Linkages:** Coordination with schools, clinics, SAPS, and municipal services ensures integrated multi-sectoral response.

iv) **Information Flow:** All field reports and incident logs flow through the DMVF Secretariat to the ACDT for validation and onward submission.

v) **Review Cycle:** The matrix shall be reviewed annually or after any major incident to update roles and contact details.

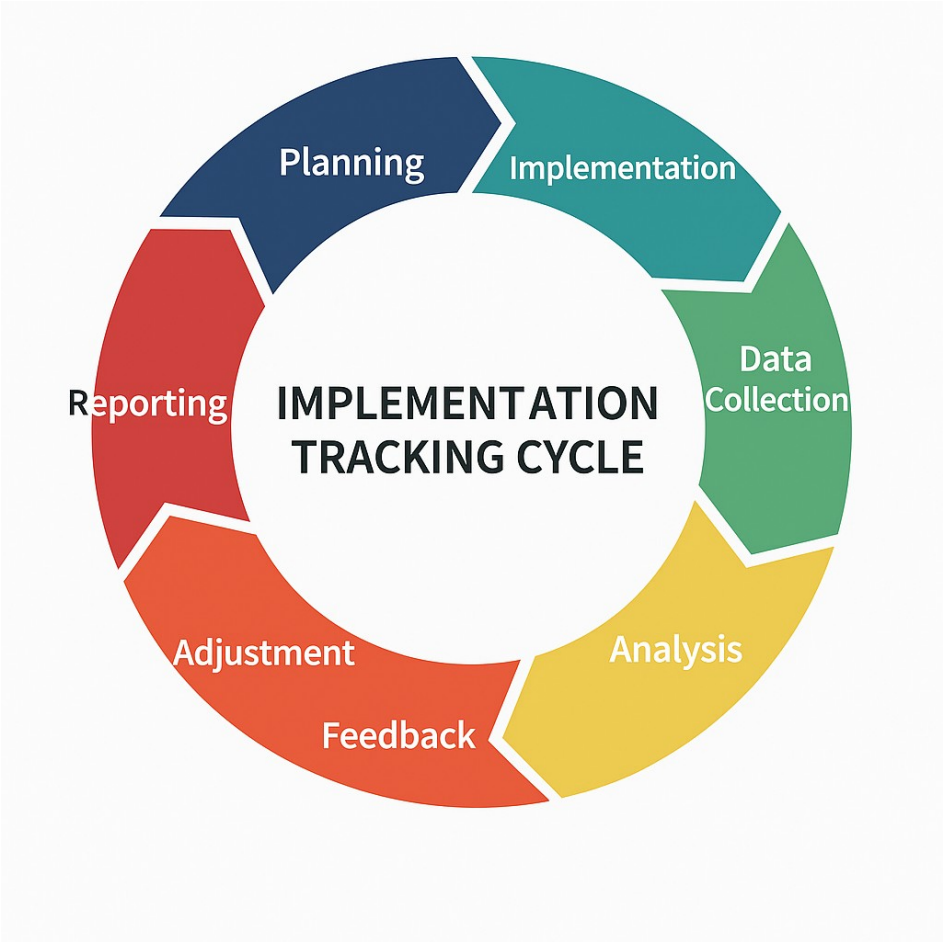
Annexure 10: Implementation Tracking Template

Referenced: Section 6.6 (Monitoring, Evaluation and Learning)

Purpose
This annexure provides a structured tool for tracking the implementation of disaster risk management interventions across all thematic areas and timeframes. It enables the Armoede Community Development Trust (ACDT), Disaster Management Volunteer Forum (DMVF), and partners to monitor progress, identify delays or bottlenecks, assess impact, and ensure that corrective measures are promptly taken. The template supports evidence-based reporting and alignment with district and national monitoring systems.

Implementation Tracking Framework
The template below provides a standard format for monitoring the progress of activities across the prevention, preparedness, response, recovery, and contingency domains. It aligns with the logical framework approach, focusing on inputs, outputs, outcomes, and impacts, while ensuring that monitoring indicators are both quantitative and qualitative.

Figure A10.1: Implementation Tracking Cycle



Below is the completed Implementation Tracking Template (Table A10.1) with realistic, context-specific examples for the Armoede Community Disaster Risk Management Plan. It demonstrates professional-level tracking aligned with community, municipal, and district roles.

Table A10.1: Implementation Tracking Template (Illustrative Example)

No.	Objective / Expected Result	Activity / Action	Lead Responsibility	Support Required	Timeframe	Progress Status	Verification / Evidence	Challenges / Risks	Corrective Actions / Follow-up	Remarks / Next Steps
1	Reduced flood and stormwater risk in high-risk zones	Clear and reopen blocked stormwater channels before rainy season	DMVF Risk Reduction Team	Mogalakwena LM – Technical Services	0–6 months	In Progress (70%)	Photos, progress report, community feedback	Limited tools and manpower	Mobilise additional volunteers and request equipment support from LM	Finalise before next rainfall season
2	Improved community water security and reduced drought impact	Install five 10,000L water tanks linked to borehole	ACDT	Waterberg DMC, local NGOs	6–12 months	Completed	Installation certificate, photos	Transport delays for tanks	Schedule early procurement in future	Plan for solar pumping integration
3	Lower incidence of household fires and electrical hazards	Conduct safety inspections and replace damaged wiring	DMVF Electrical Safety Unit	Mogalakwena LM – Electrical Department	0–6 months	In Progress (40%)	Inspection checklist, incident reports	Inconsistent household access	Awareness campaign to improve access compliance	Include follow-up inspection next quarter
4	Strengthened early warning and preparedness capacity	Conduct emergency simulation and early warning test	ACDT and DMVF Early Warning Unit	Schools, Clinic, Ward Committee	Annual	Completed	Attendance register, evaluation forms	Poor turnout in initial drill	Schedule drills on weekends	Maintain quarterly readiness drills
5	Improved public health through sanitation and hygiene	Construct simple ventilation channels at pit toilets	Households supported by DMVF	Mogalakwena LM – Sanitation Unit	6–24 months	In Progress (55%)	Site inspection records, photos	Shortage of materials	Liaise with LM for supply support	Next round of construction to focus on clinic zone
6	Reduced erosion and	Stabilise erosion gullies	Youth Green Team	ACDT, Farmers	6–24 months	Completed	Site inspection,	Initial drought	Supplemented with irrigation	Extend method to

	land degradation	using brush packing and grass planting		Group			photos	slowed grass growth	support	other risk zones
7	Enhanced data-driven monitoring and reporting	Operationalise digital DRM monitoring register	ACDT Secretariat	DMVF, Ward Committee	0–6 months	Initiated	Register template, digital copy	Limited digital literacy	Training planned for DMVF members	Link system with Waterberg DMC reporting tool

The table is to be updated quarterly and consolidated annually by the DMVF Secretariat, with inputs from all implementing partners and reporting to the ACDT.

Guidance Notes for Use

1. **Objective / Expected Result:** Clearly indicate the strategic or operational goal that the action contributes to (e.g., reduced flood risk, strengthened early warning, improved coordination).

2. **Activity / Action:** Specify the exact intervention being tracked (e.g., construction of diversion channels, community training, emergency drills).

3. **Lead Responsibility:** Identify the accountable structure (e.g., ACDT, DMVF unit, Traditional Council).

4. **Support Required:** Note supporting partners such as Mogalakwena LM, Waterberg DMC, clinics, schools, or NGOs.

5. **Progress Status:** Use qualitative or percentage measures such as “Not Started,” “In Progress (40%),” “Completed,” or “Ongoing.”

6. **Verification / Evidence:** Provide means of verification—e.g., photos, reports, minutes, site inspections, attendance registers.

7. **Challenges / Risks:** Record issues affecting implementation such as resource constraints, coordination delays, or environmental conditions.

8. **Corrective Actions / Follow-up:** Outline remedial actions and identify who will implement them and by when.

9. **Remarks / Next Steps:** Capture additional insights, lessons, or priorities for the next review cycle.

Effective tracking ensures that the Armoede Disaster Risk Management Plan remains a living instrument, that is facilitates continually learning, adapting, and responding to changing risks and realities. It strengthens institutional memory, enhances accountability, and builds the community’s long-term resilience through evidence-based governance and informed decision-making.

Annexure 11: Minimum Equipment List

Referenced in Section 5.4 (Preparedness and Early Warning), and Section 6.4 (Institutional Structure)

Purpose

This annexure outlines the **minimum equipment standards** required for the effective operation of the **Disaster Risk Reduction and Response Task Team (DRRRTT)** in Armoede. It ensures that the community has the tools necessary for preparedness, early response, and recovery, while integrating mechanisms for long-term sustainability through community-based management and limited commercial use of assets.
The standards are informed by **Sphere (2018)**, **NDMC (2017)**, and **Sendai Framework (2015–2030)** guidelines, contextualised for a **rural community of 2,028 residents (487 households)**.

Guiding Principles

- **Adequacy:** Equipment levels match risk exposure and population size.
- **Safety:** All tools must comply with minimum safety and quality standards.
- **Simplicity:** Equipment must be locally operable and serviceable.
- **Durability:** Assets must be built for rugged, field conditions.
- **Sustainability:** Equipment management must allow for self-financing and continuity through practical use and community stewardship.

Table A11.1: Minimum Equipment and Justification

Equipment Item	Minimum Standard (per households/people)	Quantity Required (Armoede: 487 households)	Operational Function / Justification
Fire Beaters	1 per 25 households	20 units	Rapid suppression of veld and grass fires before escalation.
First Aid Kits	1 per 50 households	10 kits	First-response care for injuries and medical emergencies.
Stretchers	1 per 120 households	4 units	Safe evacuation of injured or immobile individuals.
Handheld Radios	1 per 40 households	12 units	Maintains communication when networks fail; enables coordination.
Megaphones	1 per 100 households	5 units	Broadcasts emergency messages and evacuation instructions.
Reflective Vests	1 per DMVF volunteer	30 units	Improves visibility and establishes responder authority.
Emergency Lamps	1 per 50 households	10 units	Ensures visibility during night operations and power outages.
Rescue Ropes & Harnesses	1 per 150 households	4 sets	For flood rescue and stabilising hazardous structures.
Community Siren / Bell System	1 per sector	2 units	Provides wide-area early warnings for floods, fires, and severe weather.
Water Storage (10,000 L JoJo Tanks)	1 per 100 households	5 tanks	Backup water supply for firefighting, hygiene, and drought relief.
Pressurised Towable Water Bowsers	1 per 250 households	**2 units**	1 500–2 000 L pressurised trailers towed by bakkies for firefighting, drought relief, and water distribution. Dual-use model enables revenue generation through event hire to fund maintenance and replacement.
Basic Tool Kits (Spades, Pickaxes, Rakes)	1 per DMVF unit	6 sets	For drain clearing, debris removal, and minor repair work.
Protective Gloves & Masks	1 per volunteer	30 pairs	Ensures personal safety during clean-up and biohazard exposure.
Rain Gear (Ponchos and Gumboots)	1 per volunteer	30 sets	Allows safe operation in wet or stormy conditions.
Hand Extinguishers	1 per facility	6 units	Initial suppression of small fires in public facilities.
Whistles and Flashlights	1 per field responder	30 each	Low-cost signalling and night navigation tools.

Water Bowser Stationing and Rationale

To ensure equitable access, **two pressurised towable water bowsers** will be strategically stationed as tabulated below.

Stationing Point	Unit Assigned	Rationale
Community Hall	1 x Pressurised Water Bowser	Centrally located and accessible for rapid deployment to the lower and western sections of Armoede. Proximity to key community gathering points makes it ideal for firefighting, drought relief, and event hire. The Hall also hosts most major functions, enabling oversight by the ACDT and trained DMVF volunteers.
Study Centre	1 x Pressurised Water Bowser	Located towards the eastern and upper settlement zones, enabling swift deployment to outlying areas. The Study Centre already serves as a DRM coordination hub, making it suitable for water supply management and emergency dispatch.

Operational and Security Rationale:

To ensure safe custody, security personnel stationed at the Community Hall and Study Centre must be drawn from individuals who have completed the ACDT's Disaster Risk Management (DRM) First Responder training.

This ensures that:

- Security staff can respond immediately to fire or flooding incidents.
- Assets remain safeguarded by individuals who understand the operational and safety requirements of the equipment.
- Bowser operation and dispatch are always managed by trained and accountable personnel, ensuring readiness, traceability, and responsible use.
- The same personnel can assist the DMVF in water supply, hose connection, or pump activation when needed.

Procurement Standards and Technical Specifications

Equipment Item	**Technical Specification**	**Minimum Quality Standard / Certification**
Fire Beaters	Rubber paddle ≥ 250 mm; handle 1.2 m	SABS approved
First Aid Kits	Standard multi-compartment kit with antiseptic, CPR mask, and dressings	SANS 10287
Stretchers	Foldable aluminium; ≥ 120 kg load	ISO 13485
Handheld Radios	UHF 400–470 MHz; ≥ 5 km range	ICASA / IP54
Megaphones	≥ 20 W output; ≥ 500 m range	CE / SABS
Reflective Vests	High-visibility polyester with reflective strips	EN ISO 20471
Emergency Lamps	Rechargeable LED ≥ 800 lumens; ≥ 10 h runtime	SABS IEC 60598
Rescue Ropes & Harnesses	10 mm static rope; ≥ 22 kN strength	EN 1891 / EN 358
Community Siren / Bell	Solar or manual, ≥ 800 m audible range	IP65
Water Storage Tanks	10 000 L polyethylene; lockable tap	SANS 1731
Pressurised Water Bowsers	1 500–2 000 L pressurised steel tank; dual-wheel trailer; 12 V pump (≥ 2 bar); hose reel (≥ 20 m); spray nozzle; pressure gauge; rear lights; tow hitch; high-visibility colour; ACDT/DMVF markings.	SABS / NRCS certified; roadworthy; compliant with National Road Traffic Act
Basic Tool Kit	Spade, pickaxe, shovel, rake, sledgehammer	Industrial-grade
Protective Gloves & Masks	Nitrile gloves, N95 masks	SANS / EN 149
Rain Gear	PVC poncho and gumboots; ≥ 5 000 mm waterproof	SABS

Hand Extinguishers	4.5 kg ABC dry chemical	SABS 1475
Whistles & Flashlights	Metal whistle; LED torch ≥ 6 h runtime	CE / SABS

Management, Maintenance, and Deployment

- **Custodianship:** Managed by the **ACDT Disaster Desk**, with DMVF oversight.
- **Storage:** Under cover at the Community Hall and Study Centre; secured and kept filled at 70–80% capacity to maintain pump integrity.
- **Maintenance:** Monthly pressure checks, hose inspections, and trailer servicing recorded in the DMVF Equipment Register.
- **Revenue Use:** Hire income from bowsers to fund maintenance, tyre replacement, pump servicing, and repainting.
- **Training:** Refresher courses every six months for all DMVF and security staff handling the bowsers.

Compliance and Record-Keeping

- Procurement follows ACDT Procurement Policy and relevant safety standards.
- DMVF Safety Officer maintains a Procurement and Maintenance Register (serials, warranties, inspections).
- Annual joint audit by the ACDT, DMVF Secretariat, and Municipal DMC.
- Financial tracking of bowser income and expenditure ensures transparency and accountability.